HEWLETT-PACKARD

HP-67/HP-97

Stat Pac I

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Introduction

The 21 programs of Stat Pac I have been drawn from the fields of general statistics and related areas.

Each program in this pac is represented by one or more magnetic cards and a section in this manual. The manual provides a description of the program with relevant equations, a set of instructions for using the program, and one or more example problems, each of which includes a list of the actual keystrokes required for its solution. Program listings for all the programs in the pac appear at the back of this manual. Explanatory comments have been incorporated in the listings to facilitate your understanding of the actual working of each program. Thorough study of a commented listing can help you to expand your programming repertoire, since interesting techniques can often be found in this way.

On the face of each magnetic card are various mnemonic symbols which provide shorthand instructions to the use of the program. You should first familiarize yourself with a program by running it once or twice while following the complete User Instructions in the manual. Thereafter, the mnemonics on the cards themselves should provide the necessary instructions, including what variables are to be input, which user-definable keys are to be pressed, and what values will be output. A full explanation of the mnemonic symbols for magnetic cards may be found in appendix A.

If you have already worked through a few programs in the Standard Pac, you will understand how to load a program and how to interpret the User Instructions form. If these procedures are not clear to you, take a few minutes to review the sections, ''Loading a Program'' and ''Format of User Instructions,'' in your Standard Pac.

We hope that Stat Pac I will assist you in the solution of numerous problems in your discipline. We would very much appreciate knowing your reactions to the programs in this pac, and to this end we have provided a questionnaire inside the front cover of this manual. Would you please take a few minutes to give us your comments on these programs? It is in the comments we receive from you that we learn how best to increase the usefulness of programs like these.

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A WORD ABOUT PROGRAM USAGE

This application pac has been designed for both the HP-97 Programmable Printing Calculator and the HP-67 Programmable Pocket Calculator. The most significant difference between the HP-67 and the HP-97 calculators is the printing capability of the HP-97. The two calculators also differ in a few minor ways. The purpose of this section is to discuss the ways that the programs in this pac are affected by the differences in the two machines, and to suggest how you can make optimal use of your machine, be it an HP-67 or an HP-97.

Some of the computed results in this pac are output by PRINTx statements. On the HP-97, these results will be output on the printer. On the HP-67, each PRINT command will be interpreted as a PAUSE: the program will halt, display the result for about five seconds, then continue execution. The term "PRINT/PAUSE" is used to describe this output condition.

If you own an HP-67, you may want more time to copy down the number displayed by a PRINT/PAUSE. All you need to do is press down any key on the keyboard. If the command being executed is PRINTx (eight rapid blinks of the decimal point), pressing down a key will cause the program to halt. If the command being executed is PRINT STACK (two slow blinks of the decimal point), the number in the display will remain there until the depressed key is released; then the next register in the stack will be displayed, and so on. After display of all four registers, the program will halt execution if a key was pressed at any time during the display of the stack contents. In both cases, execution of the halted program may be re-initiated by pressing R/S.

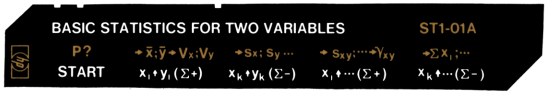
HP-97 users may also want to keep a permanent record of the values input to a certain program. A convenient way to do this is to set the Print Mode switch to NORMAL before running the program. In this mode, all input values and their corresponding user-definable keys will be listed on the printer, thus providing a record of the entire operation of the program.

Another area that could reflect differences between the HP-67 and the HP-97 is in the keystroke solutions to example problems. It is sometimes necessary in these solutions to include operations that involve prefix keys, namely, f on the HP-97 and f, g, and h on the HP-67. For example, the operation 10^x is performed on the HP-97 as f 10^x and on the HP-67 as g 10^x. In such cases, the keystroke solution omits the prefix key and indicates only the operation (as

here, [10^x]). As you work through the example problems, take care to press the appropriate prefix keys (if any) for your calculator.

Also in keystroke solutions, those values which are output by the command PRINTx will be followed by three asterisks (***).

BASIC STATISTICS FOR TWO VARIABLES



This program calculates means, standard deviations, covariance, correlation coefficient, coefficients of variation, sums of data points, sum of multiplication of data points, and sums of squares of data points derived from a set of ungrouped data points $\{(x_i, y_i), i = 1, 2, \ldots, n\}$, or grouped data points $\{(x_i, y_i, f_i), i = 1, 2, \ldots, n\}$. f_i denotes the frequency of repetition of (x_i, y_i).

$$\text{means } \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

$$\text{standard deviations } s_x = \sqrt{\frac{\Sigma x_i^2 - n\bar{x}^2}{n-1}}$$

$$\left(\text{or } s_x' = \sqrt{\frac{\Sigma x_i^2 - n\bar{x}^2}{n}}\right)$$

$$s_y = \sqrt{\frac{\Sigma y_i^2 - n\bar{y}^2}{n-1}}$$

$$\left(\text{or } s_y' = \sqrt{\frac{\Sigma y_i^2 - n\bar{y}^2}{n}}\right)$$

covariance

$$s_{xy} = \frac{1}{n-1}\left(\Sigma x_i y_i - \frac{1}{n}\Sigma x_i \Sigma y_i\right)$$

$$\left(\text{or } s_{xy}' = \frac{1}{n}\left[\Sigma x_i y_i - \frac{1}{n}\Sigma x_i \Sigma y_i\right]\right)$$

$$\text{correlation coefficient } \gamma_{xy} = \frac{s_{xy}}{s_x s_y}$$

$$\text{Coefficients of variation } V_x = \frac{s_x}{\bar{x}} \cdot 100 \quad , \; V_y = \frac{s_y}{\bar{y}} \cdot 100$$

Note: n is a positive integer and $n > 1$.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		f A	1.00
4	For grouped data points, go to 8			
5	For ungrouped data points,			
	do 6~7			
	for i=1, 2,..., n			
6	Input x_i	x_i	ENTER↑	
	y_i	y_i	B	i
7	If you made a mistake in			
	inputting x_k and y_k, then			
	correct by →	x_k	ENTER↑	
		y_k	C	i−1
8	For grouped data points			
	do 9~10			
	for i=1, 2,..., n			
9	Input x_i	x_i	ENTER↑	
	y_i	y_i	ENTER↑	
	f_i	f_i	D	Σf_i
10	If you made a mistake in			
	inputting x_k, y_k and f_k, then			
	correct by →	x_k	ENTER↑	
		y_k	ENTER↑	
		f_k	E	$\Sigma f_i - f_k$
11	Calculate means: $\bar{x}$		f B	$\bar{x}$
	$\bar{y}$		R/S	$\bar{y}$
12	Calculate coefficients of			
	variation: V_x		f B	V_x
	V_y		R/S	V_y

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
13	Calculate standard deviations:			
	s_x		f C	s_x
	s_y		R/S	s_y
	s_x'		f C	s_x'
	s_y'		R/S	s_y'
14	Calculate: covariance			
	s_{xy}		f D	s_{xy}
	s_{xy}'		R/S	s_{xy}'
15	Calculate correlation			
	coefficient γ_{xy}		f D	γ_{xy}
16	Calculate sums: Σx_i		f E	Σx_i
	Σy_i		R/S	Σy_i
	$\Sigma x_i y_i$		R/S	$\Sigma x_i y_i$
17	Calculate sums of squares			
	Σx_i^2		f E	Σx_i^2
	Σy_i^2		R/S	Σy_i^2
	For a new case, go to 2			
	*Note: to clear print mode			
	press →		CLF	
			0	

Example 1:

For the following set of data, find the means, standard deviations, covariance, correlation coefficient, coefficients of variation, and the sums.

x_i	26	30	44	50	62	68	74
y_i	92	85	78	81	54	51	40

Keystrokes: **Outputs:**

A → 0.00

f A → 1.00

26 ENTER↑ 92 B → 26.00 *** (x_1)
92.00 *** (y_1)
1.00 ***

100 ENTER↑ 100 B →	100.00 ***	(x_2)
	100.00 ***	(y_2) (error)
	2.00 ***	
100 ENTER↑ 100 C →	100.00 ***	(x_2)
	100.00 ***	(y_2) (correction)
	1.00 ***	
30 ENTER↑ 85 B →	30.00 ***	(x_2)
	85.00 ***	(y_2)
	2.00 ***	
44 ENTER↑ 78 B →	44.00 ***	(x_3)
	78.00 ***	(y_3)
	3.00 ***	
50 ENTER↑ 81 B →	50.00 ***	(x_4)
	81.00 ***	(y_4)
	4.00 ***	
62 ENTER↑ 54 B →	62.00 ***	(x_5)
	54.00 ***	(y_5)
	5.00 ***	
68 ENTER↑ 51 B →	68.00 ***	(x_6)
	51.00 ***	(y_6)
	6.00 ***	
74 ENTER↑ 40 B →	74.00 ***	(x_7)
	40.00 ***	(y_7)
	7.00 ***	
f B →	50.57 ***	($\bar{x}$)
R/S →	68.71 ***	($\bar{y}$)
f B →	36.58 ***	(V_x)
R/S →	29.10 ***	(V_y)
f C →	18.50 ***	(s_x)
R/S →	20.00 ***	(s_y)
f C →	17.13 ***	($s_{x'}$)
R/S →	18.51 ***	($s_{y'}$)

f D → −354.14 ***(s_{xy})
R/S → −303.55 ***(s'_{xy})
f D → −0.96 *** (γ_{xy})
f E → 354.00 *** (Σx_i)
R/S → 481.00 *** (Σy_i)
R/S → 22200.00***($\Sigma x_i y_i$)
f E → 19956.00***(Σx_i^2)
R/S → 35451.00***(Σy_i^2)

Example 2:

Apply the program to the following set of grouped data.

x_i	4.8	5.2	3.8	4.4	4.1
y_i	15.1	11.5	14.3	13.6	12.8
f_i	1	3	1	6	2

Keystrokes:	**Outputs:**	
A →	0.00	
f A →	1.00	
4.8 ENTER↑ 15.1 ENTER↑ 1 D →	4.80 ***	(x_1)
	15.10 ***	(y_1)
	1.00 ***	(f_1)
	1.00 ***	
5.2 ENTER↑ 11.5 ENTER↑ 3 D →	5.20 ***	(x_2)
	11.50 ***	(y_2)
	3.00 ***	(f_2)
	4.00 ***	(Σf_i)
10 ENTER↑ 10 ENTER↑ 4 D →	10.00 ***	
	10.00 ***	(error)
	4.00 ***	
	8.00 ***	
10 ENTER↑ 10 ENTER↑ 4 E →	10.00 ***	
	10.00 ***	(correction)
	4.00 ***	
	4.00 ***	
3.8 ENTER↑ 14.3 ENTER↑ 1 D →	3.80 ***	(x_3)
	14.30 ***	(y_3)
	1.00 ***	(f_3)
	5.00 ***	(Σf_i)

4.4 ENTER↑ 13.6 ENTER↑ 6 D →	4.40 ***	(x_4)
	13.60 ***	(y_4)
	6.00 ***	(f_4)
	11.00 ***	(Σf_i)
4.1 ENTER↑ 12.8 ENTER↑ 2 D →	4.10 ***	(x_5)
	12.80 ***	(y_5)
	2.00 ***	(f_5)
	13.00 ***	(Σf_i)
f B →	4.52 ***	$(\bar{x})$
R/S →	13.16 ***	$(\bar{y})$
f B →	9.93 ***	(V_x)
R/S →	8.42 ***	(V_y)
f C →	0.45 ***	(s_x)
R/S →	1.11 ***	(s_y)
f C →	0.43 ***	(s_x')
R/S →	1.07 ***	(s_y')
f D →	−0.31 ***	(s_{xy})
R/S →	−0.28 ***	(s_{xy}')
f D →	−0.62 ***	(γ_{xy})
f E →	58.80 ***	(Σx_i)
R/S →	171.10 ***	(Σy_i)
R/S →	770.22 ***	$(\Sigma x_i y_i)$
f E →	268.38 ***	(Σx_i^2)
R/S →	2266.69 ***	(Σy_i^2)

FACTORIAL, PERMUTATION AND COMBINATION

This program finds the extended range factorial (n can be greater than 69), permutation and combination. Permutation and combination are functions of the factorial, but this program will not use the factorial key, so that better accuracy and larger range can be obtained.

The equations are:

Factorial $n! = n(n-1)(n-2) \cdot \cdot \cdot 2 \cdot 1$

Permutation $${}_mP_n = \frac{m!}{(m-n)!} = m(m-1)\ldots(m-n+1)$$

Combination $${}_mC_n = \frac{m!}{(m-n)!\,n!} = \frac{m(m-1)\ldots(m-n+1)}{1 \cdot 2 \cdot \ldots \cdot n}$$

where m, n are integers and $0 \leq n \leq m$.

Notes: 1. ${}_mP_0 = 1$, ${}_mP_1 = m$, ${}_mP_m = m!$
therefore n! should be used for large m.

2. ${}_mC_0 = {}_mC_m = 1$

3. ${}_mC_1 = {}_mC_{m-1} = m$

4. ${}_mC_n = {}_mC_{m-n}$

5. In calculating n!, the accuracy will be reduced for n>69, since it is calculated by taking Log., ie
$n! = \log^{-1}\left[\log(n) + \log\left[(n-1)!\right]\right]$

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		B	1.00
4	Go to 5 or 6 or 7			
5	Calculate n!			
	(i) $n \leq 69$	n	C	n!
	(ii) $n>69$**	n	C	n
			R/S	exp. of 10
			R/S	decimal No.
6	Calculate ${}_mP_n$	m	ENTER↑	
		n	D	${}_mP_n$
7	Calculate ${}_mC_n$	m	ENTER↑	
		n	E	${}_mC_n$
	For a new case, go to 2			
	*Note: to clear print mode			
	press →		CLF	
			0	
	**In Print Mode, the 3-number			
	result will be printed out			
	automatically.			

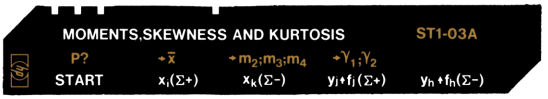
Examples:

1. $5! = 120$
2. $69! = 1.711224524 \times 10^{98}$
3. $70! = 1.197857069 \times 10^{100}$
4. $100! = 9.332622518 \times 10^{157}$
5. ${}_{27}P_5 = 9687600.00$
6. ${}_{73}C_4 = 1088430.00$

Keystrokes:	**Output:**	
A →	0.00	
B →	1.00	
5 C →	5.00 ***	
	120.00 ***	(5!)
69 C →	69.00 ***	
	1.711224524+98 ***	(69!)
70 C →	70.00***	
	100.00 ***	(10^{100})
	1.197857069 ***	(decimal no.)
100 C →	100.00 ***	
	157.00 ***	(10^{157})
	9.332622518 ***	(decimal no.)
27 ENTER↑ 5 D →	27.00 ***	
	5.00 ***	
	9687600.00 ***	(${}_{27}P_5$)
73 ENTER↑ 4 E →	73.00 ***	
	4.00 ***	
	1088430.00 ***	(${}_{73}C_4$)

Notes

MOMENTS, SKEWNESS AND KURTOSIS (FOR GROUPED OR UNGROUPED DATA)



For grouped or ungrouped data, moments are used to describe sets of data, skewness is used to measure the lack of symmetry in a distribution, and kurtosis is the relative peakness or flatness of a distribution. For a given set of data

$$\{x_1, x_2, \cdots, x_n\}:$$

1st moment $$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

2nd moment $$m_2 = \frac{1}{n} \Sigma x_i^2 - \bar{x}^2$$

3rd moment $$m_3 = \frac{1}{n} \Sigma x_i^3 - \frac{3}{n} \bar{x} \Sigma x_i^2 + 2\bar{x}^3$$

4th moment $$m_4 = \frac{1}{n} \Sigma x_i^4 - \frac{4}{n} \bar{x} \Sigma x_i^3 + \frac{6}{n} \bar{x}^2 \Sigma x_i^2 - 3\bar{x}^4$$

Moment coefficient of skewness

$$\gamma_1 = \frac{m_3}{m_2^{3/2}}$$

Moment coefficient of kurtosis

$$\gamma_2 = \frac{m_4}{m_2^2}$$

This program also provides the option for calculating those statistics for grouped data (using similar formulas as for ungrouped data):

data	y_1	y_2	...	y_m
frequency	f_1	f_2	...	f_m

Note that for this case, 1st moment

$$\bar{x} = \frac{\sum_{i=1}^{m} f_i x_i}{\sum_{i=1}^{m} f_i}$$

Reference: Theory and Problems of Statistics, M. R. Spiegel, Schaum's Outline, McGraw-Hill, 1961

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		f A	1.00
4	For grouped data, go to 12,			
5	Do 6~7 for i = 1, 2, ..., n for			
	ungrouped data			
6	Input x_i	x_i	B	i
7	If you made a mistake in			
	inputting x_k, then correct by →	x_k	C	i-1
8	Calculate: $\bar{x}$		f B	$\bar{x}$
9	Calculate: m_2		f C	m_2
	m_3		R/S	m_3
	m_4		R/S	m_4
10	Calculate: γ_1		f D	γ_1
	γ_2		R/S	γ_2
11	For a new case, go to 2			
12	Do 13~14 for j = 1, 2, ..., m			
	for grouped data			
13	Input	y_j	ENTER↑	
	and	f_j	D	j
14	If you made a mistake in			
	inputting y_h and f_h, then			
	correct by	y_h	ENTER↑	
		f_h	E	j-1
15	Go to 8			
	*Note: to clear print mode			
	press →		CLF	
			0	

Examples:

1. Ungrouped data

i	1	2	3	4	5	6	7	8	9
x_i	2.1	3.5	4.2	6.5	4.1	3.6	5.3	3.7	4.9

$\bar{x} = 4.21$, $m_2 = 1.39$, $m_3 = 0.39$, $m_4 = 5.49$

$\gamma_1 = 0.24$, $\gamma_2 = 2.84$

Keystrokes:	**Outputs:**	
A →	0.00	
f A →	1.00	
2.1 B →	2.10 ***	(x_1)
	1.00 ***	
4 B →	4.00 ***	(x_2) (error)
	2.00 ***	
4 C →	4.00 ***	(x_2) (correction)
	1.00 ***	
3.5 B →	3.50 ***	(x_2)
	2.00 ***	
4.2 B →	4.20 ***	(x_3)
	3.00 ***	
6.5 B →	6.50 ***	(x_4)
	4.00 ***	
4.1 B →	4.10 ***	(x_5)
	5.00 ***	
3.6 B →	3.60 ***	(x_6)
	6.00 ***	
5.3 B →	5.30 ***	(x_7)
	7.00 ***	

3.7 B →	3.70 ***	(x_8)
	8.00 ***	
4.9 B →	4.90 ***	(x_9)
	9.00 ***	
f B →	4.21 ***	($\bar{x}$)
f C →	1.39 ***	(m_2)
R/S →	.39 ***	(m_3)
R/S →	5.49 ***	(m_4)
f D →	0.24 ***	(γ_1)
R/S →	2.84 ***	(γ_2)

2. Grouped data

j	1	2	3	4	5
y_j	3	2	4	6	1
f_j	4	5	3	2	1

$\bar{x} = 3.13$, $m_2 = 1.98$, $m_3 = 2.14$, $m_4 = 11.05$

$\gamma_1 = 0.77$, $\gamma_2 = 2.81$

Keystrokes:	**Outputs:**	
A →	0.00	
f A →	1.00	
3 ENTER↑ 4 D →	3.00***	(y_1)
	4.00 ***	(f_1)
	1.00 ***	

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2 ENTER↑ 5 D → 2.00 *** (y_2)
5.00 *** (f_2)
2.00 ***

5 ENTER↑ 5 D → 5.00 *** (y_3)
5.00 *** (f_3) (error)
3.00 ***

5 ENTER↑ 5 E → 5.00 *** (y_3)
5.00 *** (f_3) (correction)
2.00 ***

4 ENTER↑ 3 D → 4.00 *** (y_3)
3.00 *** (f_3)
3.00 ***

6 ENTER↑ 2 D → 6.00 *** (y_4)
2.00 *** (f_4)
4.00 ***

1 ENTER↑ 1 D → 1.00 *** (y_5)
1.00 *** (f_5)
5.00 ***

f B → 3.13 *** ($\bar{x}$)
f C → 1.98 *** (m_2)
R/S → 2.14 *** (m_3)
R/S → 11.05 *** (m_4)
f D → 0.77 *** (γ_1)
R/S → 2.81 *** (γ_2)

Notes

RANDOM NUMBER GENERATOR

Random numbers are useful in a wide variety of applications, e.g., simulation, sampling, computer programming, numerical analysis and games. This program calculates (1) uniformly distributed numbers, (2) random integers, (3) normally distributed numbers, (4) exponentially distributed numbers, (5) mean, standard deviation and counter of the numbers generated.

This program calculates:

1. Uniformly distributed pseudo random numbers u_i in the range $a < u_i < b$: The multiplicative linear congruential method is used.

$$u_{i+1} = f_{i+1} (b - a) + a$$

where $i = 0, 1, 2, \ldots$ and

f_{i+1} = fractional part of $(997\ f_i)$

$f_0 = 0.5284163.$

The period has length 500000, i.e., 500000 different numbers can be generated before repeating. The least significant digits (the righthand digits) of u_i are not as random as the most significant digits (the left-hand digits). Thus random digits, if needed, should be taken from the most significant end of the numbers. This generator passes the chi-square frequency test for uniformity, serial test and run tests for randomness.

If a different sequence of numbers is desired, a different starting value f_0 $(0 < f_0 < 1)$ can be used. Some program steps (the starting value stored under LBL 0) must be changed accordingly. Note that if $10^7 \times f_0$ is not divisible by 2 or 5, then the period of the generator has length 500000. All the tests mentioned above should be applied to the new generator before using it.

2. Pseudo random integers d_i such that $1 \leq d_i \leq k$:

Suppose u_i $(i = 1, 2, \ldots)$ is a sequence of uniformly distributed pseudo random numbers between 0 and 1.

$$d_i = 1 + \text{integer part of } (ku_i)$$

3. Normally distributed pseudo random numbers n_i if the mean m and the standard deviation σ are given:

Suppose u_i ($i = 1, 2, \ldots$) is a sequence of uniformly distributed pseudo random numbers between 0 and 1.

Let

$$v_1 = (2u_i - 1) \quad , \quad v_2 = (2u_{i+1} - 1)$$
$$S = v_1^2 + v_2^2 \qquad (i = 1, 2, \ldots)$$

If $S \geqslant 1$, discard the two uniform numbers u_i, u_{i+1} and generate the next two numbers in the sequence until $S < 1$. Then compute the normally distributed numbers according to the following equations

$$n_i = \sigma v_1 \sqrt{\frac{-2 \ln S}{S}} + m$$

$$n_{i+1} = \sigma v_2 \sqrt{\frac{-2 \ln S}{S}} + m$$

4. Exponentially distributed pseudo random numbers e_i with mean μ:

Suppose u_i ($i = 1, 2, \ldots$) is a sequence of uniformly distributed pseudo random numbers between 0 and 1.

$$e_i = -\mu \ln u_i$$

5. The mean $\bar{x}$, standard deviation s and counter n of the random numbers computed:

$$\bar{x} = \sum_{i=1}^{n} x_i/n$$

$$s = \sqrt{\frac{\Sigma x_i^2 - n\bar{x}^2}{n-1}}$$

where x_i can be u_i, d_i, n_i or e_i.

Reference:

Donald E. Knuth, *The Art of Computer Programming,* Vol. 2, Addison-Wesley, 1971.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	For random integers, go to 6			
	For normal numbers, go to 9			
	For exponential numbers,			
	go to 12			
3	Input interval range for			
	uniform numbers	a	ENTER↑	
		b	f A	b
4	Perform 4 for i=1, 2, . . .		A	u_i
5	For $\bar{x}$ and s, go to 14			
6	Input maximum integer			
	desired	k	f B	k
7	Perform 7 for i=1, 2, . . .		B	d_i
8	For $\bar{x}$ and s, go to 14			
9	For normal numbers, input			
	mean	m	ENTER↑	
	and standard deviation	σ	f C	σ
10	Perform 10 for i=1, 2, . . .		C	n_i
11	For $\bar{x}$ and s, go to 14			
12	Input mean for exponential			
	numbers	μ	f D	μ
13	Perform 13 for i=1, 2, . . .		D	e_i
14	Optional: Calculate the mean		E	$\bar{x}$
	the standard deviation		R/S	s
	the counter		R/S	n
15	To continue the calculation,			
	go back to 4, 7, 10, or 13			
16	For a new case, go to 2			

Example 1:

Generate a sequence of uniform pseudo random numbers between 0 and 1.

Keystrokes:	**Outputs:**	
0 ENTER↑ 1 f A →	0.00 ***	(a)
	1.00 ***	(b)
A →	0.83 ***	
A →	0.56 ***	
A →	0.27 ***	
A →	0.04 ***	
A →	0.20 ***	
A →	0.75 ***	
A →	0.83 ***	
A →	0.95 ***	
E →	0.55 ***	(mean)
R/S →	0.34 ***	(s)
R/S →	8.00 ***	(counter)
A →	0.68 ***	
A →	0.63 ***	
A →	0.22 ***	
	etc.	

Example 2:

Use the random number generator to simulate the successive tosses of a die.

Keystrokes:	**Outputs:**	
6 f B →	6.00 ***	(k)
B →	5.00 ***	
B →	4.00 ***	
B →	2.00 ***	
B →	1.00 ***	
B →	2.00 ***	
B →	5.00 ***	
	etc.	

Example 3:

A professor decides to assign grades randomly and without bias to the students. The grades should have a normal distribution with average grade being 75 and standard deviation being 10. How can the random number generator be used for this purpose?

Keystrokes:	**Outputs:**
75 ENTER↑ 10 f C →	75.00 *** (m)
	10.00 *** (σ)
C →	87.42 ***
C →	77.17 ***
C →	67.44 ***
C →	81.23 ***
C →	89.91 ***
C →	85.32 ***
	etc.

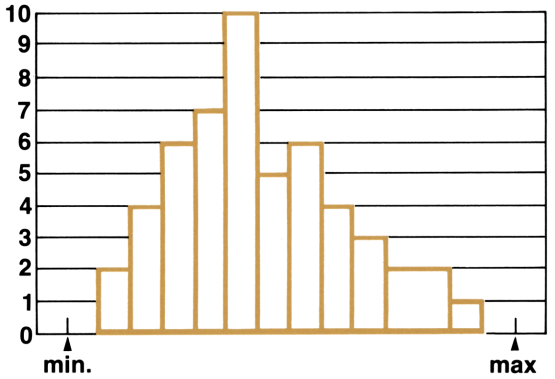
Example 4:

Suppose a radioactive substance emits alpha particles at a rate such that on the average, one particle is emitted every 5 seconds. Note that the amount of time between two successive emissions has the exponential distribution with mean 5. Generate a sequence of random numbers so that each of them can be used as the amount of time between two emissions.

Keystrokes:	**Outputs:**
5 f D →	5.00 *** (μ)
D →	0.93 ***
D →	2.92 ***
D →	6.49 ***
D →	15.93 ***
D →	8.14 ***
D →	1.44 ***
	etc.

Notes

HISTOGRAM



A histogram or bar chart can provide a meaningful way of representing tabular data or the output of an algorithm. By viewing a histogram, trends and biases can be spotted easily.

This program sorts input data into 24 intervals or bins of equal width between specified upper and lower limits.

One is added to the bin whose number is calculated. This procedure is repeated for all x values in the data set. After all data has been input, pressing f B, R/S, R/S will cause the printout of the total number of inputs, the mean of the inputs, and the standard deviation of the inputs. Pressing f A gives the number of inputs in each bin and a representation of the histogram. The bins are arranged in maximum x value to minimum x value order.

The 24 intervals are stored three at a time in registers $R_1 \sim R_8$.

Incorrect values may be deleted at any time by keying them in and pressing E. However, if the value is out of bounds, then "Error" will be displayed. Press CL x and continue.

To start the program you must specify the minimum expected value x_{min} and the maximum expected value x_{max}.

Equations:

For the histogram:

$$\text{mean} = \frac{\Sigma x}{n}$$

$$\text{standard deviation} = \sqrt{\frac{\Sigma x^2 - n\,\bar{x}^2}{n-1}}$$

$$y_i = 1 + \text{INT}\left[\, 24\,\frac{x_i - x_{min}}{x_{max} - x_{min}} \,\right]$$

where

y_i = interval number
x_i = input data
x_{min} = lower limit of histogram
x_{max} = upper limit of histogram
INT = integer part of

Remark:

Because each interval is represented by only three digits, overflow from one interval to the next lower interval will occur for most of the intervals if there are more than 999 counts in the interval.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		B	1.00
4	Input minimum value	x_{min}	ENTER↑	
	maximum value	x_{max}	C	x_{min}, x_{max}
5	Do 6~7 for i=1, 2, . . . , n			
6	Input x_i	x_i	D	i
7	If you made a mistake in			
	inputting x_k, then correct by →	x_k	E	i–1
8	List histogram		f A	LIST

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
9	Print n, $\bar{x}$, s		f B	n
			R/S	$\bar{x}$
			R/S	s
10	For a new case, go to 2			
	*Note: to clear print mode			
	press→		CLF	
			0	

Example 1:

Compute a histogram of the following data with $x_{min}=0$, $x_{max}=24$.(18.1, 14.3, 8.4, 0.7, 20.2, 14, 17.2, 24, 8.8, 5.7, 13.2, 22.1, 15.7, 18.9, 23).

Keystrokes:	**Outputs:**	
A →	0.00	
B →	1.00	
0 ENTER↑ 24 C →	0.00 ***	(x_{min})
	24.00 ***	(x_{max})
18.1 D →	18.10 ***	
	1. ***	
14.3 D →	14.30 ***	
	2. ***	
8.4 D →	8.40 ***	
	3. ***	
0.7 D →	0.70 ***	
	4. ***	
9.9 D →	9.90 ***	(error)
	5. ***	
9.9 E →	9.90 ***	(correction)
	4. ***	
20.2 D →	20.20 ***	
	5. ***	
14.0 D →	14.00 ***	
	6. ***	
17.2 D →	17.20 ***	
	7. ***	

24 D	24.00 ***	
	8. ***	
8.8 D	8.80 ***	
	9. ***	
5.7 D	5.70 ***	
	10. ***	
13.2 D	13.20 ***	
	11. ***	
22.1 D	22.10 ***	
	12. ***	
15.7 D	15.70 ***	
	13. ***	
18.9 D	18.90 ***	
	14. ***	
23 D	23.00 ***	
	15. ***	
f B	15.00 ***	(n)
R/S	14.95 ***	($\bar{x}$)
R/S	6.71 ***	(s)
f A	0.00 ***	
	1.00 ***	
	1. ***	
	1.00 ***	
	2.00 ***	
	0. ***	
	2.00 ***	
	3.00 ***	
	0. ***	
	3.00 ***	
	4.00 ***	
	0. ***	
	4.00 ***	
	5.00 ***	
	0. ***	

5.00 ***
6.00 ***
1. ***

6.00 ***
7.00 ***
0. ***

7.00 ***
8.00 ***
0. ***

8.00 ***
9.00 ***
2. ***

9.00 ***
10.00 ***
0. ***

10.00 ***
11.00 ***
0. ***

11.00 ***
12.00 ***
0. ***

12.00 ***
13.00 ***
0. ***

13.00 ***
14.00 ***
1. ***

14.00 ***
15.00 ***
2. ***

15.00 ***
16.00 ***
1. ***

16.00 ***
17.00 ***
0. ***

17.00 ***
18.00 ***
1. ***

18.00 ***
19.00 ***
2. ***

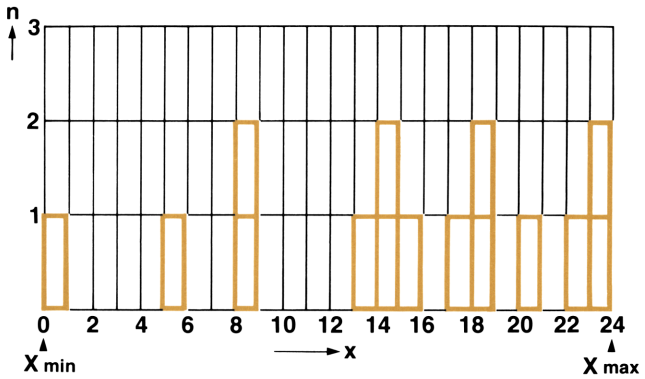
19.00 ***
20.00 ***
0. ***

20.00 ***
21.00 ***
1. ***

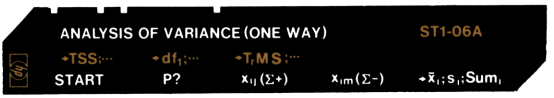
21.00 ***
22.00 ***
0. ***

22.00 ***
23.00 ***
1. ***

23.00 ***
24.00 ***
2. ***



ANALYSIS OF VARIANCE (ONE WAY)



The one-way analysis of variance is used to test if observed differences among k sample means can be attributed to chance or whether they are indicative of actual differences among the corresponding population means. Suppose the i^{th} sample has n_i observations (samples may have equal or unequal number of observations). The null hypothesis we want to test is that the k population means are all equal. This program generates the complete ANOVA table.

1. Mean of observations in the i^{th} sample (i=1, 2, - - -, k)

$$\bar{x}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} x_{ij}$$

2. Standard deviation of observations in the i^{th} sample

$$s_i = \left[\left(\sum_{j=1}^{n_i} x_{ij}^2 - n_i \bar{x}_i^2 \right) \Big/ (n_i - 1) \right]^{1/2}$$

3. Sum of observations in the i^{th} sample

$$\text{Sum}_i = \sum_{j=1}^{n_i} x_{ij}$$

4. Total sum of squares

$$TSS = \sum_{i=1}^{k} \sum_{j=1}^{n_i} x_{ij}^2 - \frac{\left(\sum_{i=1}^{k} \sum_{j=1}^{n_i} x_{ij} \right)^2}{\sum_{i=1}^{k} n_i}$$

5. Treatment sum of squares

$$TrSS = \sum_{i=1}^{k} \frac{\left(\sum_{j=1}^{n_i} x_{ij} \right)^2}{n_i} - \frac{\left(\sum_{i=1}^{k} \sum_{j=1}^{n_i} x_{ij} \right)^2}{\sum_{i=1}^{k} n_i}$$

6. Error sum of squares

$$ESS = TSS - TrSS$$

7. Treatment degrees of freedom

$$df_1 = k - 1$$

8. Error degrees of freedom

$$df_2 = \sum_{i=1}^{k} n_i - k$$

9. Total degrees of freedom

$$df_3 = df_1 + df_2 = \sum_{i=1}^{k} n_i - 1$$

10. Treatment mean square

$$TrMS = \frac{TrSS}{df_1}$$

11. Error mean square

$$EMS = \frac{ESS}{df_2}$$

12. The F ratio

$$F = \frac{TrMS}{EMS} \quad \text{(with degrees of freedom } df_1, df_2\text{)}$$

Reference:

J. E. Freund, *Mathematical Statistics*, Prentice Hall, 1962.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		B	1.00
4	Do 4 ~ 8 for i=1, 2, ..., k			
5	Do 6 for j=1, 2, ..., n_i			
6	Input x_{ij}	x_{ij}	C	j
7	If you made a mistake in			
	inputting x_{im}, then correct by	x_{im}	D	j−1
8	Calculate: mean $\bar{x}_i$		E	$\bar{x}_i$
	standard deviation s_i		R/S	s_i
	sum Sum_i		R/S	Sum_i
9	Calculate: total sum of squares		f A	TSS
	treatment sum of squares		R/S	TrSS
	error sum of squares		R/S	ESS
10	Calculate degrees of freedom			
	df_1		f B	df_1
	df_2		R/S	df_2
	df_3		R/S	df_3
11	Calculate:			
	treatment mean square		f C	TrMS
	error mean square		R/S	EMS
	F ratio		R/S	F
12	For a new case, go to 2			
	*Note: to clear print mode			
	press →		CLF	
			0	

Example:

The following are the scores obtained in an achievement test by random samples of students from four different schools:

i \ j	1	2	3	4	5	6	7
School 1	88	99	96	68	85		
School 2	78	62	98	83	61	88	
School 3	80	61	74	92	78	54	77
School 4	71	65	90	46			

Calculate the ANOVA table and test the null hypothesis that the differences among the sample means can be attributed to chance. Use significance level $\alpha = 0.01$.

Keystrokes:	**Outputs:**	
A →	0.00	
B →	1.00	
88 C →	88.00 ***	
	1.00 ***	
99 C →	99.00 ***	
	2.00 ***	
96 C →	96.00 ***	
	3.00 ***	
68 C →	68.00 ***	
	4.00 ***	
85 C →	85.00 ***	
	5.00 ***	
E →	87.20 ***	($\bar{x}_1$)
R/S →	12.15 ***	(s_1)
R/S →	436.00 ***	(Sum_1)
78 C →	78.00 ***	
	1.00 ***	

06-05

62 C → 62.00 ***
2.00 ***

98 C → 98.00 ***
3.00 ***

83 C → 83.00 ***
4.00 ***

61 C → 61.00 ***
5.00 ***

88 C → 88.00 ***
6.00 ***

E → 78.33 *** ($\bar{x}_2$)
R/S → 14.62 *** (s_2)
R/S → 470.00 *** (Sum_2)

80 C → 80.00 ***
1.00 ***

61 C → 61.00 ***
2.00 ***

74 C → 74.00 ***
3.00 ***

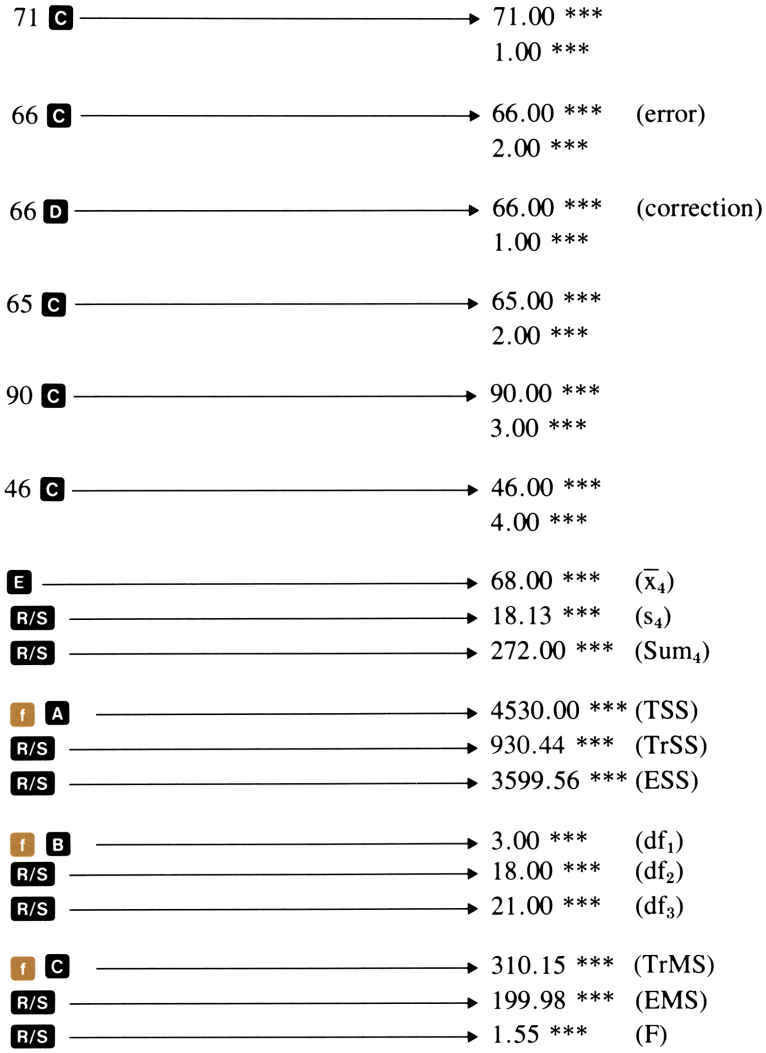
92 C → 92.00 ***
4.00 ***

78 C → 78.00 ***
5.00 ***

54 C → 54.00 ***
6.00 ***

77 C → 77.00 ***
7.00 ***

E → 73.71 *** ($\bar{x}_3$)
R/S → 12.61 *** (s_3)
R/S → 516.00 *** (Sum_3)



ANOVA Table

	SS	df	MS	F
Treatments	930.44	3	310.15	1.55
Error	3599.56	18	199.98	
Total	4530.00	21		

Since $F = 1.55$ does not exceed $F_{.01,3,18} = 5.09$, the null hypothesis can not be rejected. We conclude that the means of the scores for the four schools are not significantly different.

TWO WAY ANALYSIS OF VARIANCE (NO REPLICATIONS)

The analysis of variance is the analysis of the total variability of a set of data (measured by their total sum of squares) into components which can be attributed to different sources of variation.

The two way analysis of variance tests the row effects and the column effects independently. This program will generate the ANOVA table for the case such that (1) each cell only has one observation and (2) the row and column effects do not interact.

Equations:

1. Sums

$$\text{Row } RS_i = \sum_j x_{ij} \qquad i = 1, 2, \ldots, r$$

$$\text{Column } CS_j = \sum_i x_{ij} \qquad j = 1, 2, \ldots, c$$

2. Sums of squares

$$\text{Total TSS} = \Sigma\Sigma x_{ij}^2 - (\Sigma\Sigma x_{ij})^2/rc$$

$$\text{Row RSS} = \sum_i \left(\sum_j x_{ij} \right)^2/c - (\Sigma\Sigma x_{ij})^2/rc$$

$$\text{Column CSS} = \sum_j \left(\sum_i x_{ij} \right)^2/r - (\Sigma\Sigma x_{ij})^2/rc$$

$$\text{Error ESS} = \text{TSS} - \text{RSS} - \text{CSS}$$

3. Degrees of freedom

$$\text{Row } df_1 = r - 1$$

$$\text{Column } df_2 = c - 1$$

$$\text{Error } df_3 = (r - 1)(c - 1)$$

4. F ratios

$$\text{Row } F_1 = \frac{RSS}{df_1} \Big/ \frac{ESS}{df_3}$$

$$\text{Column } F_2 = \frac{CSS}{df_2} \Big/ \frac{ESS}{df_3}$$

Reference:

Dixon and Massey, *Introduction to Statistical Analysis,* McGraw-Hill, 1969.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		B	1.00
4	Input: number of rows r	r	ENTER↑	
	number of columns c	c	C	c
5	Do 5~9 for i=1, 2, . . . , r			
	Do 6 for j=1, 2, . . . , c			
6	Input x_{ij}	x_{ij}	D	j
7	If you made a mistake in			
	inputting x_{im}, then correct by	x_{im}	E	j−1
8	Calculate row sums RS_i		f A	RS_i
9	Re-initialize for columns		f B	0.00
10	Do 11~14 for j=1, 2, . . . , c			
11	Do 12 for i=1, 2, ..., r			
12	Input x_{ij}	x_{ij}	D	i
13	If you made a mistake in			
	inputting x_{hj}, then correct by	x_{hj}	E	i−1
14	Calculate column sums CS_i		f C	CS_i
15	Calculate F ratios: Row F_1		f D	F_1
	Column F_2		R/S	F_2

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
16	Calculate degrees of freedom:			
	row df_1		f D	df_1
	column df_2		R/S	df_2
	error df_3		R/S	df_3
17	Calculate sums of squares:			
	row RSS		f E	RSS
	column CSS		R/S	CSS
	error ESS		R/S	ESS
	total TSS		R/S	TSS
18	For a new case, go to 2			
	*Note: to clear print mode			
	press →		CLF	
			0	

Example:

Apply this program to analyze the following set of data.

	i \ j	Column 1	2	3	4
Row	1	7	6	8	7
	2	2	4	4	4
	3	4	6	5	3

Keystrokes:	Outputs:	
A →	0.00	
B →	1.00	
3 ENTER↑ 4 C →	3.00 ***	(r)
	4.00 ***	(c)
7 D →	7.00 ***	
	1.00 ***	
6 D →	6.00 ***	
	2.00 ***	
8 D →	8.00 ***	
	3.00 ***	

7 D →	7.00 ***	
	4.00 ***	
f A →	28.00 ***	(RS_1)
2 D →	2.00 ***	
	1.00 ***	
4 D →	4.00 ***	
	2.00 ***	
4 D →	4.00 ***	
	3.00 ***	
4 D →	4.00 ***	
	4.00 ***	
f A →	14.00 ***	(RS_2)
4 D →	4.00 ***	
	1.00 ***	
7 D →	7.00 ***	(error)
	2.00 ***	
7 E →	7.00 ***	(correction)
	1.00 ***	
6 D →	6.00 ***	
	2.00 ***	
5 D →	5.00 ***	
	3.00 ***	
3 D →	3.00 ***	
	4.00 ***	
f A →	18.00 ***	(RS_3)
f B →	0.00 ***	
7 D →	7.00 ***	
	1.00 ***	
2 D →	2.00 ***	
	2.00 ***	
4 D →	4.00 ***	
	3.00 ***	
f C →	13.00 ***	(CS_1)
6 D →	6.00 ***	
	1.00 ***	
4 D →	4.00 ***	
	2.00 ***	

6 D	→	6.00 ***	
		3.00 ***	
f C	→	16.00 ***	(CS_2)
8 D	→	8.00 ***	
		1.00 ***	
4 D	→	4.00 ***	
		2.00 ***	
5 D	→	5.00 ***	
		3.00 ***	
f C	→	17.00 ***	(CS_3)
7 D	→	7.00 ***	
		1.00 ***	
4 D	→	4.00 ***	
		2.00 ***	
3 D	→	3.00 ***	
		3.00 ***	
f C	→	14.00 ***	(CS_4)
f D	→	11.70 ***	(F_1)
R/S	→	1.00 ***	(F_2)
f D	→	2.00 ***	(df_1)
R/S	→	3.00 ***	(df_2)
R/S	→	6.00 ***	(df_3)
f E	→	26.00 ***	(RSS)
R/S	→	3.33 ***	(CSS)
R/S	→	6.67 ***	(ESS)
R/S	→	36.00 ***	(TSS)

ANOVA

	SS	df	F ratio
Row	26.00	2	11.70
Column	3.33	3	1.00
Error	6.67	6	
Total	36.00		

Notes

ANALYSIS OF COVARIANCE (ONE WAY)

The one way analysis of covariance program tests the effect of one variable separately from the effect of a second variable, if the second variable represents an actual measurement for each individual (rather than a category).

Suppose (x_{ij}, y_{ij}) represents the j^{th} observation from the i^{th} population ($i = 1,2, \ldots, k$, $j = 1,2, \ldots, n_i$). Note that samples may have equal or unequal number of observations. The analysis of covariance tests for a difference in means of residuals. The residuals are the differences of the observations and a regression quantity based on the associated second variable. The analysis of covariance procedure is based on the separations of the sums of squares and the sums of products into several portions. This program will generate the complete ANOCOV table.

Equations:

1. Sums and sums of squares

$$Sx_i = \sum_j x_{ij} \quad (i = 1,2, \ldots, k)$$

$$TSSx = \Sigma\,\Sigma x_{ij}^2 - \frac{(\Sigma\Sigma x_{ij})^2}{\sum_i n_i}$$

$$ASSx = \sum_i \frac{\left(\sum_j x_{ij}\right)^2}{n_i} - \frac{(\Sigma\Sigma x_{ij})^2}{\sum_i n_i}$$

$$WSSx = TSSx - ASSx$$

2. Degrees of freedom

$$df_1 = k - 1$$

$$df_2 = \sum_i n_i - k$$

3. Mean squares and F statistic

$$AMSx = \frac{ASSx}{df_1}$$

$$WMSx = \frac{WSSx}{df_2}$$

$$F_x = \frac{AMSx}{WMSx} \text{ with degrees of freedom } df_1, df_2$$

By changing x_{ij} to y_{ij}, similar formulas for y_{ij} can be obtained.

4. Sums of products

$$TSP = \Sigma\Sigma x_{ij}\ y_{ij} - \frac{(\Sigma\Sigma x_{ij})\ (\Sigma\Sigma y_{ij})}{\sum_i n_i}$$

$$ASP = \sum_i \frac{\left(\sum_j x_{ij}\right)\left(\sum_j y_{ij}\right)}{n_i} - \frac{(\Sigma\Sigma x_{ij})(\Sigma\Sigma y_{ij})}{\sum_i n_i}$$

$$WSP = TSP - ASP$$

5. Residual sums of squares

$$TSS\hat{y} = TSSy - \frac{(TSP)^2}{TSSx}$$

$$WSS\hat{y} = WSSy - \frac{(WSP)^2}{WSSx}$$

$$ASS\hat{y} = TSS\hat{y} - WSS\hat{y}$$

6. Residual degrees of freedom

$$df_3 = k - 1$$

$$df_4 = \sum_i n_i - k - 1$$

7. Residual mean squares and F statistic

$$AMS\hat{y} = \frac{ASS\hat{y}}{df_3}$$

$$WMS\hat{y} = \frac{WSS\hat{y}}{df_4}$$

$$F = \frac{AMS\hat{y}}{WMS\hat{y}} \text{ with degrees of freedom } df_3, df_4$$

ANOCOV Table

					Residuals			
	degrees of freedom	SSx	SP	SSy	degrees of freedom	SS$\hat{y}$	MS$\hat{y}$	F statistic
Among means	df_1	ASSx	ASP	ASSy	df_3	ASS$\hat{y}$	AMS$\hat{y}$	F
Within groups	df_2	WSSx	WSP	WSSy	df_4	WSS$\hat{y}$	WMS$\hat{y}$	
Total		TSSx	TSP	TSSy		TSS$\hat{y}$		

Remarks:

1. F_x can be used to test if the X means are equal (ANOVA for X).
2. F_y can be used to test if the Y means (not making use of the X values) are equal (ANOVA for unadjusted Y).

Reference:

Dixon and Massey, *Introduction to Statistical Analysis,* McGraw-Hill, 1969.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2 of card 1			
2	Initialize		A	0.00
3	To set print mode*		f E	1.00
4	Do 5~9 for i=1, 2, . . . , k			
5	Initialize for new i		B	i
6	Do 6 for j=1, 2, . . . , n_j			
7	Input x_{ij} and y_{ij}	x_{ij}	ENTER↑	
		y_{ij}	C	j
8	If you made a mistake in			
	inputting x_{im} and y_{im}, then			
	correct by	x_{im}	ENTER↑	
		y_{im}	D	j−1
9	Calculate the i^{th} sums Sx_i		E	Sx_i
	Sy_i		R/S	Sy_i
10	Calculate: the total sum of			
	squares for x		f A	TSSx
	Among means sum of squares			
	for x		R/S	ASSx
	Within groups sum of squares			
	for x		R/S	WSSx
11	Calculate: the total sum of			
	squares for y		f A	TSSy
	Among means sum of squares			
	for y		R/S	ASSy
	Within groups sum of squares			
	for y		R/S	WSSy
12	Calculate: F_x		f B	F_x
	F_y		R/S	F_y
	degrees of freedom df_1		R/S	df_1
	df_2		R/S	df_2

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
13	Load side 1 of card 2			
14	Calculate: the total sum of			
	products (SP)		f C	TSP
	among means SP		R/S	ASP
	within group SP		R/S	WSP
15	Calculate: TSS$\hat{y}$		f D	TSS$\hat{y}$
	WSS$\hat{y}$		R/S	WSS$\hat{y}$
	ASS$\hat{y}$		R/S	ASS$\hat{y}$
16	Calculate: residual mean			
	squares		f E	AMS$\hat{y}$
	WMS$\hat{y}$		R/S	WMS$\hat{y}$
	the F statistic		R/S	F
	the degrees of freedom df_3		R/S	df_3
	df_4		R/S	df_4
17	For a new case, go to 1			
	*Note: to clear print mode			
	press →		CLF	
			0	

Example:

			j			
			1	2	3	4
i	1	x	3	2	1	2
		y	10	8	8	11
	2	x	4	3	3	5
		y	12	12	10	13
	3	x	1	2	3	1
		y	6	5	8	7

($k = 3$, $n_1 = n_2 = n_3 = 4$)

Keystrokes:

Load side 1 and side 2 of card 1

Keystrokes:	Outputs:	
A →	0.00	
f E →	1.00	
B →	1.00 ***	
3 ENTER↑ 10 C →	3.00 ***	
	10.00 ***	
	1.00 ***	
2 ENTER↑ 8 C →	2.00 ***	
	8.00 ***	
	2.00 ***	
5 ENTER↑ 5 C →	5.00 ***	
	5.00 ***	(error)
	3.00 ***	
5 ENTER↑ 5 D →	5.00 ***	
	5.00 ***	(correction)
	2.00 ***	
1 ENTER↑ 8 C →	1.00 ***	
	8.00 ***	
	3.00 ***	
2 ENTER↑ 11 C →	2.00 ***	
	11.00 ***	
	4.00 ***	
E →	8.00 ***	(Sx_1)
R/S →	37.00 ***	(Sy_1)
B →	2.00 ***	
4 ENTER↑ 12 C →	4.00 ***	
	12.00 ***	
	1.00 ***	
3 ENTER↑ 12 C →	3.00 ***	
	12.00 ***	
	2.00 ***	
3 ENTER↑ 10 C →	3.00 ***	
	10.00 ***	
	3.00 ***	

5 ENTER↑ 13 C →	5.00 ***	
	13.00 ***	
	4.00 ***	
E →	15.00 ***	(Sx_2)
R/S →	47.00 ***	(Sy_2)
B →	3.00 ***	
1 ENTER↑ 6 C →	1.00 ***	
	6.00 ***	
	1.00 ***	
2 ENTER↑ 5 C →	2.00 ***	
	5.00 ***	
	2.00 ***	
3 ENTER↑ 8 C →	3.00 ***	
	8.00 ***	
	3.00 ***	
1 ENTER↑ 7 C →	1.00 ***	
	7.00 ***	
	4.00 ***	
E →	7.00 ***	(Sx_3)
R/S →	26.00 ***	(Sy_3)
f A →	17.00 ***	(TSSx)
R/S →	9.50 ***	(ASSx)
R/S →	7.50 ***	(WSSx)
f A →	71.67 ***	(TSSy)
R/S →	55.17 ***	(ASSy)
R/S →	16.50 ***	(WSSy)
f B →	5.70 ***	(Fx)
R/S →	15.05 ***	(Fy)
R/S →	2.00 ***	(df_1)
R/S →	9.00 ***	(df_2)

Load side 1 of card 2

f C → 27.00 *** (TSP)
R/S → 20.75 *** (ASP)
R/S → 6.25 *** (WSP)

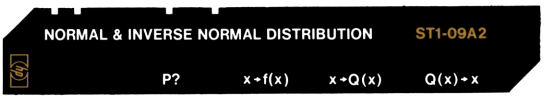
f D → 28.78 *** (TSS$\hat{y}$)
R/S → 11.29 *** (WSS$\hat{y}$)
R/S → 17.49 *** (ASS$\hat{y}$)

f E → 8.75 *** (AMS$\hat{y}$)
R/S → 1.41 *** (WMS$\hat{y}$)
R/S → 6.20 *** (F)
R/S → 2.00 *** (df_3)
R/S → 8.00 *** (df_4)

ANOCOV Table

					Residuals			
	df	SSx	SP	SSy	df	SS$\hat{y}$	MS$\hat{y}$	F
Among means	2	9.50	20.75	55.17	2	17.49	8.75	6.20
Within groups	9	7.50	6.25	16.50	8	11.29	1.41	
Total		17.00	27.00	71.67		28.78		

NORMAL AND INVERSE NORMAL DISTRIBUTION



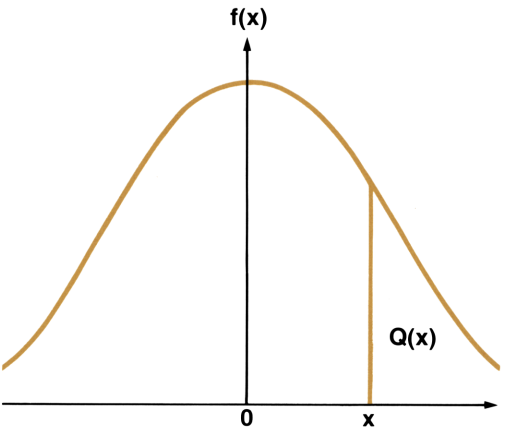
This program evaluates the standard normal density function f(x) and the normal integral Q(x) for given x. If Q is given, x can also be found.

The standard normal distribution has mean 0 and standard deviation 1.

Equations:

1. Standard normal density

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$



2. Normal integral

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-\frac{t^2}{2}} dt$$

Polynomial approximation is used to compute Q(x) for given x.

Define $R = f(x)\,(b_1 t + b_2 t^2 + b_3 t^3 + b_4 t^4 + b_5 t^5) + \epsilon(x)$

where $|\epsilon(x)| < 7.5 \times 10^{-8}$

$$t = \frac{1}{1 + r|x|}, \quad r = 0.2316419$$

$b_1 = .319381530, \quad b_2 = -.356563782$

$b_3 = 1.781477937, \quad b_4 = -1.821255978$

$b_5 = 1.330274429$

$$\text{Then } Q(x) = \begin{cases} R & \text{if } x \geq 0 \\ 1-R & \text{if } x < 0 \end{cases}$$

3. Inverse normal

For a given $Q > 0$, x can be found such that

$$Q = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-\frac{t^2}{2}}\, dt.$$

The following rational approximation is used:

$$\text{Define } y = t - \frac{c_0 + c_1 t + c_2 t^2}{1 + d_1 t + d_2 t^2 + d_3 t^3} + \epsilon(Q)$$

where $|\epsilon(Q)| < 4.5 \times 10^{-4}$

$$t = \begin{cases} \sqrt{\ln \frac{1}{Q^2}} & \text{if } 0 < Q \leq 0.5 \\ \sqrt{\ln \frac{1}{(1-Q)^2}} & \text{if } 0.5 < Q < 1 \end{cases}$$

$c_0 = 2.515517$ $\quad d_1 = 1.432788$
$c_1 = 0.802853$ $\quad d_2 = 0.189269$
$c_2 = 0.010328$ $\quad d_3 = 0.001308$

Then $x = \begin{cases} y & \text{if } 0 < Q \leq 0.5 \\ -y & \text{if } 0.5 < Q < 1 \end{cases}$

Reference:

Abramowitz and Stegun, *Handbook of Mathematical Functions,* National Bureau of Standards, 1970.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2 of			
	card 1			
2	Initialize		A	0.00
3	Load side 1 and side 2 of			
	card 2			
4	To set print mode*		B	1.00
	Optional: Step 5			
5	Input x to compute f(x)	x	C	f(x)
6	Input x to compute Q(x)	x	D	Q(x)
	For a new case of x, go to			
	5 or 6			
7	Input Q(x) to compute x	Q(x)	E	x
	For a new case of Q(x), go			
	to 7			
	*Note: to clear print mode			
	press →		0	
			STO	
			A	
			STO	
			B	

Example 1:

Find f(x) and Q(x) for x = 1.18 and x = −2.28.

Keystrokes:		**Outputs:**	
Load side 1 and side 2 of card 1			
A	→	0.00	
Load side 1 and side 2 of card 2			
B	→	1.00	
1.18 C	→	1.18 ***	
		0.20 ***	(f(1.18))
1.18 D	→	1.18 ***	
		0.12 ***	(Q(1.18))
2.28 CHS D	→	-2.28 ***	
		0.99 ***	(Q(−2.28))
2.28 CHS C	→	-2.28 ***	
		0.03 ***	(f(−2.28))

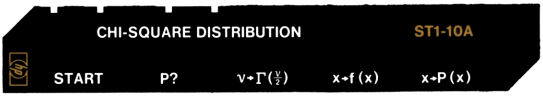
Example 2:

Given Q = 0.12 and Q =0.95, find x.

(If you have run through Example 1, then you can proceed; otherwise you have to load programs as described in Example 1).

Keystrokes:		**Outputs:**	
0.12 E	→	0.12 ***	
		1.18 ***	(x)
0.95 E	→	0.95 ***	
		-1.65 ***	(x)

CHI-SQUARE DISTRIBUTION

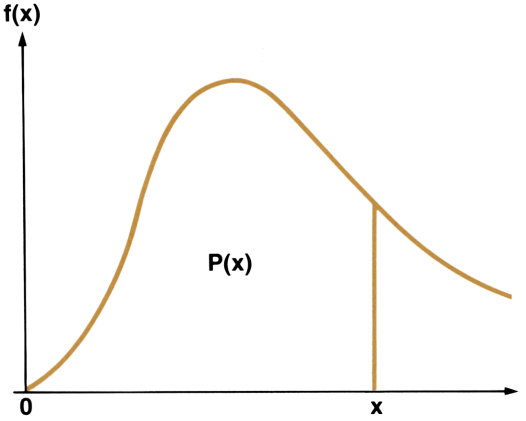


This program evaluates the chi-square density

$$f(x) = \frac{1}{2^{\frac{\nu}{2}}\,\Gamma\left(\frac{\nu}{2}\right)}\, x^{\frac{\nu}{2}-1}\, e^{-\frac{x}{2}}$$

where $x \geq 0$

ν is the degrees of freedom.



Series approximation is used to evaluate the cumulative distribution

$$P(x) = \int_0^x f(t)\, dt$$

$$= \left(\frac{x}{2}\right)^{\frac{\nu}{2}} \frac{e^{-\frac{x}{2}}}{\Gamma\left(\frac{\nu+2}{2}\right)} \left[1 + \sum_{k=1}^{\infty} \frac{x^k}{(\nu+2)(\nu+4)\ldots(\nu+2k)}\right]$$

The program computes successive partial sums of the above series. When two consecutive partial sums are equal, the value is used as the sum of the series.

Notes: 1. Program requires $\nu < 141$. If $\nu > 141$, erroneous overflow will result.

2. If both x and ν are large, f(x) may overflow the machine.

3. If ν is even,

$$\Gamma\left(\frac{\nu}{2}\right) = \left(\frac{\nu}{2} - 1\right)!$$

If ν is odd,

$$\Gamma\left(\frac{\nu}{2}\right) = \left(\frac{\nu}{2} - 1\right)\left(\frac{\nu}{2} - 2\right) \cdots \left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)$$

4. $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

Reference:

Abramowitz and Stegun, *Handbook of Mathematical Functions,* National Bureau of Standards, 1970.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		B	1.00
4	Input degrees of freedom ν	ν	C	$\Gamma(\nu/2)$
	Optional: Step 5			
5	Input x to compute f(x)	x	D	f(x)
6	Input x to compute P(x)	x	E	P(x)
	(i) For a new case with the			
	same ν go to 5 or 6			
	(ii) For a new case with			
	different ν, go to 2			
	*Note: to clear print mode			
	press →		CLF	
			0	

Example 1:

If degrees of freedom $\nu = 20$, find f(x), P(x) for x = 9.6 and x = 15.

Keystrokes:	**Outputs:**	
A →	0.00	
B →	1.00	
20 C →	20.00 ***	
	362880.00 ***	(Γ(20/2))
9.6 D →	9.60 ***	
	0.02 ***	(f(9.6))
9.6 E →	9.60 ***	
	0.03 ***	(P(9.6))
15 E →	15.00 ***	
	0.22 ***	(P(15))
15 D →	15.00 ***	
	0.06 ***	(f(15))

Example 2:

If $\nu = 3$, find f(x) and P(x) for x =7.82.

Keystrokes:	**Outputs:**	
A →	0.00	
B →	1.00	
3 C →	3.00 ***	
	0.89 ***	(Γ(3/2))
7.82 D →	7.82 ***	
	0.02 ***	(f(7.82))
7.82 E →	7.82 ***	
	0.95 ***	(P(7.82))

Notes

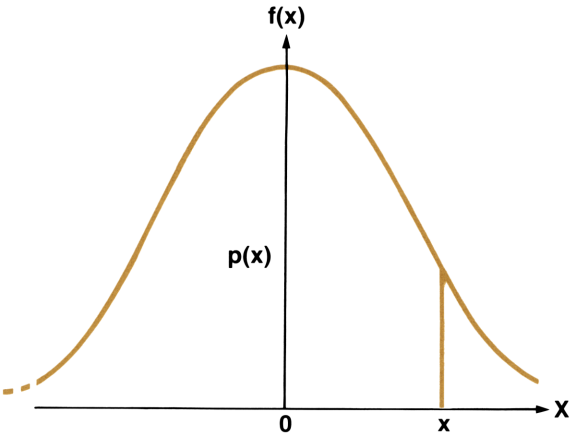
t DISTRIBUTION

This program evaluates the t density function f(x) and the cumulative distribution P(x) for a given x and degrees of freedom ν.

Equations:

1. Density function

$$f(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$



2. Cumulative distribution function

$$P(x) = \int_{-\infty}^{x} f(y)dy$$

$$\text{Let} \quad \theta = \tan^{-1}\left(\frac{|x|}{\sqrt{\nu}}\right)$$

(a) ν even

$$\text{Let} \quad R = \sin\theta \left\{ 1 + \frac{1}{2}\cos^2\theta + \frac{1\cdot 3}{2\cdot 4}\cos^4\theta + \ldots + \frac{1\cdot 3\cdot 5 \ \ldots \ (\nu\text{-}3)}{2\cdot 4\cdot 6 \ \ldots \ (\nu\text{-}2)}\cos^{\nu\text{-}2}\theta \right\}$$

(b) ν odd

$$\text{Let} \quad R = \begin{cases} \dfrac{2\theta}{\pi} & \text{if } \nu = 1 \\ \dfrac{2\theta}{\pi} + \dfrac{2}{\pi}\cos\theta \left\{ \sin\theta \left[1 + \dfrac{2}{3}\cos^2\theta + \ldots + \dfrac{2\cdot 4 \ \ldots \ (\nu\text{-}3)}{1\cdot 3 \ \ldots \ (\nu\text{-}2)}\cos^{\nu-3}\theta \right] \right\} & \text{if } \nu > 1 \end{cases}$$

$$\text{Then} \quad P(x) = \begin{cases} \dfrac{1+R}{2} & \text{if } x > 0 \\ \dfrac{1-R}{2} & \text{if } x \leq 0 \end{cases}$$

Remark:

The program requires $\nu < 141$ for f(x), otherwise erroneous overflow will result.

Reference:

Abramowitz and Stegun, *Handbook of Mathematical Functions*, National Bureau of Standards, 1970.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		B	1.00
4	Input degrees of freedom ν	ν	C	ν
	Optional: Step 5			
5	Input x to compute f(x)	x	D	f(x)
6	Input x to compute P(x)	x	E	P(x)
7	(i) For a new case with the			
	same ν go to 5 or 6			
	(ii) For a new case with a			
	different ν, go to 2			
	*Note: to clear print mode			
	press →		CLF	
			0	

Example 1:

Find f(x) and P(x) for x = 2.2, ν = 11.

Keystrokes:		**Outputs:**	
A	→	0.00	
B	→	1.00	
11 C	→	11.00 ***	(ν)
2.2 E	→	2.20 ***	(x)
		0.97***	(P(2.2))
2.2 D	→	2.20 ***	(x)
		0.04 ***	(f(2.2))

Example 2:

Find f(x) and P(x) for x = −1.75, ν = 30.

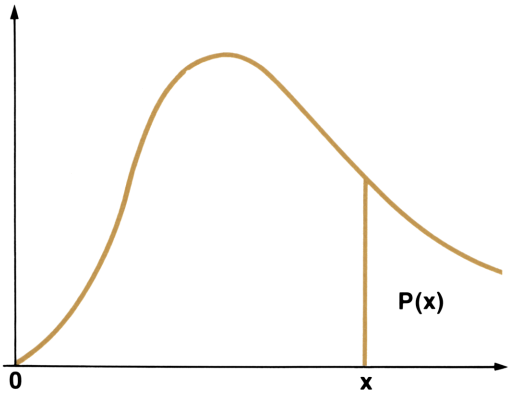
Keystrokes:	**Outputs:**	
A →	0.00	
B →	1.00	
30 C →	30.00 ***	(ν)
1.75 CHS D →	−1.75 ***	(x)
	0.09 ***	(f(−1.75))
1.75 CHS E →	−1.75 ***	(x)
	0.05 ***	(P(−1.75))

F DISTRIBUTION

This program evaluates the integral of the F distribution

$$P(x) = \int_x^{\infty} \frac{\Gamma\left(\frac{\nu_1 + \nu_2}{2}\right) y^{\frac{\nu_1}{2} - 1} \left(\frac{\nu_1}{\nu_2}\right)^{\frac{\nu_1}{2}}}{\Gamma\left(\frac{\nu_1}{2}\right)\Gamma\left(\frac{\nu_2}{2}\right)\left(1 + \frac{\nu_1}{\nu_2} y\right)^{\frac{\nu_1 + \nu_2}{2}}} dy$$

for given values of x (>0), degrees of freedom ν_1, ν_2, provided either ν_1 or ν_2 is even.



The integral is evaluated by means of the following series:

(1) ν_1 even

$$P(x) = t^{\frac{\nu_2}{2}} \left[1 + \frac{\nu_2}{2}(1 - t) + \ldots + \frac{\nu_2(\nu_2 + 2) \ldots (\nu_2 + \nu_1 - 4)}{2 \cdot 4 \ldots (\nu_1 - 2)} (1-t)^{\frac{\nu_1 - 2}{2}} \right]$$

(2) ν_2 even

$$P(x) = 1 - (1 - t)^{\frac{\nu_1}{2}} \left[1 + \frac{\nu_1}{2} t + \ldots + \frac{\nu_1(\nu_1 + 2) \ldots (\nu_2 + \nu_1 - 4)}{2 \cdot 4 \ldots (\nu_2 - 2)} t^{\frac{\nu_2 - 2}{2}} \right]$$

$$\text{where } t = \frac{\nu_2}{\nu_2 + \nu_1 x}$$

Note: Usually ν_1 is identified as the degree of freedom for numerator, and ν_2 is identified as the degree of freedom for denominator.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		B	1.00
4	Input ν_1	ν_1	C	ν_1
5	Input ν_2	ν_2	D	ν_2
6	Input x to calculate P(x)	x	E	P(x)
7	For a new case go to 2			
	*Note: to clear print mode			
	press →		CLF	
			0	

Examples:

1. $\nu_1 = 7$, $\nu_2 = 6$
   P(4.21) = 0.05
2. $\nu_1 = 4$, $\nu_2 = 20$
   P(2.25) = 0.10

Keystrokes:	**Outputs:**	
A →	0.00	
B →	1.00	
7 C →	7.00 ***	(ν_1)
6 D →	6.00 ***	(ν_2)
4.21 E →	4.21 ***	(x)
	0.05 ***	(P(x))
4 C →	4.00 ***	(ν_1)
20 D →	20.00 ***	(ν_2)
2.25 E →	2.25 ***	(x)
	0.10 ***	(P(x))

MULTIPLE LINEAR REGRESSION

For a set of data points $\{(x_i, y_i, z_i), i = 1, 2, \ldots, n\}$ this program fits a linear equation of the form

$$z = a + bx + cy$$

by the least squares method.
Regression coefficients a, b, c can be found by solving the normal equations:

$$\left.\begin{aligned} \Sigma z_i &= an + b\,\Sigma x_i + c\,\Sigma y_i \\ \Sigma x_i z_i &= a\,\Sigma x_i + b\,\Sigma x_i^2 + c\,\Sigma x_i y_i \\ \Sigma y_i z_i &= a\,\Sigma y_i + b\,\Sigma x_i y_i + c\,\Sigma y_i^2 \end{aligned}\right\} \quad i = 1, 2, \ldots, n$$

$$c = \frac{A - B}{\left[n\Sigma x_i^2 - (\Sigma x_i)^2\right]\left[n\Sigma y_i^2 - (\Sigma y_i)^2\right] - \left[n\Sigma x_i y_i - (\Sigma x_i)(\Sigma y_i)\right]^2}$$

where $A = \left[n\Sigma x_i^2 - (\Sigma x_i)^2\right]\left[n\Sigma y_i z_i - (\Sigma y_i)(\Sigma z_i)\right]$

$$B = \left[n\Sigma x_i y_i - (\Sigma x_i)(\Sigma y_i)\right]\left[n\Sigma x_i z_i - (\Sigma x_i)(\Sigma z_i)\right]$$

$$b = \frac{\left[n\Sigma x_i z_i - (\Sigma x_i)(\Sigma z_i)\right] - c\left[n\Sigma x_i y_i - (\Sigma x_i)(\Sigma y_i)\right]}{n\Sigma x_i^2 - (\Sigma x_i)^2}$$

$$a = \frac{1}{n}\left(\Sigma z_i - c\,\Sigma y_i - b\,\Sigma x_i\right)$$

$$R^2 = \frac{a\,\Sigma z_i + b\,\Sigma x_i z_i + c\,\Sigma y_i z_i - \frac{1}{n}(\Sigma z_i)^2}{(\Sigma z_i^2) - \frac{(\Sigma z_i)^2}{n}}$$

Reference: Introduction to the Theory of Statistics, Mood and Graybill, McGraw-Hill, 1963

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		B	1.00
4	Do 5~6 for i = 1,2, . . . ,n			
5	Input x_i	x_i	ENTER↑	
	y_i	y_i	ENTER↑	
	z_i	z_i	C	i
6	If you made a mistake in			
	inputting x_k, y_k, and z_k,			
	then correct by →	x_k	ENTER↑	
		y_k	ENTER↑	
		z_k	D	i−1
7	Calculate coefficients: a		E	a
	b		R/S	b
	c		R/S	c
8	Calculate the square of			
	multiple correlation			
	coefficient R^2		f A	R^2
9	Calculate estimated z from			
	regression, input: x	x	ENTER↑	
	y	y	f B	$\hat{z}$
10	Repeat step 9 for different			
	(x, y)'s			
11	Recall sums: Σx_i		f C	Σx_i
	Σy_i		R/S	Σy_i
	Σz_i		R/S	Σz_i
12	Recall sums of squares: Σx_i^2		f D	Σx_i^2
	Σy_i^2		R/S	Σy_i^2
	Σz_i^2		R/S	Σz_i^2

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
13	Recall sums of cross			
	products: $\Sigma x_i y_i$		f E	$\Sigma x_i y_i$
	$\Sigma x_i z_i$		R/S	$\Sigma x_i z_i$
	$\Sigma y_i z_i$		R/S	$\Sigma y_i z_i$
14	For a new case, go to 2			
	*Note: to clear print mode			
	press →		CLF	
			0	

Example:

A set of data points are given as the following

i	1	2	3	4
x_i	1.5	0.45	1.8	2.8
y_i	0.7	2.3	1.6	4.5
z_i	2.1	4.0	4.1	9.4

Find the regression line, coefficients a, b, c, R^2, $\hat{z}$, sums, sums of squares, and sums of products.

Keystrokes: **Outputs:**

A → 0.00

B → 1.00

1.5 ENTER↑ 0.7 ENTER↑ 2.1 C → 1.50 ***
0.70 ***
2.10 ***
1.00 ***

9 ENTER↑ 9 ENTER↑ 9 C → 9.00 ***
9.00 *** (error)
9.00 ***
2.00 ***

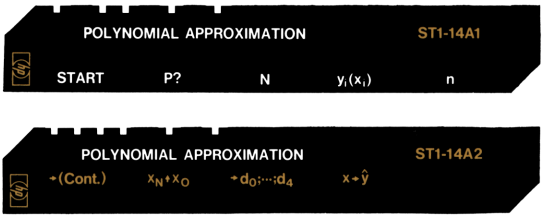
9 [ENTER↑] 9 [ENTER↑] 9 [D] →	9.00 ***	
	9.00 ***	(correction)
	9.00 ***	
	1.00 ***	
0.45 [ENTER↑] 2.3 [ENTER↑] 4 [C] →	0.45 ***	
	2.30 ***	
	4.00 ***	
	2.00 ***	
1.8 [ENTER↑] 1.6 [ENTER↑] 4.1 [C] →	1.80 ***	
	1.60 ***	
	4.10 ***	
	3.00 ***	
2.8 [ENTER↑] 4.5 [ENTER↑] 9.4 [C] →	2.80 ***	
	4.50 ***	
	9.40 ***	
	4.00 ***	
[E] →	−0.10 ***	(a)
[R/S] →	0.79 ***	(b)
[R/S] →	1.63 ***	(c)
[f] [A] →	1.00 ***	(R^2)
[DSP] [9] [PRINT x] →	0.998411259 ***	
[DSP] [2]		
2 [ENTER↑] 3 [f] [B] →	2.00 ***	
	3.00 ***	
	6.37 ***	($\hat{z}$)
[f] [C] →	6.55 ***	(Σx_i)
[R/S] →	9.10 ***	(Σy_i)
[R/S] →	19.60 ***	(Σz_i)
[f] [D] →	13.53 ***	(Σx_i^2)
[R/S] →	28.59 ***	(Σy_i^2)
[R/S] →	125.58 ***	(Σz_i^2)
[f] [E] →	17.57 ***	($\Sigma x_i y_i$)
[R/S] →	38.65 ***	($\Sigma x_i z_i$)
[R/S] →	59.53 ***	($\Sigma y_i z_i$)

Regression line is

$$z = -0.10 + 0.79x + 1.63y$$

For $x = 2$ and $y = 3$, $\hat{z} = 6.37$

POLYNOMIAL APPROXIMATION



Suppose $x_0, x_1, \ldots, x_N$ are equally spaced points ($x_0 < x_N$) at which the corresponding values $f(x_0), f(x_1), \ldots, f(x_N)$ of a function f(x) are known.

This program approximates in the least squares sense the function f(x) by a polynomial of degree m, where $2 \leq m \leq 4$. The special Chebyshev polynomials for discrete intervals are used.

Equations:

Let $f_n(x)$ be the orthogonal polynomials ($x = 0, 1, 2, \ldots, N$) such that

$$f_0(x) = 1$$

$$f_1(x) = 1 - \frac{2x}{N} \text{ and}$$

$$(n+1)(N-n)\, f_{n+1}(x) = (2n+1)(N-2x)\, f_n(x) - n(N+n+1)\, f_{n-1}(x)$$

where

$$n = 1, 2, \ldots, m-1.$$

Then let

$$(f_n, f_n) = \frac{(N+n+1)!\,(N-n)!}{(2n+1)\,(N!)^2}$$

$$(f, f_n) = \sum_{j=0}^{n} f_n(j)\, f(x_j)$$

and

$$a_n = \frac{(f, f_n)}{(f_n, f_n)}.$$

This program computes all values of (f, f_n) for $n = 0, 1, 2, 3, 4$. If the degree $m = 4$, all terms are used; if $m = 3$, (f, f_4) is replaced by zero in later calculations; and if $m = 2$, (f, f_4) and (f, f_3) are both replaced by zero.

Let $g_n(u)$ be the symmetrical form of the orthogonal polynomial in the domain $-1 < u < 1$ such that

$$g_0(u) = 1 \qquad g_1(u) = u$$

and

$$g_{n+1}(u) = \frac{(2 + 1)\, N}{(n + 1)\, (N - n)} u g_n(u) - \frac{n(N + n + 1)}{(n + 1)\, (N - n)} g_{n-1}(u)$$

where

$$n = 1, 2, \ldots, m - 1.$$

The program computes the coefficients of the polynomial

$$\sum_{n=0}^{N} a_n\, g_n(u) = b_0 + b_1\, u + b_2\, u^2 + b_3\, u^3 + b_4\, u^4. \qquad (1)$$

Then $g_n(u)$ is shifted to a proper interval between x_0 and x_N by letting

$$u = \beta + \alpha x$$

where

$$\alpha = \frac{-2}{x_N - x_0}$$

$$\beta = \frac{x_N + x_0}{x_N - x_0}$$

The transformation is done in two steps. First, let $z = u - \beta$, thus (1) becomes

$$c_0 + c_1\, z + c_2\, z^2 + c_3\, z^3 + c_4\, z^4 \qquad (2)$$

where

$$c_0 = b_0 + b_1 \beta + b_2 \beta^2 + b_3 \beta^3 + b_4 \beta^4$$

$$c_1 = b_1 + 2b_2 \beta + 3b_3 \beta^2 + 4b_4 \beta^3$$

$$c_2 = b_2 + 3b_3 \beta + 6b_4 \beta^2$$

$$c_3 = b_3 + 4\, b_4 \beta$$

$$c_4 = b_4.$$

Then set $z = \alpha x$ and (2) becomes

$$d_0 + d_1 x + d_2 x^2 + d_3 x^3 + d_4 x^4 \qquad (3)$$

where

$$d_i = \alpha^i c_i \ (i = 0, 1, 2, 3, 4).$$

(3) is the polynomial approximation for the function f(x).

Note: $N \geqslant 4$ has to be satisfied in order to make the program work.

Reference:

Abramowitz and Stegun, *Handbook of Mathematical Functions,* National Bureau of Standards, 1970.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2 of card 1			
2	Initialize		A	0.00
3	To set print mode*		B	1.00
4	Input N**	N	C	N
5	Do 6 for i = 0,1,2, . . . , N			
6	Input y_i (x_i)	$y_i(x_i)$	D	i
7	Input n for nth order fit	n	E	0.00
8	Load side 1 and side 2 of card 2			
9	To continue execution of			
	program		f A	1.00

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
10	Input x_N	x_N	ENTER↑	
	and x_0	x_0	f B	
11	To obtain the coefficients: d_i		f C	d_0
			R/S	d_1
			R/S	d_2
			R/S	d_3
			R/S	d_4
12	To evaluate y from the	x	f D	$\hat{y}$
	polynomial			
13	For a new case, go to 1			
	**N = No. of data −1			
	*Note: to clear print mode			
	press →		CLF	
			0	

Example:

Find a third order polynomial approximation for the following data.

x	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3
f(x)	2.72	3.49	4.48	5.75	7.39	9.49	12.18	15.64	20.09

(Note: $f(x) = e^x$)

Keystrokes: **Outputs:**

Load side 1 and side 2 of Card 1

A → 0.00

B → 1.00

8 C → 8.00 *** (N)

2.72 D → 2.72 ***

1.00 ***

3.49 D → 3.49 ***

2.00 ***

4.48 D → 4.48***
3.00 ***

5.75 D → 5.75 ***
4.00 ***

7.39 D → 7.39 ***
5.00 ***

9.49 D → 9.49 ***
6.00 ***

12.18 D → 12.18 ***
7.00 ***

15.64 D → 15.64 ***
8.00 ***

20.09 D → 20.09 ***
9.00 ***

3 E → 3.00 *** (n)

Load side 1 and side 2 of Card 2

f A → 1.00
3 ENTER↑ 1 f B → 3.00 *** (x_N)
1.00 *** (x_o)

f C → −1.79 *** (d_0)

R/S → 7.03 *** (d_1)

R/S → −3.85 *** (d_2)

R/S → 1.31 *** (d_3)

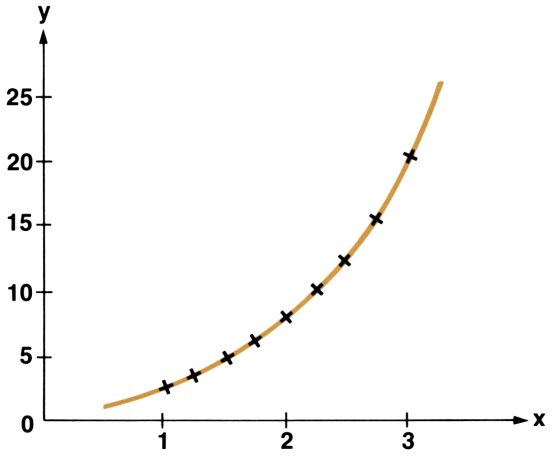
R/S → 0.00 *** (d_4)

2 f D → 2.00 ***
7.35 *** ($\hat{y}$)

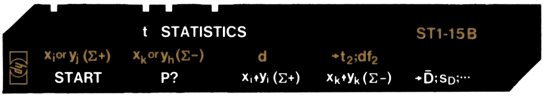
3 f D → 3.00 ***
20.06 *** (ŷ)

1 f D → 1.00 ***
2.69 *** (ŷ)

The polynomial is $-1.79 + 7.03\ x - 3.85\ x^2 + 1.31\ x^3$.



t STATISTICS



I. Paired t Statistic

Given a set of paired observations from two normal populations with means μ_1, μ_2 (unknown)

x_i	x_1	x_2	...	x_n
y_i	y_1	y_2	...	y_n

let

$$D_i = x_i - y_i$$

$$\bar{D} = \frac{1}{n} \sum_{i=1}^{n} D_i$$

$$s_D = \sqrt{\frac{\Sigma D_i^2 - \frac{1}{n} (\Sigma D_i)^2}{n - 1}}$$

$$s_{\bar{D}} = \frac{s_D}{\sqrt{n}}$$

The test statistic

$$t = \frac{\bar{D}}{s_{\bar{D}}}$$

which has n – 1 degrees of freedom (df) can be used to test the null hypothesis

$$H_0: \mu_1 = \mu_2$$

Reference:

Statistics in Research, B. Ostle, Iowa State University Press, 1963.

II. t Statistic For Two Means

Suppose $\{x_1, x_2, \ldots, x_{n1}\}$ and $\{y_1, y_2, \ldots, y_{n2}\}$ are independent random samples from two normal populations having means μ_1, μ_2 (unknown) and the same unknown variance σ^2.

We want to test the null hypothesis

H_0: $\mu_1 - \mu_2 = d$

Define

$$\bar{x} = \frac{1}{n_1} \sum_{i=1}^{n_1} x_i$$

$$\bar{y} = \frac{1}{n_2} \sum_{i=1}^{n_2} y_i$$

$$t = \frac{\bar{x} - \bar{y} - d}{\sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \quad \sqrt{\frac{\Sigma x_i^2 - n_1\bar{x}^2 + \Sigma y_i^2 - n_2\bar{y}^2}{n_1 + n_2 - 2}}}$$

We can use this t statistic which has the t distribution with $n_1 + n_2 - 2$ degrees of freedom (df) to test the null hypothesis H_0.

Note: n_2, Σy_i, Σy_i^2, n_1, Σx_i, Σx_i^2 are in registers R_1 through R_6.

Reference:

Statistical Theory and Methodology in Science and Engineering, K. A. Brownlee, John Wiley & Sons, 1965.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		B	1.00
4	For paired t statistic, go to 6			
5	For t statistic for two means,			
	go to 11			

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
	Paired t statistic:			
6	Do 7~8 for i = 1,2, . . . , n			
7	Input: x_i	x_i	ENTER↑	
	y_i	y_i	C	i
8	If you made a mistake in			
	inputting x_k, y_k then correct by	x_k	ENTER↑	
		y_k	D	i−1
9	Calculate: $\bar{D}$		E	$\bar{D}$
	s_D		R/S	s_D
	test statistic:		R/S	t
	degrees of freedom		R/S	df
10	For a new case, go to 2			
	t statistic for two means:			
11	Do 12 ~ 13 for i = 1,2, . . ., n_1			
12	Input x_i	x_i	f A	i
13	If you made a mistake in			
	inputting x_k, then correct by →	x_k	f B	i−1
14	Null hypothsis test	d	f C	d
15	Do 16~17 for j = 1,2, . . . , n_2	y_j	f A	j
16	If you made a mistake in			
	inputting y_k, then correct by →	y_k	f B	j−1
17	Calculate: t		f D	t
	df		R/S	df
18	For a different value of d	d	f C	d
	Calculate: t		f D	t
	df		R/S	df
19	For a new case, go to 2			
	*Note: to clear print mode			
	press →		CLF	
			0	

Example 1:

x_1	14	17.5	17	17.5	15.4
y_1	17	20.7	21.6	20.9	17.2

$\overline{D}$ = -3.20
s_D = 1.00
t = -7.16
df = 4.00

Keystrokes:	**Outputs:**	
A →	0.00	
B →	1.00	
14 ENTER↑ 17 C →	14.00 ***	
	17.00 ***	
	1.00 ***	
17 ENTER↑ 15 C →	17.00 ***	
	15.00 ***	(error)
	2.00 ***	
17 ENTER↑ 15 D →	17.00 ***	
	15.00 ***	(correction)
	1.00 ***	
17.5 ENTER↑ 20.7 C →	17.50 ***	
	20.70 ***	
	2.00 ***	
17 ENTER↑ 21.6 C →	17.00 ***	
	21.60 ***	
	3.00 ***	
17.5 ENTER↑ 20.9 C →	17.50 ***	
	20.90 ***	
	4.00 ***	
15.4 ENTER↑ 17.2 C →	15.40 ***	
	17.20 ***	
	5.00 ***	

E →	−3.20 ***	($\overline{D}$)
R/S →	1.00 ***	(s_D)
R/S →	−7.16 ***	(t)
R/S →	4.00 ***	(df)

Example 2:

x: 79, 84, 108, 114, 120, 103, 122, 120
y: 91, 103, 90, 113, 108, 87, 100, 80, 99, 54
n_1 = 8
n_2 = 10
If d = 0 (i.e., H_0: $\mu_1 = \mu_2$)
then t = 1.73, df = 16.00

Keystrokes:	**Outputs:**	
A →	0.00	
B →	1.00	
79 f A →	79.00 ***	
	1.00 ***	
84 f A →	84.00 ***	
	2.00 ***	
99 f A →	99.00 ***	
	3.00 ***	(error)
99 f B →	99.00 ***	
	2.00 ***	(correction)
108 f A →	108.00 ***	
	3.00 ***	
114 f A →	114.00 ***	
	4.00 ***	
120 f A →	120.00 ***	
	5.00 ***	
103 f A →	103.00 ***	
	6.00 ***	
122 f A →	122.00 ***	
	7.00 ***	

120 f A → 120.00 ***
8.00 ***

0 f C → 0.00 *** (d)

91 f A → 91.00 ***
1.00 ***

103 f A → 103.00 ***
2.00 ***

90 f A → 90.00 ***
3.00 ***

113 f A → 113.00 ***
4.00 ***

108 f A → 108.00 ***
5.00 ***

87 f A → 87.00 ***
6.00 ***

100 f A → 100.00 ***
7.00 ***

80 f A → 80.00 ***
8.00 ***

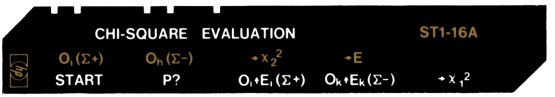
99 f A → 99.00 ***
9.00 ***

54 f A → 54.00 ***
10.00 ***

f D → 1.73 *** (t)

R/S → 16.00 *** (df)

CHI-SQUARE EVALUATION



This program calculates the value of the χ^2 statistic for the goodness of fit test by the equation

$$\chi_1^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

where O_i = observed frequency
E_i = expected frequency

If the expected values are equal

$$\left(E = E_i = \frac{\Sigma O_i}{n} \text{ for all i} \right)$$

then

$$\chi_2^2 = \frac{n\Sigma O_i^2}{\Sigma O_i} - \Sigma O_i$$

Note: In order to apply the goodness of fit test to a set of given data, combining some classes may be necessary to make sure that each expected frequency is not too small (say, not less than 5).

Reference: Mathematical Statistics, J. E. Freund, Prentice Hall, 1962

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		B	1.00
4	For equal expected values,			
	go to 10			
5	Do 6~7 for i = 1,2, ... , n			
6	Input: O_i	O_i	ENTER↑	
	E_i	E_i	C	i

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
7	If you made a mistake in			
	inputting O_k and E_k, then			
	correct by →	O_k	ENTER↑	
		E_k	D	i–1
8	Calculate χ_1^2		E	χ_1^2
9	For a new case, go to 2			
10	Do 11~12 for i = 1,2, . . . , n			
	for equal expected values			
11	Input: O_i	O_i	f A	i
12	If you made a mistake in			
	inputting O_h, then correct by →	O_h	f B	i–1
13	Calculate: χ_2^2		f C	χ_2^2
	E		f D	E
14	For a new case, go to 2			
	*Note: to clear print mode			
	press →		CLF	
			0	

Examples 1:

Find the value of χ^2 statistic for the goodness of fit for the following data set:

O_i	8	50	47	56	5	14
E_i	9.6	46.75	51.85	54.4	8.25	9.15

$\chi_1^2 = 4.84$

Keystrokes: **Outputs:**

A → 0.00

B → 1.00

8 ENTER↑ 9.6 C → 8.00 ***

9.60 ***

1.00 ***

50 ENTER↑ 46.75 C → 50.00 ***
46.75 ***
2.00 ***

47 ENTER↑ 51.85 C → 47.00 ***
51.85 ***
3.00 ***

56 ENTER↑ 54.4 C → 56.00 ***
54.40 ***
4.00 ***

5 ENTER↑ 8.25 C → 5.00 ***
8.25 ***
5.00 ***

100 ENTER↑ 100 C → 100.00 *** (error)
100.00 ***
6.00 ***

100 ENTER↑ 100 D → 100.00 *** (correction)
100.00 ***
5.00 ***

14 ENTER↑ 9.15 C → 14.00 ***
9.15 ***
6.00 ***

E → 4.84 *** (χ_1^2)

Example 2:

The following table shows the observed frequencies in tossing a die 120 times. χ^2 can be used to test if the die is fair.

Note: Assume that the expected frequencies are equal.

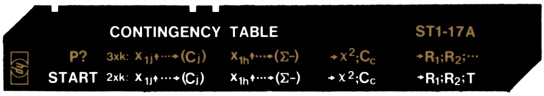
number	1	2	3	4	5	6
frequency O_i	25	17	15	23	24	16

$\chi_2^2 = 5.00$

$E = 20.00$

Keystrokes:	Outputs:	
A	0.00	
B	1.00	
25 f A	25.00 *** 1.00 ***	
17 f A	17.00 *** 2.00 ***	
19 f A	19.00 *** 3.00 ***	(error)
19 f B	19.00 *** 2.00 ***	(correction)
15 f A	15.00 *** 3.00 ***	
23 f A	23.00 *** 4.00 ***	
24 f A	24.00 *** 5.00 ***	
16 f A	16.00 *** 6.00 ***	
f C	5.00 ***	(χ_2^2)
f D	20.00 ***	(E)

CONTINGENCY TABLE



Contingency tables can be used to test the null hypothesis that two variables are independent.

I. 2 x k CONTINGENCY TABLE

i \ j	1	2	...	k	Totals
1	x_{11}	x_{12}	...	x_{1k}	R_1
2	x_{21}	x_{22}	...	x_{2k}	R_2
Totals	C_1	C_2	...	C_k	T

Test statistic

$$\chi^2 = \frac{T}{R_1} \sum_{i=1}^{k} \frac{x_{1i}^2}{C_i} + \frac{T}{R_2} \sum_{i=1}^{k} \frac{x_{2i}^2}{C_i} - T$$

Degrees of freedom df = k - 1

Pearson's coefficient of contingency C_c measures the degree of association between the two variables

$$C_c = \sqrt{\frac{\chi^2}{T + \chi^2}}$$

II. 3 x k CONTINGENCY TABLE

i \ j	1	2	...	k	Totals
1	x_{11}	x_{12}	...	x_{1k}	R_1
2	x_{21}	x_{22}	...	x_{2k}	R_2
3	x_{31}	x_{32}	...	x_{3k}	R_3
Totals	C_1	C_2	...	C_k	T

This program computes the χ^2 statistic (with 2(k – 1) degrees of freedom) for testing the independence of the two variables. Also Pearson's coefficient of contingency C_c, which measures the degree of association between the two variables, is calculated.

Equations:

$$\text{Row sum} \quad R_i = \sum_{j=1}^{k} x_{ij} \qquad i = 1, 2, 3$$

$$\text{Column sum} \quad C_j = \sum_{i=1}^{3} x_{ij} \qquad j = 1, 2, \ldots, k$$

$$\text{Total} \quad T = \sum_{i=1}^{3} \sum_{j=1}^{k} x_{ij}$$

Chi-square statistic

$$\chi^2 = \sum_{i=1}^{3} \sum_{j=1}^{k} \frac{(x_{ij} - E_{ij})^2}{E_{ij}}$$

$$= T \left(\sum_{i=1}^{3} \sum_{j=1}^{k} \frac{x_{ij}^2}{R_i C_j} \right) - T$$

Where the expected frequency

$$E_{ij} = \frac{R_i C_j}{T}$$

Contingency coefficient

$$C_c = \sqrt{\frac{\chi^2}{T + \chi^2}}$$

Reference:

B. Ostle, *Statistics in Research,* Iowa State University Press, 1972.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		f A	1.00
4	For 2 x k go to 5			
	For 3 x k go to 11			
5	Do 6~7 for j = 1,2, . . . , k			
	for 2 × k			
6	Input: x_{1j}	x_{1j}	ENTER↑	
	x_{2j}	x_{2j}	B	j
	(Optional) Calculate column			
	sum C_j		R/S	C_j
7	If you made a mistake in			
	inputting x_{1h} and x_{2h}, then			
	correct by →	x_{1h}	ENTER↑	
		x_{2h}	C	j−1
	(Optional) Calculate column			
	sum C_h (correction)		R/S	$-C_h$
8	Calculate: χ^2		D	χ^2
	C_c		R/S	C_c
9	Calculate: row sums R_1		E	R_1
	R_2		R/S	R_2
	total T		R/S	T
10	For a new case go to 2			
11	Do 12~13 for j =1,2, . . . , k			
	for 3 X k			
12	Input x_{1j}	x_{1j}	ENTER↑	
	x_{2j}	x_{2j}	ENTER↑	
	x_{3j}	x_{3j}	f B	j
	(Optional) Calculate column			
	sum C_j		R/S	C_j

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
13	If you made a mistake in			
	inputting X_{1h} , X_{2h} and X_{3h},			
	then correct by →	x_{1h}	ENTER↑	
		x_{2h}	ENTER↑	
		x_{3h}	f C	j−1
	(Optional) Calculate column			
	sum C_h (correction)		R/S	$-C_h$
14	Calculate: χ^2		f D	χ^2
	C_c		R/S	C_c
15	Calculate: row sums R_1		f E	R_1
	R_2		R/S	R_2
	R_3		R/S	R_3
	total T		R/S	T
16	For a new case go to 2			
	*Note: to clear print mode			
	press →		CLF	
			0	

Example 1:

A random sample of 250 men and 250 women were polled as to their desires concerning the ownership of television sets. The data in the following table resulted. Apply the program to analyze the result of the poll, i.e., can the hypothesis that the desire to own a television set is independent of sex be rejected?

Results of Sample Poll on Television Ownership

Classification	Men	Women	Total
Want television	80	120	200
Don't want television	170	130	300
Total	250	250	500

Keystrokes:	Outputs:
A →	0.00
f A →	1.00
80 ENTER↑ 170 B →	80.00 ***
	170.00 ***
	1.00 ***
120 ENTER↑ 130 B →	120.00 ***
	130.00 ***
	2.00 ***
D →	13.33 *** (χ^2)

$$\chi^2 = 13.33 > \chi^2_{.99(1)} = 6.63$$

Thus, the hypothesis that desire to own a television set is independent of sex is rejected.

Example 2:

Find test statistic χ^2 and coefficient of contingency C_c for the following set of data.

	1	2	3
A	2	5	4
B	3	8	7

Keystrokes:	Outputs:
A →	0.00
f A →	1.00
2 ENTER↑ 3 B →	2.00 ***
	3.00 ***
	1.00 ***
R/S →	5.00 *** (C_1)
5 ENTER↑ 8 B →	5.00 ***
	8.00 ***
	2.00 ***

R/S	13.00 ***	(C_2)
6 ENTER↑ 9 B	6.00 ***	(error)
	9.00 ***	
	3.00 ***	
R/S	15.00 ***	(C_3)
6 ENTER↑ 9 C	6.00 ***	(correction)
	9.00 ***	
	2.00 ***	
R/S	−15.00 ***	($-C_3$)
4 ENTER↑ 7 B	4.00 ***	
	7.00 ***	
	3.00 ***	
R/S	11.00 ***	(C_3)
D	0.02 ***	(χ^2)
R/S	0.03 ***	(C_c)
E	11.00 ***	(R_1)
R/S	18.00 ***	(R_2)
R/S	29.00 ***	(T)

Example 3:

Find test statistic χ^2 and coefficient of contingency C_c for the following set of data.

i \ j	1	2	3	4
1	36	67	49	58
2	31	60	49	54
3	58	87	80	68

Keystrokes:	Outputs:	
A →	0.00	
f A →	1.00	
36 ENTER↑ 31 ENTER↑ 58 f B →	36.00 ***	
	31.00 ***	
	58.00 ***	
	1.00 ***	
R/S →	125.00 ***	(C_1)
67 ENTER↑ 60 ENTER↑ 87 f B →	67.00 ***	
	60.00 ***	
	87.00 ***	
	2.00 ***	
R/S →	214.00 ***	(C_2)
4 ENTER↑ 49 ENTER↑ 80 f B →	4.00 ***	(error)
	49.00 ***	
	80.00 ***	
	3.00 ***	
R/S →	133.00 ***	(C_3)
49 ENTER↑ 49 ENTER↑ 80 f B →	49.00 ***	
	49.00 ***	
	80.00 ***	
	4.00 ***	
R/S →	178.00 ***	(C_4)
4 ENTER↑ 49 ENTER↑ 80 f C →	4.00 ***	(correction)
	49.00 ***	
	80.00 ***	
	3.00 ***	
R/S →	−133.00 ***	($-C_3$)

58 ENTER↑ 54 ENTER↑ 68 f B →	58.00 ***	
	54.00 ***	
	68.00 ***	
	4.00 ***	
R/S →	180.00 ***	(C_4)
f D →	3.36 ***	(χ^2)
R/S →	0.07 ***	(C_c)
f E →	210.00 ***	(R_1)
R/S →	194.00 ***	(R_2)
R/S →	293.00 ***	(R_3)
R/S →	697.00 ***	(T)

SPEARMAN'S RANK CORRELATION COEFFICIENT

Spearman's rank correlation coefficient is a measure of rank correlation under the following circumstance: n individuals are ranked from 1 to n according to some specified characteristic by 2 observers, and we wish to know if the 2 rankings are substantially in agreement with one another.

Spearman's rank correlation coefficient is defined by

$$r_s = 1 - \frac{6\sum_{i=1}^{n} D_i^2}{n(n^2 - 1)}$$

where n = number of paired observations (x_i, y_i)
D_i = rank (x_i) − rank (y_i) = $R_i - S_i$

If the X and Y random variables from which these n pairs of observations are derived are independent, then r_s has zero mean and a variance equals to

$$\frac{1}{n - 1}$$

A test for the null hypothesis

H_0: X, Y are independent

is using

$$z = r_s\sqrt{n - 1}$$

which is approximately a standardized normal variable (for large n, say $n \geq 10$).

If the null hypothesis of independence is not rejected, we can infer that the population correlation coefficient $\rho(x, y) = 0$, but dependence between the variables does not necessarily imply that $\rho(x, y) \neq 0$.

Note: $-1 \leq r_s \leq 1$
$r_s = 1$ indicates complete agreement in order of the ranks and $r_s = -1$ indicates complete agreement in the opposite order of the ranks.

Reference: Nonparametric Statistical Inference, J. D. Gibbons, McGraw Hill, 1971

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1			
2	Initialize		A	0.00
3	To set print mode*		B	1.00
4	Do 5~6 for i=1, 2, - - - - , n			
5	Input R_i	R_i	ENTER↑	
	S_i	S_i	C	i
6	If you made a mistake in			
	inputting R_k and S_k, then			
	correct by →	R_k	ENTER↑	
		S_k	D	i−1
7	Calculate: r_s		E	r_s
	z		R/S	z
8	For a new case, go to 2			
	*Note: to clear print mode			
	press →		CLF	
			0	

Example:

The following data set is the result of two tests in a class; find r_s and z.

Student	x_i Math Grade	y_i Stat Grade	R_i Rank of x_i	S_i Rank of y_i
1	82	81	6	7
2	67	75	14	11
3	91	85	3	4
4	98	90	1	2
5	74	80	11	8
6	52	60	15	15
7	86	94	4	1
8	95	78	2	9
9	79	83	9	6
10	78	76	10	10
11	84	84	5	5
12	80	69	8	13
13	69	72	13	12
14	81	88	7	3
15	73	61	12	14

Keystrokes:	**Outputs:**
A →	0.00
B →	1.00
6 ENTER↑ 7 C →	6.00 ***
	7.00 ***
	1.00 ***
14 ENTER↑ 11 C →	14.00 ***
	11.00 ***
	2.00 ***
3 ENTER↑ 4 C →	3.00 ***
	4.00 ***
	3.00 ***
1 ENTER↑ 2 C →	1.00 ***
	2.00 ***
	4.00 ***

11 ENTER↑ 8 C → 11.00 ***
8.00 ***
5.00 ***

5 ENTER↑ 5 C → 5.00 *** (errors)
5.00 ***
6.00 ***

5 ENTER↑ 5 D → 5.00 *** (correction)
5.00 ***
5.00 ***

15 ENTER↑ 15 C → 15.00 ***
15.00 ***
6.00 ***

4 ENTER↑ 1 C → 4.00 ***
1.00 ***
7.00 ***

2 ENTER↑ 9 C → 2.00 ***
9.00 ***
8.00 ***

9 ENTER↑ 6 C → 9.00 ***
6.00 ***
9.00 ***

10 ENTER↑ 10 C → 10.00 ***
10.00 ***
10.00 ***

5 ENTER↑ 5 C → 5.00 ***
5.00 ***
11.00 ***

8 ENTER↑ 13 C → 8.00 ***
13.00 ***
12.00 ***

18-05

13 ENTER↑ 12 C → 13.00 ***
12.00 ***
13.00 ***

7 ENTER↑ 3 C → 7.00 ***
3.00 ***
14.00 ***

12 ENTER↑ 14 C → 12.00 ***
14.00 ***
15.00 ***

E → 0.76 *** (r_s)
R/S → 2.85 *** (z)

Notes

$\bar{x}$ AND R CONTROL CHARTS

In quality control, a chart is used to decide periodically whether a process is in statistical control. The use of such a chart facilitates the detection and elimination of assignable causes of process variation, thereby reducing rejects and rework, improving product quality, and lowering inspection cost. The x chart and R chart are two of the most frequently encountered, they deal with measurement data.

Suppose x_{ij} represents the j^{th} data point from the i^{th} sample, i = 1, 2, ..., m and j = 1, 2, ..., n. This program computes (1) the sample mean $\bar{x}$ and the sample range R_i, (2) the over-all mean $\bar{\bar{x}}$ and the average range $\bar{R}$, (3) the upper control limit $U_{\bar{x}}$ and the lower control limit $L_{\bar{x}}$ for $\bar{x}$, and (4) the upper control limit U_R and the lower control limit L_R for R.

Equations:

1.
$$\bar{x}_i = \sum_{j=1}^{n} x_{ij}/n$$

$$R_i = x_{max} - x_{min}$$

where x_{max} is the maximum of the x values and x_{min} is the minimum of the x values in the i^{th} sample.

2.
$$\bar{\bar{x}} = \sum_{i=1}^{m} \bar{x}_i/m$$

$$\bar{R} = \sum_{i=1}^{m} R_i/m$$

3.
$$L_{\bar{x}} = \bar{\bar{x}} - A_2\bar{R}$$

$$U_{\bar{x}} = \bar{\bar{x}} + A_2\bar{R}$$

where A_2 is the factor for the $\bar{x}$ chart, which can be found in the following table.

4.
$$L_R = D_3\bar{R}$$

$$U_R = D_4\bar{R}$$

D_3 and D_4 are factors for the R chart, which can be found in the table.

Factors for determining from R the 3-sigma control limits for x and R charts.

Number of observations in subgroup n	Factor for x chart A_2	Factors for R chart	
		Lower limit D_3	Upper limit D_4
2	1.88	0	3.27
3	1.02	0	2.57
4	0.73	0	2.28
5	0.58	0	2.11
6	0.48	0	2.00
7	0.42	0.08	1.92
8	0.37	0.14	1.86
9	0.34	0.18	1.82
10	0.31	0.22	1.78
11	0.29	0.26	1.74
12	0.27	0.28	1.72
13	0.25	0.31	1.69
14	0.24	0.33	1.67
15	0.22	0.35	1.65
16	0.21	0.36	1.64
17	0.20	0.38	1.62
18	0.19	0.39	1.61
19	0.19	0.40	1.60
20	0.18	0.41	1.59

All factors are based on the normal distribution.

The table is reproduced from *Statistical Quality Control,* by Grant and Leavenworth, 1972, with permission of McGraw-Hill Book Company.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		B	1.00
4	Do 5~ 9 for i=1, 2, . . . , m			
5	Do 6~7 for j=1, 2, . . . , n			
6	Input x_{ij}	x_{ij}	C	j
7	If you made a mistake in			
	inputting x_{ik}, then correct			
	by ** →	x_{ik}	D	j−1
8	Calculate: x_{max}		E	x_{imax}
	x_{min}		E	x_{imin}
9	Calculate: the mean $\bar{x}_i$		f A	$\bar{x}_i$
	the range R_i		f A	R_i
10	Calculate: $\bar{\bar{x}}$		f B	$\bar{\bar{x}}$
	$\bar{R}$		f B	$\bar{R}$
11	Calculate the $\bar{x}$ limits:			
	the lower	A_2	f C	$L_{\bar{x}}$
	the upper		f C	$U_{\bar{x}}$
12	Calculate L_R	D_3	f D	L_R
13	Calculate U_R	D_4	f E	U_R
14	For a new case, go to 2			
	*Note: to clear print mode			
	press →		0	
			STO	
			E	
	**Note: If there are two or more			
	x_{ik}'s entered incorrectly (one			
	follows the other), then do not			
	try to correct them, go to step 2.			

Example:

For the following set of data, find the lower and upper control limits for $\bar{x}$ and R.

	i \ j	1	2	3	4	5
Sample	1	10.04	10.00	10.02	10.01	10.02
	2	10.00	10.01	10.03	10.02	10.01
	3	10.02	10.02	10.02	10.04	10.01

(Note: n = 5, A_2 = 0.58, D_3 = 0, D_4 = 2.11)

Keystrokes:	**Outputs:**	
A →	0.00	
B →	1.00	
10.04 C →	10.04 ***	
	1.00 ***	
10 C →	10.00 ***	
	2.00 ***	
10.02 C →	10.02 ***	
	3.00 ***	
11.11 C →	11.11 ***	(error)
	4.00 ***	
11.11 D →	11.11 ***	(correction)
	3.00 ***	
10.01 C →	10.01 ***	
	4.00 ***	
10.02 C →	10.02 ***	
	5.00 ***	
E →	10.04 ***	($x_{1\ max}$)
E →	10.00 ***	($x_{1\ min}$)
f A →	10.02 ***	($\bar{x}_1$)
f A →	0.04 ***	(R_1)
10 C →	10.00 ***	
	1.00 ***	
10.01 C →	10.01 ***	
	2.00 ***	
10.03 C →	10.03 ***	
	3.00 ***	
10.02 C →	10.02 ***	
	4.00 ***	

10.01 C → 10.01 ***
5.00 ***

E → 10.03 *** ($x_{2\ max}$)
E → 10.00 *** ($x_{2\ min}$)
A → 10.01 *** ($\bar{x}_2$)
A → 0.03 *** (R_2)

10.02 C → 10.02 ***
1.00 ***
10.02 C → 10.02 ***
2.00 ***
10.04 C → 10.04 *** (error)
3.00 ***
10.04 D → 10.04 *** (correction)
2.00 ***
10.02 C → 10.02 ***
3.00 ***
10.04 C → 10.04 ***
4.00 ***
10.01 C → 10.01 ***
5.00 ***

E → 10.04 *** ($x_{3\ max}$)
E → 10.01 *** ($x_{3\ min}$)
f A → 10.02 *** ($\bar{x}_3$)
f A → 0.03 *** (R_3)

f B → 10.02 *** ($\bar{\bar{x}}$)
f B → 0.03 *** ($\bar{R}$)

0.58 f C → 10.00 *** ($L_{\bar{x}}$)
f C → 10.04 *** ($U_{\bar{x}}$)

0 f D → 0.00 *** (L_R)

2.11 f E → 0.07 *** (U_R)

Reference:

Grant and Leavenworth, *Statistical Quality Control,* McGraw-Hill, 1972

Notes

OPERATING CHARACTERISTIC CURVES

This program evaluates the probability P_a of acceptance for a single sampling plan with finite or infinite lot size.

Equations:

1. Finite lot size

The hypergeometric distribution is used to evaluate the probability P_a. The lot size N, sample size n and the acceptance number c (maximum allowable number of defectives in the sample) should be given. The probability P_a, which is the ordinate of the type A operating characteristic curve, can be computed for the different values of the fraction defective p in the lot.

$$P_a = \sum_{x=0}^{c} f(x)$$

$$f(x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}$$

where f(x) is the hypergeometric density function, M is the number of defectives in a lot which is calculated as the integer part of Np.

The recursive relation

$$f(x+1) = \frac{(x-M)(x-n)}{(x+1)(N-M-n+x+1)} f(x)$$

$$(x = 0, 1, 2, \ldots, n-1)$$

is used to find the probability

$$P_a = \sum_{x=0}^{c} f(x)$$

with starting value

$$f(0) = \frac{\binom{N-M}{n}}{\binom{N}{n}}$$

The binomial coefficient $\binom{N}{n}$ is computed by the formula

$$\binom{N}{n} = \frac{N(N-1)\ \ldots\ (N-n+1)}{1 \cdot 2 \cdot \ldots \cdot n}$$

2. Infinite lot size

The binomial distribution is used to evaluate the probability P_a. The sample size n and the acceptance number c should be given. The probability P_a, which is the ordinate of the type B operating characteristic curve, can be computed for different values of the fraction defective p.

$$P_a = \sum_{x=0}^{c} f(x)$$

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

where $0 \leq p < 1$.

The recursive relation

$$f(x+1) = \frac{p(n-x)}{(x+1)(1-p)} f(x)$$

$$(x = 0, 1, 2, \ldots, n-1)$$

is used to find the probability

$$P_a = \sum_{x=0}^{c} f(x)$$

with starting value

$$f(0) = (1-p)^n$$

Remarks:

1. The program requires that $0 \leq p < 1$.
2. For the type A curve (finite lot size), if $c = 0$, $P_a = f(0)$.

3. For certain combinations of N, n ,and c (usually when they are large), an overflow condition will occur. In that case, the program halts and the display shows all 9's.
4. If the lot size is finite (type A), the execution time mainly depends on the sample size n and the acceptance number c; the larger they are, the longer it takes.
5. The type A OC curve for finite lot sizes is really a set of discrete points, since defectives can occur only as whole numbers. For very large lot sizes, these points come very close together, giving a practically continuous curve.

   Type B OC curves can be considered as suitable approximations to type A OC curves, provided the sample size n is small compared with the lot size N (in general, if $n/N \leq 0.1$).
6. The lot size N has a relatively small effect on the type A OC curve as long as n/N is not large. The absolute sample size n is a much more controlling factor in determining the type A OC curve.

7. The acceptance number *c* affects drastically the probability of acceptance for the type *B OC* curve for any given fraction defective *p*.

References:

1. Dodge and Romig, *Sampling Inspection Tables,* John Wiley and Sons, 1959.
2. Grant and Leavenworth, *Statistical Quality Control,* McGraw-Hill, 1972.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	Initialize		A	0.00
3	To set print mode*		B	1.00
4	For infinite lot size (type B),			
	go to 11			
5	For finite lot size (type A), do 6~ 9			
6	Input lot size	N	C	N
7	Input: sample size	n	ENTER↑	n
	acceptance number	c	D	c
8	Calculate probability P_a	p	E	P_a

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
9	For a different p, go to 8			
10	For a new case, go to 2			
11	Input: sample size	n	ENTER↑	n
	acceptance number	c	f D	c
12	Calculate probability P_a	p	f E	P_a
13	For a different p, go to 12			
14	For a new case, go to 2			
	*Note: to clear print mode			
	press →		CLF	
			0	

Example 1:

Find the type A OC curve for the sampling plan with N = 200, n = 20 and c = 0 (compute P_a for p = 0, 0.02, 0.04, 0.06, 0.08, 0.1, 0.12, 0.14).

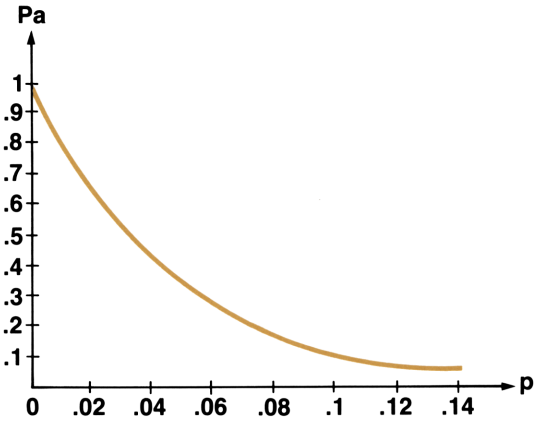
Keystrokes:	**Outputs:**	
A →	0.00	
B →	1.00	
200 C →	200.00 ***	(N)
20 ENTER↑ 0 D →	20.00 ***	(n)
	0.00 ***	(c)
0 E →	0.00 ***	
	1.00 ***	
0.02 E →	0.02 ***	
	0.65 ***	
0.04 E →	0.04 ***	
	0.42 ***	
0.06 E →	0.06 ***	
	0.27 ***	

0.08 E → 0.08 ***
0.17 ***

0.1 E → 0.10 ***
0.11 ***

0.12 E → 0.12 ***
0.07 ***

0.14 E → 0.14 ***
0.04 ***



Example 2:

Find the type B OC curve for the sampling plan with n = 200, c = 1 (compute P_a for p = 0, 0.01, 0.02, 0.03 and 0.04).

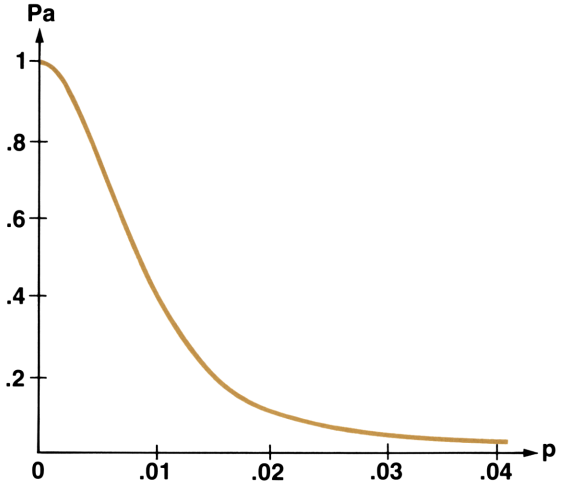
Keystrokes:	**Outputs:**	
A →	0.00	
B →	1.00	
200 ENTER↑ 1 f D →	200.00 ***	(n)
	1.00 ***	(c)
0 f E →	0.00 ***	
	1.00 ***	

0.01 f E → 0.01 ***
0.40 ***

0.02 f E → 0.02 ***
0.09 ***

0.03 f E → 0.03 ***
0.02 ***

0.04 f E → 0.04 ***
2.656338303–03 ***



SINGLE- AND MULTI-SERVER QUEUES

I. Infinite Customers

Suppose there are n ($n \geq 1$) identical stations available to service calls from an infinite number of customers. Let λ be the arrival rate of customers (Poisson input), μ be the service rate of each server (exponential service), and let the service discipline be first-come, first-served. Assume all customers wait in a single line and are directed to whichever station is available. Assume further that, no customers are lost from the queue.

This program computes the following values for given n, λ and μ.

Equations:

1. The intensity factor

$$\rho = \frac{\lambda}{\mu}$$

(ρ must be less than n)

2. The probability that all servers are idle

$$P_0 = \left[\sum_{k=0}^{n-1} \frac{\rho^k}{k!} + \frac{\rho^n}{n! \left(1 - \frac{\rho}{n} \right)} \right]^{-1}$$

3. The probability that all servers are busy

$$P_b = \frac{\rho^n P_0}{n! \left(1 - \frac{\rho}{n} \right)}$$

4. The average number of customers in the queue

$$L_q = \frac{\rho P_b}{n - \rho}$$

5. The average number of customers in the system (waiting or being served)

$$L = L_q + \rho$$

6. The average waiting time in the queue

$$T_q = \frac{L_q}{\lambda}$$

7. The average flow time through the system

$$T = \frac{L}{\lambda}$$

8. The probability of waiting longer than time t

$$P(t) = P_b \, e^{-(n\mu - \lambda)\, t}$$

Remarks:

1. n must be an integer greater than or equal to 1.
2. $\rho < n$, otherwise the queue increases without bound.
3. λ and μ are rates, that is, numbers per unit time.

II. Finite Customers

Suppose there are n ($n \geq 1$) identical stations available to service calls. This program handles the case in which demand arises from a finite rather than an infinite population of customers.

Let the number of customers m be fixed; let a be the mean time between service calls; and s be the mean time to serve one customer. Given m, n, s and a, this program computes the following values.

Equations:

1. The average number of customers in the system (waiting or being served)

$$L = \frac{\sum_{k=0}^{m} k \, Q_k}{\sum_{k=0}^{m} Q_k}$$

where

$$Q_0 = 1$$

$$(m - k + 1)\,\rho\,Q_{k-1} = \begin{cases} kQ_k & \text{if } 1 \leq k \leq n \\ nQ_k & \text{if } n < k \leq m \end{cases}$$

and

$$\rho = \frac{s}{a}$$

2. The average flow time through the system

$$T = aL$$

3. The average number of customers in the queue

$$L_q = m\left[(\rho + 1)\left(\frac{L}{M} - 1\right) + 1\right]$$

4. The average waiting time in the queue

$$T_q = aL_q$$

5. The over-all efficiency factor of the system

$$F = -(\rho + 1)\left(\frac{L}{m} - 1\right)$$

Remarks:

1. For large values of m and/or small values of ρ, the calculation of Q_k in the routine under f C may underflow. To avoid this, the program tests to see if $Q_k < 10^{-90}$. If it does, the program will halt its recursive solution for Q_k and go directly to the calculation of L. This should not affect the calculated value of L.

2. For certain combinations of m, n, s and a, an overflow condition will occur. In that case, the program halts and the display shows all 9's.

3. The execution time for L depends on m; the larger m is, the longer it takes. A rough estimate of the time for this routine (f C) is given by m/30 minutes.

4. Suppose instead of knowing s and a, the service rate μ of each server and the arrival rate λ are given. Then the following formulas can be used to compute s and a in order to run this program.

$$s = \frac{1}{\mu}$$

$$a = \frac{1}{\lambda}$$

Note that $$\rho = \frac{\lambda}{\mu}$$

References:

1. H. M. Wagner, *Principles of Operations Research with Applications to Managerial Decisions,* Prentice-Hall, 1969.

2. James Martin, *Systems Analysis for Data Transmission,* Prentice-Hall, 1972.

3. Hillier and Lieberman, *Introduction to Operations Research,* Holden-Day, 1970.

4. Peck and Hazelwood, *Finite Queuing Tables,* John Wiley and Sons, 1958.

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
1	Load side 1 and side 2			
2	For finite customers go to 11			
3	Do 4 ~ 9 for infinite customers			
4	Input μ	μ	ENTER↑	
	λ	λ	ENTER↑	
	n	n	A	ρ
5	Calculate P_o		B	P_o
	P_b		B	P_b
6	Calculate: L_q		C	L_q
	L		C	L

STEP	INSTRUCTIONS	INPUT DATA/UNITS	KEYS	OUTPUT DATA/UNITS
7	Calculate: T_q		D	T_q
	T		D	T
8	Input t to calculate P(t)	t	E	P(t)
9	For a different t, go to 8			
10	For a new case, go to 2			
11	Do 12 ~ 16 for finite customers			
12	Input: number of customers	m	ENTER↑	
	number of servers	n	f A	m
13	Input: service time	s	ENTER↑	
	arrival time	a	f B	ρ
14	Calculate: customers in system		f C	L
	time through system		f C	T
15	Calculate: queue length		f D	L_q
	waiting time in queue		f D	T_q
16	Calculate efficiency factor F		f E	F
17	For a new case, go to 2			

Example 1:

Bank customers arrive at a bank on the average of 1.2 customers per minute. They join a common queue for 3 tellers, each teller serves at a rate of 30 customers per hour. Find ρ, P_0, P_b, L_q, L, T_q, T and the probability P(2) that a customer will have to wait for more than 2 minutes.

$$\left(\begin{array}{l}\text{Note: Service rate } \mu = \dfrac{30}{60} = 0.5 \text{ customers per minute} \\ \text{Arrival rate } \lambda = 1.2 \text{ customers per minute}\end{array}\right)$$

Keystrokes:

.5 ENTER↑ 1.2 ENTER↑ 3 A ⟶

Outputs:

0.5 *** (μ)
1.20 *** (λ)
3.00 *** (n)
2.40 *** (ρ)

B →	0.06 ***	(P_0)
B →	0.65 ***	(P_b)
C →	2.59 ***	(L_q)
C →	4.99 ***	(L)
D →	2.16 ***	(T_q)
D →	4.16 ***	(T)
2 E →	2.00 ***	(t)
	0.36 ***	(P(t))

Example 2:

A laundromat has 12 washers which require an average of 4 hours of service after every 60 hours of operation. If there is only one service person in the laundromat, find ρ, L, T, L_q, T_q and F.

Keystrokes:	**Outputs:**	
12 ENTER↑ 1 f A →	12.00 ***	(m)
	1.00 ***	(n)
4 ENTER↑ 60 f B →	4.00 ***	(s)
	60.00 ***	(a)
	0.07 ***	(ρ)
f C →	1.64 ***	(L)
f C →	98.66 ***	(T)
f D →	0.95 ***	(L_q)
f D →	57.24 ***	(T_q)
f E →	0.92 ***	(F)

Notes

PROGRAM LISTINGS

The following listings are included for your reference. A table of keycodes and keystrokes corresponding to the symbols used in the listings can be found in Appendix E of your Owners Handbook.

Program		Page
1.	Basic Statistics for Two Variables	**L01-01**
2.	Factorial, Permutation and Combination	**L02-01**
3.	Moments, Skewness and Kurtosis (For Grouped or Ungrouped Data)	**L03-01**
4.	Random Number Generator	**L04-01**
5.	Histogram	**L05-01**
6.	Analysis of Variance (One Way)	**L06-01**
7.	Two Way Analysis of Variance	**L07-01**
8.	Analysis of Covariance (One Way) Card 1 Card 2	**L08-01**
9.	Normal and Inverse Normal Distribution Card 1 Card 2	**L09-01**
10.	Chi-Square Distribution	**L10-01**
11.	t Distribution	**L11-01**
12.	F Distribution	**L12-01**
13.	Multiple Linear Regression	**L13-01**
14.	Polynomial Approximation Card 1 Card 2	**L14-01**
15.	t Statistics	**L15-01**
16.	Chi-Square Evaluation	**L16-01**
17.	Contingency Table	**L17-01**
18.	Spearman's Rank Correlation Coefficient	**L18-01**
19.	$\bar{x}$ and R Control Chart	**L19-01**
20.	Operating Characteristic Curves	**L20-01**
21.	Single- and Multi-Server Queues	**L21-01**

Basic Statistics for Two Variables

001	*LBLA		057	STO9	
002	CLRG		058	P≠S	
003	CF0	Initialize	059	GSB9	
004	CF1		060	GSB8	
005	CF2		061	RTN	
006	P≠S		062	*LBLE	
007	CLRG		063	SF1	Correction for x_k, y_k, f_k.
008	P≠S		064	GSBD	
009	0		065	CF1	
010	RTN		066	RTN	
011	*LBLa	Set flag 0 for print.	067	*LBL3	
012	SF0		068	F1?	
013	1		069	GTO4	
014	RTN		070	Σ+	
015	*LBLC	Correction for x_k, y_k.	071	RTN	
016	F0?		072	*LBL4	
017	GSB0		073	Σ-	
018	SF1		074	RTN	
019	X≠Y		075	*LBLb	
020	Σ-		076	$\bar{x}$	$\bar{x}$, $\bar{y}$
021	GSB9		077	GSB9	
022	GSB8		078	R/S	
023	CF1		079	X≠Y	
024	RTN		080	GSB9	
025	*LBLB		081	GSB8	
026	F0?	Input x_i, y_i.	082	RTN	
027	GSB0		083	*LBLb	
028	X≠Y		084	$\bar{x}$	
029	Σ+		085	STO0	
030	GSB9		086	X≠Y	
031	GSB8		087	P≠S	
032	RTN		088	STO0	
033	*LBLD	Input x_i, y_i, f_i.	089	P≠S	V_x, V_y
034	STOC		090	S	
035	F1?		091	EEX	
036	CHS		092	2	
037	ST+9		093	x	
038	R↓		094	X≠Y	
039	STOB		095	LSTX	
040	R↓		096	x	
041	STOA		097	X≠Y	
042	R↑		098	RCL0	
043	F0?		099	÷	
044	GSB0		100	GSB9	
045	R↑		101	R/S	
046	ABS		102	X≠Y	
047	GSB9		103	P≠S	
048	STO1		104	RCL0	
049	*LBL2		105	P≠S	
050	RCLB		106	÷	
051	RCLA		107	GSB9	
052	GSB3		108	GSB8	
053	DSZ1		109	RTN	
054	GTO2		110	*LBLc	
055	RCL9		111	S	
056	P≠S		112	GSB9	

REGISTERS

0 $\bar{x}$	1	2	3	4	5	6	7	8	9 Σf_i
S0 $\bar{y}$	S1	S2	S3	S4 Σx_i	S5 Σx_i^2	S6 Σy_i	S7 Σy_i^2	S8 $\Sigma x_i y_i$	S9 n
A x_i	B y_i	C f_i	D	E	I				

113	R/S		169	GSB9		
114	X≷Y	s_x, s_y	170	RTN		
115	GSB9		171	*LBLd		
116	GSB8		172	S		
117	RTN		173	RCLE		
118	*LBLc		174	÷		
119	S		175	×		γ_{xy}
120	*LBL1		176	1/X		
121	P≷S	s_x', s_y'	177	GSB9		
122	RCL9		178	GSB8		
123	P≷S		179	RTN		
124	ENT↑		180	*LBLe		
125	X≷Y		181	RCLΣ		
126	1		182	GSB9		Σx_i, Σy_i
127	-		183	R/S		
128	÷		184	X≷Y		$\Sigma x_i y_i$
129	√X		185	GSB9		
130	÷		186	R/S		
131	GSB9		187	P≷S		
132	F2?		188	RCL8		
133	GSB8		189	P≷S		
134	CF2		190	GSB9		
135	R/S		191	GSB8		
136	LSTX		192	RTN		
137	S		193	*LBLe		
138	X≷Y		194	P≷S		
139	SF2		195	RCL7		
140	GTO1		196	RCL5		Σx_i^2, Σy_i^2
141	RTN		197	P≷S		
142	*LBLd		198	GSB9		
143	x̄		199	R/S		
144	X≷Y		200	X≷Y		
145	P≷S		201	GSB9		
146	STO0		202	GSB8		
147	RCL8		203	RTN		
148	RCL4		204	*LBL0		
149	RCL0		205	X≷Y		
150	×		206	PRTX		
151	-	s_{xy}, s_{xy}'	207	X≷Y		Print x_i, y_i.
152	RCL9		208	PRTX		
153	1		209	RTN		
154	-		210	*LBL9		
155	÷		211	F0?		Subroutine for print.
156	P≷S		212	PRTX		
157	STOE		213	RTN		
158	GSB9		214	*LBL8		
159	R/S		215	F0?		
160	P≷S		216	SPC		Subroutine for space.
161	RCL9		217	RTN		
162	P≷S					
163	ENT↑					
164	X≷Y					
165	1					
166	-					
167	÷					
168	÷					

LABELS

A Start	B x_i↑y_i (Σ+)	C x_k↑y_k (Σ−)	D x_i, y_i, f_i	E x_k, y_k, f_k
a Print	b Used	c Used	d Used	e Σx_i. . . .
0 Print x_i, y_i	1 Used	2 Used	3 Used	4 Used
5	6	7	8 Space	9 Print

FLAGS

0 Print	
1 Σ−(Correct.)	
2 S_y'	
3	

SET STATUS

FLAGS	ON	OFF
0	☐	☒
1	☐	☒
2	☐	☒
3	☐	☒

TRIG	
DEG	☒
GRAD	☐
RAD	☐

DISP	
FIX	☒
SCI	☐
ENG	☐
n	2

Factorial, Permutation and Combination

001	*LBLA		057	LSTX			
002	CLRG	Initialize	058	X≤Y?			
003	CF0		059	GSB6			
004	0		060	STO7			
005	RTN		061	1			
006	*LBLB	Set flag 0 for print.	062	STOI			
007	SF0		063	+			
008	1		064	STO6			
009	RTN		065	CLX			
010	*LBL0		066	X=Y?			
011	X≠Y		067	GTO3			
012	GSBa	Input m, n for $_mP_n$.	068	*LBL0			
013	X≠Y		069	R↓			
014	GSBa		070	1			
015	X>Y?		071	RCL1			
016	GTO2		072	+			
017	ENT↑		073	STOI			
018	0		074	X>Y?			
019	X=Y?		075	GTO5			
020	GTO3		076	RCL7			
021	CLX		077	X≠Y			
022	1		078	+			
023	X=Y?		079	LSTX			
024	GTO4		080	÷			
025	-		081	RCL6			
026	STOI		082	x			
027	R↓		083	STO6			
028	STO7		084	GTO0			
029	*LBL1		085	*LBL4			
030	RCL7		086	R↓			
031	1		087	R↓			
032	-		088	GSBa		$_mP_1 = m$	
033	STO7		089	GSBb			
034	x		090	RTN			
035	DSZI		091	*LBL6			
036	GTO1		092	STO6			
037	GSBa		093	X≠Y			
038	GSBb		094	RTN			
039	RTN		095	*LBL5			
040	*LBL2		096	RCL6			
041	0		097	GSBa			
042	÷		098	GSBb			
043	*LBL3	Error	099	RTN			
044	ENT↑		100	*LBLC			
045	1		101	GSBa			
046	GSBa		102	STO1			
047	GSBb		103	STO3			
048	RTN		104	6		Input n for n!	
049	*LBLE		105	9			
050	X≠Y		106	X≠Y			
051	GSBa		107	X≤Y?			
052	X≠Y	Input m, n for $_mC_n$	108	GTO9			
053	GSBa		109	X≠Y			
054	X>Y?		110	-			
055	GTO2		111	STOI			
056	-		112	LSTX			

REGISTERS

0	1	2	3	4	5	6	7	8	9
	m, n	Log(69!)+....	(n – i)			Used	m	n – 1	
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9

A	B	C	D	E	I
					m – 69

```
113    N!
114    LOG
115    STO2
116    RCL1
117    ENT↑
118    LOG
119    ST+2
120  *LBL7
121    DSZI
122    GTO8
123    GTO7
124  *LBL7
125    RCL2
126    INT
127    GSBa
128    RCL2
129    FRC
130    10ˣ
131    DSP9
132    GSBa
133    DSP2
134    RCL1
135    GSBb
136    R/S
137    R↓
138    R↓
139    R/S
140    R↑
141    RTN
142  *LBL8
143    RCL3
144      1
145      -
146    STO3
147    LOG
148    ST+2
149    GTO7
150  *LBL9
151    N!
152    GSBa
153    GSBb
154    RTN
155  *LBLa
156    FØ?
157    PRTX
158    RTN
159  *LBLb
160    FØ?
161    SPC
162    RTN
```

Take log (n − 1).

For n ⩽ 69.

Subroutine for print.

Subroutine for space.

LABELS					FLAGS	SET STATUS		
[A] Start	[B] Print?	[C] n→n!	[D] m↑n→$_mP_n$	[E] m↑n→$_mC_n$	[0] Print	FLAGS	TRIG	DISP
[a] Print	[b] Space	[c]	[d]	[e]	[1]	ON OFF		
[0] $_mC_n$	[1] m!	[2] Error	[3] $_mP_0$, $_mC_0$	[4] $_mP_1$, $_mC_1$	[2]	0 ☐ ☒ 1 ☐ ☒ 2 ☐ ☒	DEG ☒ GRAD ☐ RAD ☐	FIX ☒ SCI ☐ ENG ☐
[5] Output	[6] x⇄y	[7] n > 69	[8] n > 69	[9] n ⩽ 69	[3]	3 ☐ ☒		n 2

L03-01

Moments, Skewness and Kurtosis (For Grouped or Ungrouped Data)

001	*LBLA		057	F1?	
002	CLRG		058	CHS	
003	P≠S	Initialize	059	ST+1	
004	CLRG		060	X≠Y	
005	P≠S		061	×	
006	CF1		062	ST+2	
007	CF0		063	LSTX	
008	0		064	×	
009	RTN		065	ST+3	
010	*LBLa	Set flag 0 for print.	066	LSTX	
011	SF0		067	×	
012	1		068	ST+4	
013	RTN		069	LSTX	
014	*LBLB		070	×	
015	STOA		071	ST+5	
016	GSB9	Input x_i for ungrouped data.	072	RCL0	
017	Σ+		073	1	
018	*LBL1		074	F1?	
019	P≠S		075	CHS	
020	RCL4		076	+	
021	RCL5		077	STO0	
022	RCL9		078	GSB9	
023	P≠S		079	GSB8	
024	STO1		080	RTN	
025	R↓		081	*LBLE	
026	STO3		082	SF1	Correction for y_h, f_h.
027	R↓		083	GSBD	
028	STO2		084	CF1	
029	RCLA		085	RTN	
030	3		086	*LBLb	
031	Y^x		087	RCL2	
032	F1?		088	RCL1	
033	CHS		089	÷	
034	ST+4		090	STO6	$\bar{x}$
035	RCLA		091	GSB9	
036	4		092	GSB8	
037	Y^x		093	RTN	
038	F1?		094	*LBLc	
039	CHS		095	RCL3	
040	ST+5		096	RCL1	
041	RCL1		097	÷	
042	GSB9		098	RCL6	
043	GSB8		099	X²	
044	RTN		100	STO8	
045	*LBLC		101	−	
046	GSB9		102	STO7	
047	SF1	Correction for x_k.	103	GSB9	m_2
048	Σ−		104	R/S	
049	GSB1		105	RCL4	
050	CF1		106	RCL3	
051	RTN		107	RCL6	
052	*LBLD		108	×	m_3
053	X≠Y		109	3	
054	GSB9	Input y_i, f_i for grouped data.	110	^x	
055	X≠Y		111	−	
056	GSB9		112	RCL1	

REGISTERS

0	1 n or Σf_i	2 Σx_i or $\Sigma f_i y_i$	3 Σx_i^2 or $\Sigma f_i y_i^2$	4 Σx_i^3 or $\Sigma f_i y_i^3$	5 Σx_i^4 or $\Sigma f_i y_i^4$	6 $\bar{x}$, m_4	7 m_2	8 $\bar{x}^2$	9 m_3
S0	S1	S2	S3	S4 Σx_i	S5 Σx_i^2	S6 Σy_i	S7 Σy_i^2	S8 Σ_{xy}	S9 n
A x_i	B	C n	D	E	I				

113	÷		169	F0?	
114	RCL6		170	SPC	Subroutine for space.
115	RCL8		171	RTN	
116	×				
117	2				
118	×				
119	+				
120	STO9				
121	GSB9				
122	R/S				
123	RCL5				
124	RCL6				
125	RCL4				
126	×	m_4			
127	4				
128	×				
129	−				
130	RCL8				
131	RCL3				
132	×				
133	6				
134	×				
135	+				
136	RCL1				
137	÷				
138	RCL8				
139	X^2				
140	3				
141	×				
142	−				
143	STO6				
144	GSB9				
145	GSB8				
146	RTN				
147	*LBLd				
148	RCL9				
149	RCL7	γ_1			
150	1				
151	.				
152	5				
153	Y^x				
154	÷				
155	GSB9				
156	R/S				
157	RCL6				
158	RCL7				
159	X^2				
160	÷	γ_2			
161	GSB9				
162	GSB8				
163	RTN				
164	*LBL9	Subroutine for print.			
165	F0?				
166	PRTX				
167	RTN				
168	*LBL0				

LABELS					FLAGS	SET STATUS		
A Start	B x_i (Σ+)	C x_k (Σ−)	D $y_i \uparrow f_i$ (Σ+)	E $y_k \uparrow f_k$ (Σ−)	0 Print	FLAGS	TRIG	DISP
a Print?	b $\rightarrow \bar{x}$	c $\rightarrow m_2, m_3, m_4$	d $\rightarrow \gamma_1 ; \gamma_2$	e	1 Correction	ON OFF 0 ☐ ☒ 1 ☐ ☒	DEG ☒	FIX ☒
0	1 Used	2	3	4	2	2 ☐ ☒	GRAD ☐	SCI ☐
5	6	7	8 Space	9 Print	3	3 ☐ ☒	RAD ☐	ENG ☐
								n 2

Random Number Generator

001	*LBL$_a$	Input a, b, clear secondary registers.	057	*LBL5	
002	P≠S		058	*LBLC	
003	CLRG		059	GSB7	
004	P≠S		060	STO7	
005	GTOc		061	GSB7	
006	*LBLA		062	2	Normal number generator.
007	GSB7		063	×	
008	RCLC		064	1	
009	RCLD		065	−	
010	−		066	STO2	
011	×	u_i	067	RCL7	
012	RCLD		068	2	
013	+		069	×	
014	*LBL2		070	1	
015	PRTX		071	−	
016	STO9		072	STO1	
017	Σ+		073	→P	
018	RCLA		074	X^2	
019	+		075	1	
020	STOA		076	X≤Y?	
021	RCL9		077	GTO5	
022	X^2		078	R↓	
023	RCLB		079	ENT↑	
024	+		080	LN	
025	STOB		081	2	
026	1		082	×	
027	RCLI		083	CHS	
028	+		084	X≠Y	
029	STOI		085	÷	
030	RCL9		086	√X	
031	RTN		087	STO8	
032	*LBLb		088	RCL1	
033	STOD		089	GSB6	
034	GSB0	Input k.	090	RTN	
035	RCLD		091	*LBLC	
036	*LBL3		092	RCL8	
037	PRTX		093	RCL2	
038	SPC		094	*LBL6	
039	RTN		095	×	
040	*LBLB		096	RCLC	
041	GSB7		097	×	
042	RCLD	Random digit generator.	098	RCLD	
043	×		099	+	
044	INT		100	GTO2	
045	1		101	*LBLd	
046	+		102	GTOb	
047	GTO2		103	*LBLD	
048	*LBLc		104	GSB7	
049	STOC		105	LN	
050	X≠Y	Input m, σ.	106	CHS	
051	STOD		107	RCLD	Input μ, exponential random number.
052	GSB0		108	×	
053	RCLD		109	GTO2	
054	GSB4		110	*LBLE	
055	RCLC		111	SPC	$\bar{x}$
056	GTO3		112	$\bar{x}$	

REGISTERS

0	1 V_1	2 V_2	3	4	5	6	7 u_i	8 $\sqrt{-2 \ln S/S}$	9
S0	S1	S2	S3	S4 Σx_i	S5 Σx_i^2	S6	S7	S8	S9 n

A	B	C	D	E	I
Used	Used	b or σ	a or k or m or μ	f_{i+1}	index n

113	GSB4			
114	R/S			
115	S	s		
116	GSB4			
117	R/S			
118	PSE			
119	RCL9			
120	PSE			
121	GSB4	n		
122	SPC			
123	RTN			
124	*LBL4	Subroutine to print.		
125	PRTX			
126	RTN			
127	*LBL0			
128	.			
129	5			
130	2			
131	8			
132	4	u_0		
133	1			
134	6			
135	3			
136	STOE			
137	0			
138	STOA			
139	STOB			
140	STOI			
141	SPC			
142	RTN			
143	*LBL7			
144	RCLE			
145	9			
146	9			
147	7			
148	x	Random number generator.		
149	FRC			
150	STOE			
151	RTN			

LABELS					FLAGS
A $\rightarrow u_i$	B $\rightarrow d_i$	C $\rightarrow n_i$	D $\rightarrow e_i$	E $\rightarrow \bar{x}$; s; n	0
a a↑b→	b k→	c m↑σ→	d $\mu\rightarrow$	e	1
0 u_0 to R_E	1 Used	2	3 Print, space	4 Print	2
5 Used	6 n_i, n_{i+1}	7 997 x u_i	8	9	3

SET STATUS			
FLAGS		TRIG	DISP
	ON OFF		
0	☐ ☒	DEG ☒	FIX ☒
1	☐ ☒	GRAD ☐	SCI ☐
2	☐ ☒	RAD ☐	ENG ☐
3	☐ ☒		n 2

Histogram

001	*LBLA	Initialize	057	RCLD	
002	CLRG		058	X≷Y	
003	P≷S		059	X=Y?	
004	CLRG		060	GSB2	
005	P≷S		061	RCLC	
006	CF0		062	−	
007	CF1		063	RCLA	
008	0		064	1/X	
009	RTN		065	×	
010	*LBLC		066	INT	
011	R↓		067	1	
012	GSB8	Input x_{min}, x_{max}.	068	X≷Y	
013	STOC		069	+	
014	R↑		070	LSTX	
015	STOD		071	3	
016	GSB8		072	÷	
017	GSB7		073	INT	
018	GSB7		074	1	
019	X≷Y		075	+	
020	−		076	STOI	
021	2		077	1	
022	4		078	−	
023	STOE		079	3	
024	÷		080	×	
025	STOA		081	−	
026	RTN		082	GTO1	
027	*LBLE		083	*LBL9	
028	0		084	RCL9	
029	STOI		085	DSP0	
030	X≷Y		086	GSB8	
031	GSB8	Correction	087	GSB7	
032	Σ−		088	DSP2	
033	SF1		089	CF1	
034	GSBc		090	R/S	
035	CF1		091	RTN	
036	RTN		092	*LBL1	
037	*LBLD		093	GSBd	
038	STO0		094	F1?	
039	0		095	CHS	
040	STOI	Input x_i.	096	ST+i	
041	R↓		097	GTO9	
042	GSB8		098	*LBLd	
043	RCLC		099	3	
044	X>Y?		100	CHS	
045	GTO0		101	×	
046	R↓		102	10^	
047	RCLD		103	RTN	
048	X≷Y		104	*LBLe	
049	X>Y?		105	SPC	
050	GTO0		106	0	Listing
051	0		107	STOI	
052	X≷Y		108	RCLC	
053	Σ+		109	STOB	
054	*LBLc		110	*LBL5	
055	STO9		111	ISZI	
056	RCL0		112	2	

REGISTERS

0 x_i	1 1, 2, 3	2 4, 5, 6	3 7, 8, 9	4 10, 11, 12	5 13, 14, 15	6 16, 17, 18	7 19, 20, 21	8 22, 23, 24	9 n
S0	S1	S2	S3	S4 Σx_i	S5 Σx_i^2	S6 Σy_i	S7 Σy_i^2	S8 $\Sigma x_i y_i$	S9 n

A $(x_{max} - x_{min})/24$	B x_{min}, last range	C x_{min}	D x_{max}	E 24	I Counter 1–8

```
113  STO9
114  GSBe
115  RCLi
116   EEX
117     3
118     x
119   INT
120  DSP0
121  PRTX
122   SPC
123  DSP2
124     1
125  STO9
126  GSBe
127  RCLi
128   EEX
129     3
130     x
131   FRC
132   EEX
133     3
134     x
135   INT
136  DSP0
137  PRTX
138   SPC
139  DSP2
140     0
141  STO9
142  GSBe
143  RCLi
144   EEX
145     6
146     x
147   FRC
148   EEX
149     3
150     x
151   INT
152  DSP0
153  PRTX
154   SPC
155  DSP2
156  RCLI
157     8
158  X>Y?
159  GTO5
160   RTN
161 *LBLe
162  RCLB
163  PRTX
164  RCLC
165  RCLA
166  RCLI
167     3
168     x
```

Listing

```
169  RCL9
170     -
171     x
172     +
173  PRTX
174  STOB
175   RTN
176 *LBLB
177   SF0
178     1
179   RTN
180 *LBL6
181  GSB7
182  GSB7
183   F?S
184  RCL9
185   F?S
186  GSB8
187   R/S
188     x̄
189  GSB8
190   R/S
191     S
192  GSB8
193  GSB7
194   RTN
195 *LBL8
196   F0?
197  PRTX
198   RTN
199 *LBL7
200   F0?
201   SPC
202   RTN
203 *LBL2
204  RCLA
205     2
206     ÷
207     -
208   RTN
```

Set flag 0 for print.

For n, $\bar{x}$, s

Subroutine for print.

Subroutine for space.

Correction for input equals to x_{max}.

LABELS					FLAGS	SET STATUS		
A Start	B Print	C x_{max}, x_{min}	D Input	E Correct	0 Print	FLAGS ON OFF	TRIG	DISP
a List	b n; $\bar{x}$; s	c y	d 10^x	e Used	1 Correction	0 ☐ ☒	DEG ☒	FIX ☒
0 Error	1 Sorting	2 Cor. for x_{max}	3	4	2	1 ☐ ☒	GRAD ☐	SCI ☐
5 Listing	6	7 Space	8 Print	9 Print index	3	2 ☐ ☒	RAD ☐	ENG ☐
						3 ☐ ☒		n 2

Analysis of Variance (One Way)

001	*LBLA		057	RTN	
002	CLRG		058	*LBL9	
003	PSE		059	PSE	Clear registers for new i.
004	CLRG	Initialize	060	CLRG	
005	PSE		061	PSE	
006	CF0		062	RTN	
007	CF1		063	*LBLD	
008	CF2		064	GSB3	Correction
009	0		065	Σ-	
010	RTN		066	GSB8	
011	*LBLC	Input x_{ij}.	067	RTN	
012	F2?		068	*LBLB	Set flag 0 for print.
013	GSB9		069	SF0	
014	GSB3		070	1	
015	Σ+		071	RTN	
016	*LBL8		072	*LBLa	
017	PSE		073	RCL4	
018	RCL4		074	RCL7	
019	RCL5		075	X²	
020	PSE		076	RCL6	
021	STOB		077	÷	TSS
022	R↓		078	STO8	
023	STOA		079	-	
024	R↓		080	STO0	
025	GSB3		081	GSB3	TrSS
026	GSB0		082	R/S	
027	RTN		083	RCL5	
028	*LBLE		084	RCL8	
029	1		085	-	
030	ST+9		086	STO1	
031	SF2		087	GSB3	
032	RCLA		088	R/S	
033	ST+7		089	RCL0	ESS
034	RCLB		090	RCL1	
035	ST+4		091	-	
036	PSE		092	STO2	
037	RCL9	$\bar{x}_i$	093	GSB3	
038	PSE		094	GSB0	
039	ST+6		095	RTN	
040	RCLA		096	*LBL0	Subroutine for space.
041	X²		097	F0?	
042	PSE		098	SPC	
043	RCL9		099	RTN	
044	PSE		100	*LBL3	Subroutine to print.
045	÷		101	F0?	
046	ST+5		102	PRTX	
047	x̄		103	RTN	
048	GSB3		104	*LBL6	
049	R/S		105	RCL9	
050	S	s_i	106	1	df_1
051	GSB3		107	-	
052	R/S		108	STO3	
053	RCLA		109	GSB3	
054	GSB3		110	R/S	
055	GSB0	Sum$_i$	111	RCL6	df_2
056	GSB0		112	RCL9	

REGISTERS

0 TSS	1 T_rSS	2 ESS	3 df_1	4 $\Sigma\Sigma x_{ij}^2$	5 $\Sigma(\Sigma x_{ij})^2/n_i$	6 Σn_i	7 $\Sigma\Sigma x_{ij}$	8 Σx_{ij}	9 k
S0	S1	S2	S3	S4 Σx_i	S5 Σx_i^2	S6 Σy_i	S7 Σy_i^2	S8 $\Sigma x_i y_i$	S9 n
A Σx_{ij}, df_2	B Σx_{ij}^2, F	C	D	E	I				

113 -			
114 STOA			
115 GSB3			
116 R/S			
117 RCLA			
118 RCL3	df_3		
119 +			
120 GSB3			
121 GSB0			
122 RTN			
123 *LBL c			
124 RCL1	T_rMs		
125 RCL3			
126 ÷			
127 GSB3			
128 R/S			
129 RCL2	EMS		
130 RCLA			
131 ÷			
132 GSB3			
133 R/S			
134 ÷			
135 GSB3			
136 GSB6	F		
137 STOB			
138 RTN			

LABELS					FLAGS	SET STATUS		
						FLAGS	TRIG	DISP
A Start	B Print?	C x_{ij} $(\Sigma+)\rightarrow$	D x_{im} $(\Sigma-)\rightarrow$	E $\bar{x}$, s_i, sum_i	0 Print	ON OFF 0 ☐ ☒ 1 ☐ ☒ 2 ☐ ☒ 3 ☐ ☒	DEG ☒ GRAD ☐ RAD ☐	FIX ☒ SCI ☐ ENG ☐ n 2
a TSS, ...	b df_1, ...	c T_rMs, ...	d	e	1 Correction			
0 Space	1	2	3 Print	4	2 New data			
5	6	7	8 Σx_i, Σx_i^2	9 CLREG	3			

Two Way Analysis of Variance

001	*LBLA			057	RCL6		
002	CLRG	Initialize		058	×		
003	CF0			059	÷		
004	0			060	STO7		
005	RTN			061	CHS		
006	*LBLC			062	RCL2		
007	X≷Y	Input r, c.		063	+		
008	STO5			064	STO1		
009	GSB9			065	RCL3		
010	R↓			066	RCL6		
011	STO6			067	÷		
012	GSB9			068	RCL7		
013	GSB8			069	−		
014	RTN			070	STO2		
015	*LBLD			071	RCL4		
016	ST+7	Input x_{ij}.		072	RCL5		
017	GSB9			073	÷		
018	X²			074	RCL7		
019	ST+2			075	−	F_1, F_2	
020	RCLA			076	STO3		
021	1			077	RCL2		
022	+			078	+		
023	STOA			079	CHS		
024	GSB9			080	RCL1		
025	RTN			081	+		
026	*LBLa			082	STO4		
027	RCL7			083	RCL5		
028	ST+1			084	1		
029	X²	Calculate RS_i.		085	−		
030	ST+3			086	STO5		
031	*LBL0			087	RCL6		
032	RCL7			088	1		
033	0			089	−		
034	STOA			090	STO6		
035	STO7			091	×		
036	X≷Y			092	STO7		
037	GSB9			093	÷		
038	GSB8			094	STO8		
039	RTN			095	RCL2		
040	*LBLb			096	RCL5		
041	RCLA	Re-initialize for column.		097	÷		
042	STOB			098	RCL8		
043	0			099	÷		
044	STO2			100	GSB8		
045	STOA			101	GSB9		
046	GSB8			102	R/S		
047	RTN			103	RCL3		
048	*LBLc			104	RCL6		
049	RCL7			105	÷		
050	X²	CS_i		106	RCL8		
051	ST+4			107	÷		
052	GTO0			108	GSB9		
053	*LBLd			109	GSB8		
054	RCL1	F_1, F_2		110	RTN		
055	X²			111	*LBLd		
056	RCL5			112	RCL5		

REGISTERS

0	1 $\Sigma\Sigma x_{ij}$, TSS	2 $\Sigma\Sigma x_{ij}^2$, RSS	3 Used	4 Used	5 r, r − 1	6 c, c − 1	7 Used	8 Used	9 0
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9

A i, j	B r	C	D	E	I

113 GSB9			
114 R/S	df_1, df_2, df_3		
115 RCL6			
116 GSB9			
117 R/S			
118 RCL7			
119 GSB9			
120 GSB8			
121 RTN			
122 *LBLe	RSS, CSS, ESS, TSS		
123 RCL2			
124 GSB9			
125 R/S			
126 RCL3			
127 GSB9			
128 R/S			
129 RCL4			
130 GSB9			
131 R/S			
132 RCL1			
133 GSB9			
134 GSB8			
135 RTN			
136 *LBLE	For correction of x_{im} or x_{hj}.		
137 GSB9			
138 CHS			
139 ST+7			
140 X²			
141 ST-2			
142 RCLA			
143 1			
144 -			
145 STOA			
146 GSB9			
147 RTN			
148 *LBLB	Set flag 0 for print.		
149 SF0			
150 1			
151 RTN			
152 *LBL9	Subroutine for print.		
153 F0?			
154 PRTX			
155 RTN			
156 *LBL8	Subroutine for space.		
157 F0?			
158 SPC			
159 RTN			

LABELS					FLAGS	SET STATUS		
A Start	B Print?	C r↑c→	D x_{ij} (Σ+)→	E x_{im} (Σ−)→	0 Print	FLAGS	TRIG	DISP
a →RS_i	b Re-int.	c →CS_i	d→$F_1, F_2 \ldots$	e→RSS; . . .	1	ON OFF 0 ☐ ☒ 1 ☐ ☒ 2 ☐ ☒ 3 ☐ ☒	DEG ☒ GRAD ☐ RAD ☐	FIX ☒ SCI ☐ ENG ☐ n 2
0 0 to R_A, R_7	1	2	3	4	2			
5	6	7	8 Space	9 Print	3			

Analysis of Covariance (One Way) (Card 1)

001	*LBLA		057	RCLA	
002	CLRG		058	+	
003	PSE	Initialize	059	STOA	
004	CLRG		060	RCL5	
005	PSE		061	RCLB	
006	CF0		062	+	
007	0		063	STOB	
008	STO8		064	RCL6	
009	RTN		065	RCLC	
010	*LBLB		066	+	
011	PSE		067	STOC	
012	0	For new i.	068	RCL7	
013	STO4		069	RCLD	
014	STO5		070	+	
015	STO6		071	STOD	
016	STO7		072	RCL8	
017	STO8		073	RCLE	
018	STO9		074	+	
019	PSE		075	STOE	
020	ISZI		076	PSE	
021	RCLI		077	RCL9	Σn_i
022	GSB0		078	ST+0	
023	GSB1		079	PSE	
024	RTN		080	ST+0	
025	*LBL9	Subroutine to print x_{ij},	081	RCL4	$\Sigma \frac{(\Sigma x_{ij})^2}{n_i}$
026	R↓	y_{ij}.	082	PSE	
027	PRTX		083	GSB8	
028	R↑		084	PSE	
029	PRTX		085	RCL6	$\Sigma \frac{(\Sigma y_{ij})^2}{n_i}$
030	RTN		086	GSB8	
031	*LBLe	Set flag 0 for print.	087	RCL4	
032	SF0		088	RCL6	
033	1		089	x	
034	STO8		090	RCL9	
035	RTN		091	÷	
036	*LBLD	Correction for x_{im}, y_{im}.	092	PSE	
037	F0?		093	ST+5	
038	GSB9		094	PSE	
039	XEY		095	RCL6	
040	Σ-		096	RCL4	
041	STO9		097	PSE	
042	GSB0		098	GSB0	s_{xi}
043	GSB1		099	R/S	
044	RTN		100	R↓	
045	*LBLC	Input for x_{ij}, y_{ij}.	101	GSB0	
046	F0?		102	GSB1	s_{yi}
047	GSB9		103	R/S	
048	XEY		104	*LBL8	
049	Σ+		105	X²	
050	STO9		106	RCL9	
051	GSB0		107	÷	
052	GSB1		108	ST+2	
053	RTN		109	RTN	
054	*LBLE		110	*LBLa	TSS_x
055	PSE		111	RCLB	
056	RCL4		112	RCLA	

REGISTERS

0 Σn_i	1 TSS_x	2 Used	3 WSS_x	4 TSP_i	5 Used	6 WSP	7 $WSS_{\hat{y}}$	8 1 or 0	9 j
S0 Σn_i	S1 TSS_y	S2 Used	S3 WSS_y	S4 Σx_{ij}	S5 Σx_{ij}^2	S6 Σy_{ij}	S7 Σy_{ij}^2	S8 $\Sigma x_{ij}y_{ij}$	S9 $n_{i \neq j}$
A $\Sigma\Sigma x_{ij}$	B $\Sigma\Sigma x_{ij}^2$	C $\Sigma\Sigma y_{ij}$	D $\Sigma\Sigma y_{ij}^2$	E $\Sigma\Sigma x_{ij}y_{ij}$	I i = 1, 2, ... k				

113	GSB7		169	-			
114	R/S	ASS_x	170	÷			
115	RCL2		171	÷			
116	RCLA		172	GSB0			
117	GSB6		173	R/S			
118	R/S	WSS_x	174	P≠S			
119	-		175	RCL2			
120	STO3		176	RCLI			
121	GSB0		177	1			
122	GSB1		178	-			
123	R/S		179	÷			
124	*LBL6	TSS_y	180	RCL3			
125	RCLD		181	P≠S			
126	RCLC		182	RCL0			
127	P≠S		183	RCLI			
128	GSB7		184	-			
129	P≠S		185	÷			
130	R/S	ASS_y	186	÷	F_y		
131	P≠S		187	GSB0			
132	RCL2		188	R/S			
133	RCLC		189	RCLI			
134	GSB6		190	1			
135	P≠S		191	-	df_1		
136	R/S		192	GSB0			
137	-	WSS_y	193	R/S			
138	P≠S		194	RCL0			
139	STO3		195	RCLI			
140	P≠S		196	-	df_2		
141	GSB0		197	GSB0			
142	GSB1		198	GSB1			
143	R/S		199	RTN			
144	*LBL7		200	*LBL0			
145	X^2	For $(TSS)_{x \text{ or } y}$	201	F0?	Subroutine for print.		
146	RCL0		202	PRTX			
147	÷		203	RTN			
148	-		204	*LBL1			
149	STO1		205	F0?	Subroutine for space.		
150	GSB0		206	SPC			
151	RTN		207	RTN			
152	*LBL6						
153	X^2	For $(ASS)_{x \text{ or } y}$					
154	RCL0						
155	÷						
156	-						
157	STO2						
158	GSB0						
159	RTN						
160	*LBL6						
161	RCL2						
162	RCLI						
163	1						
164	-						
165	÷						
166	RCL3						
167	RCL0						
168	RCLI	F_x					

LABELS					FLAGS
A Start	B New i	C x_{ij}↑y_{ij} (Σ+)	D x_{im}↑y_{im} (Σ−)	E s_{xi}; s_{yi}	0 Print
a TSS_x; ...	b F_x; ...	c	d	e Print	1
0 Print	1 Space	2	3	4	2
5	6 ASS_x	7 TSS_x	8 $\Sigma\Sigma x_{ij}/n_i$	9 Print x_{ij}, y_{ij}	3

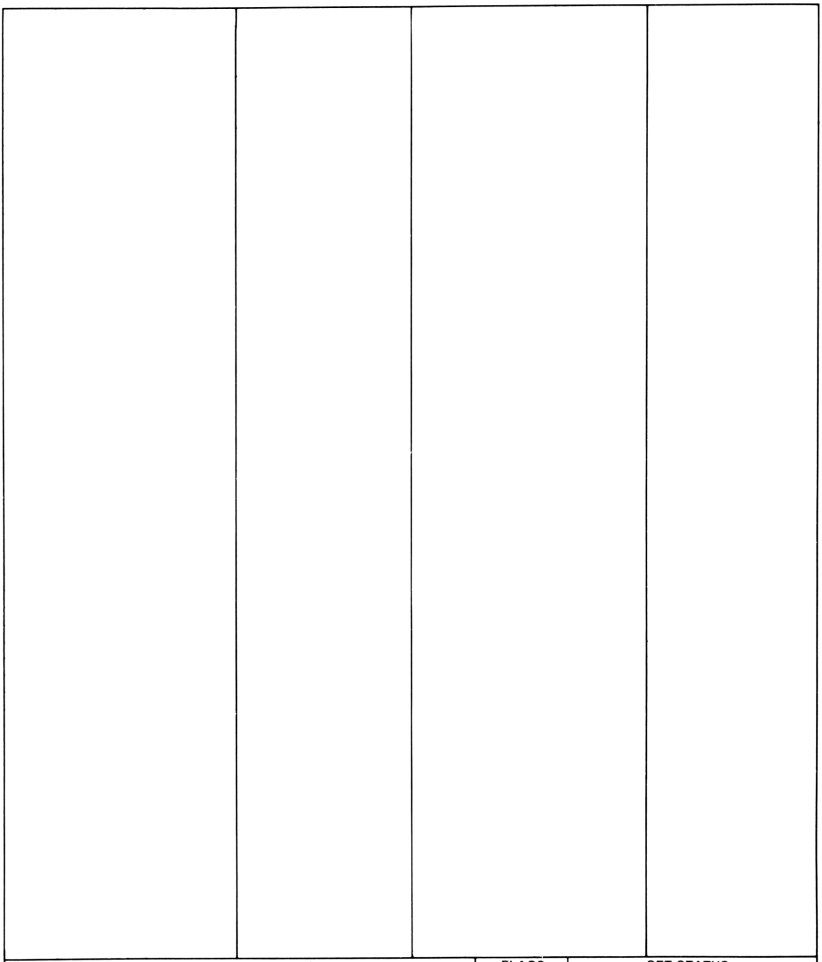
SET STATUS					
FLAGS		TRIG		DISP	
	ON OFF				
0	☐ ☒	DEG	☒	FIX	☒
1	☐ ☒	GRAD	☐	SCI	☐
2	☐ ☒	RAD	☐	ENG	☐
3	☐ ☒			n	2

(Card 2)

001	*LBLc		057	÷	
002	GSB2		058	GSB0	AMS$\hat{y}$
003	RCLE		059	R/S	
004	RCLA		060	RCL7	
005	RCLC		061	RCL0	
006	×		062	RCLI	WMS$\hat{y}$
007	RCL0		063	−	
008	÷		064	1	
009	−	TSP	065	−	
010	STO4		066	÷	
011	GSB0		067	GSB0	
012	R/S		068	R/S	F
013	RCL5		069	÷	
014	RCLA		070	GSB0	
015	RCLC		071	R/S	
016	×		072	RCLI	
017	RCL0		073	1	df_3
018	÷	ASP	074	−	
019	−		075	GSB0	
020	GSB0		076	R/S	
021	R/S		077	RCL0	
022	−	WSP	078	RCLI	
023	STO6		079	−	df_4
024	GSB0		080	1	
025	GSB1		081	−	
026	RTN		082	GSB0	
027	*LBLd		083	GSB1	
028	FIX		084	RTN	
029	RCL1		085	*LBL0	Subroutine for print.
030	FIX		086	F0?	
031	RCL4		087	PRTX	
032	X^2		088	RTN	
033	RCL1		089	*LBL1	
034	÷	TSS$\hat{y}$	090	F0?	Subroutine for space.
035	−		091	SPC	
036	GSB0		092	RTN	
037	R/S		093	*LBL2	
038	FIX		094	CF0	Subroutine for setting flag 0 for print.
039	RCL3		095	RCL8	
040	FIX		096	1	
041	RCL6		097	X=Y?	
042	X^2		098	SF0	
043	RCL3		099	RTN	
044	÷				
045	−	WSS$\hat{y}$			
046	STO7				
047	GSB0				
048	R/S				
049	−	ASS$\hat{y}$			
050	GSB0				
051	GSB1				
052	R/S				
053	*LBLe				
054	RCLI				
055	1				
056	−				

REGISTERS

0 Σn_i	1 TSS_x	2 Used	3 WSS_x	4 TSP_i	5 Used	6 WSP	7 $WSS_{\hat{y}}$	8 1 or 0	9 j
S0 Σn_i	S1 TSS_y	S2 Used	S3 WSS_y	S4 Σx_{ij}	S5 Σx_{ij}^2	S6 Σy_{ij}	S7 Σy_{ij}^2	S8 $\Sigma x_{ij}y_{ij}$	S9 $n_{i \neq j}$
A $\Sigma\Sigma x_{ij}$	B $\Sigma\Sigma x_{ij}^2$	C $\Sigma\Sigma y_{ij}$	D $\Sigma\Sigma y_{ij}^2$	E $\Sigma\Sigma x_{ij}y_{ij}$	I i = 1, 2, ... k				



LABELS					FLAGS
A	B	C	D	E	0 Print
a	b	c TSP; ...	d TSS◊; ...	e AMS◊; ...	1
0 Print	1 Space	2 Used	3	4	2
5	6	7	8	9	3

SET STATUS

FLAGS			TRIG		DISP	
	ON	OFF				
0	☐	☒	DEG	☒	FIX	☒
1	☐	☒	GRAD	☐	SCI	☐
2	☐	☒	RAD	☐	ENG	☐
3	☐	☒			n	2

Normal and Inverse Normal Distribution (Card 1)

```
001  *LBLA
002     .
003     2
004     3
005     1
006     6
007     4
008     1
009     9
010   STO3
011     1
012     .
013     3
014     3
015     0
016     2
017     7
018     4
019     4
020     2
021     9
022   STO4
023     1
024     .
025     8
026     2
027     1
028     2
029     5
030     5
031     9
032     7
033     8
034   CHS
035   STO5
036     1
037     .
038     7
039     8
040     1
041     4
042     7
043     7
044     9
045     3
046     7
047   STO6
048     .
049     3
050     5
051     6
052     5
053     6
054     3
055     7
056     8
```

Store constants for normal distribution.

```
057     2
058   CHS
059   STO7
060     .
061     3
062     1
063     9
064     3
065     8
066     1
067     5
068     3
069   STO8
070   P≠S
071     2
072     .
073     5
074     1
075     5
076     5
077     1
078     7
079   STO1
080     .
081     8
082     0
083     2
084     8
085     5
086     3
087   STO2
088     .
089     0
090     1
091     0
092     3
093     2
094     8
095   STO3
096     1
097     .
098     4
099     3
100     2
101     7
102     8
103     8
104   STO4
105     .
106     1
107     8
108     9
109     2
110     6
111     9
112   STO5
```

Store constants for inverse N.D.

REGISTERS

0	1	2	3 r	4 b_5	5 b_4	6 b_3	7 b_2	8 b_1	9
S0	S1 C_0	S2 C_1	S3 C_2	S4 d_1	S5 d_2	S6 d_3	S7	S8	S9
A	B	C	D	E	I				

```
113      .
114      0
115      0
116      1
117      3
118      0
119      8
120   STO6
121    P≠S
122      0
123   STOA
124   STOB
125    RTN
```

LABELS					FLAGS
A Start	B	C	D	E	0
a	b	c	d	e	1
0	1	2	3	4	2
5	6	7	8	9	3

SET STATUS

FLAGS			TRIG	DISP
	ON	OFF		
0	☐	☒	DEG ☒	FIX ☐
1	☐	☒	GRAD ☐	SCI ☐
2	☐	☒	RAD ☐	ENG ☐
3	☐	☒		n 2

(Card 2)

001	*LBLB		057	x	
002	1		058	RCL2	
003	STOA	Store 1 in R_A for print.	059	x	
004	RTN		060	F0?	
005	*LBLC		061	GSB9	
006	GSB9		062	F0?	
007	STO1		063	GSB6	
008	ENT↑		064	RTN	
009	x	Input x and compute f(x).	065	*LBL1	
010	2		066	CF0	
011	÷		067	RCL1	
012	CHS		068	CHS	
013	e^x		069	STO1	
014	Pi		070	GSB6	
015	2		071	1	
016	x		072	X≷Y	
017	√X		073	-	
018	÷		074	STO9	
019	STO2		075	GSB9	
020	GSB9		076	GSB6	
021	GSB6		077	RCL9	
022	GSB3		078	RTN	
023	RCL2		079	*LBLE	
024	RTN		080	GSB9	
025	*LBLD		081	X<0?	Input Q(x) and calculate x.
026	GSB9		082	GTO0	
027	STO1		083	1	
028	GSB5		084	X≤Y?	
029	GSBC		085	GTO0	
030	RCL1		086	R↓	
031	X<0?		087	.	
032	GTO1	Input x and calculate Q(x).	088	5	
033	SF0		089	X≷Y	
034	*LBLC		090	X>Y?	
035	1		091	GSB8	
036	RCL1		092	ENT↑	
037	RCL3		093	x	
038	x		094	1/X	
039	+		095	LN	
040	1/X		096	√X	
041	ENT↑		097	P≷S	
042	ENT↑		098	STO7	
043	ENT↑		099	RCL3	
044	RCL4		100	x	
045	x		101	RCL2	
046	RCL5		102	+	
047	+		103	RCL7	
048	x		104	x	
049	RCL6		105	RCL1	
050	+		106	+	
051	x		107	RCL7	
052	RCL7		108	RCL6	
053	+		109	x	
054	x		110	RCL5	
055	RCL8		111	+	
056	+		112	RCL7	

REGISTERS

0	1 x	2 f(x)	3 r	4 b_5	5 b_4	6 b_3	7 b_2	8 b_1	9 Used
S0	S1 C_0	S2 C_1	S3 C_2	S4 d_1	S5 d_2	S6 d_3	S7 t	S8	S9

A 1 or 0	B 1 or 0	C	D	E	I

113 x 114 RCL4 115 + 116 RCL7 117 x 118 1 119 + 120 ÷ 121 RCL7 122 X≠Y 123 - 124 P≠S 125 F1? 126 CHS 127 GSB9 128 GSB6 129 CF1 130 RTN 131 *LBL8 132 SF1 133 1 134 - 135 CHS 136 RTN 137 *LBL9 138 RCLA 139 X>0? 140 GSB7 141 R↓ 142 RTN 143 *LBL7 144 R↓ 145 PRTX 146 R↑ 147 RTN 148 *LBL6 149 RCLA 150 X>0? 151 SPC 152 R↓ 153 RTN 154 *LBL5 155 RCLA 156 X>0? 157 GSB4 158 R↓ 159 RTN 160 *LBL4 161 STOB 162 CLX 163 STOA 164 RTN 165 *LBL3 166 RCLB 167 X>0? 168 GSB2	For (1 - Q) Subroutine to print. Subroutine for space. Clear R_A for calculating Q(x).	169 RTN 170 *LBL2 171 1 172 STOA 173 RTN	Restore 1 to R_A.

LABELS					FLAGS	SET STATUS		
A	B Print?	C x→f(x)	D x→Q(x)	E Q(x)→x	0 x positive	FLAGS	TRIG	DISP
a	b	c Q(x)	d	e	1 Q(x) > .5	ON OFF 0 ☐ ☒ 1 ☐ ☒ 2 ☐ ☒ 3 ☐ ☒	DEG ☒ GRAD ☐ RAD ☐	FIX ☒ SCI ☐ ENG ☐ n 2
0 Error	1 x < 0	2 1→R_A	3 R_B > 0?	4 0→R_A	2			
5 R_A→R_B	6 Space	7 Print	8	9 Print	3			

Chi-Square Distribution

001	*LBLA		057	Y^x	
002	CLRG		058	RCL2	
003	CF0	Initialize	059	2	
004	CF1		060	÷	
005	0		061	CHS	
006	RTN		062	e^x	
007	*LBLB	Set flag for print.	063	x	
008	SF0		064	2	
009	1		065	RCL1	
010	RTN		066	Y^x	
011	*LBLC		067	÷	
012	GSB9	Input ν.	068	RCL3	
013	1		069	÷	
014	STO3		070	STO5	
015	X≠Y	Calculate $\Gamma\left(\frac{\nu}{2}\right)$	071	F1?	
016	2		072	GSB9	
017	÷		073	F1?	
018	STO1		074	GSB8	
019	INT		075	CF1	
020	LSTX		076	RTN	
021	X≠Y?		077	*LBLE	
022	GTO1		078	GSBe	Input x, calculate P(x).
023	1		079	RCL2	
024	-		080	RCL1	
025	N!		081	÷	
026	GSB9		082	STx5	
027	GSB8		083	2	
028	STO3		084	RCL1	
029	R/S		085	x	
030	*LBL1		086	STO6	
031	.		087	1	
032	5		088	STO4	
033	X=Y?		089	*LBL3	
034	GTO2		090	RCL2	
035	X≠Y		091	RCL6	
036	1		092	2	
037	-		093	+	
038	STx3		094	STO6	
039	GTO1		095	÷	
040	*LBL2		096	RCL4	
041	Pi		097	x	
042	√X		098	STO4	
043	RCL3		099	+	
044	x		100	X≠Y?	
045	STO3		101	GTO3	
046	GSB9		102	RCL5	
047	GSB8		103	x	
048	R/S		104	GSB9	
049	*LBLD		105	GSB8	
050	SF1		106	RTN	
051	*LBLe		107	*LBL9	
052	GSB9	Input x, calculate f(x).	108	F0?	Subroutine for print.
053	STO2		109	PRTX	
054	RCL1		110	RTN	
055	1		111	*LBL8	
056	-		112	F0?	

REGISTERS

0	1 $\nu/2$	2 x	3 1, $\Gamma(\nu/2)$	4 Used	5 f(x)	6 Used	7	8	9
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9

A	B	C	D	E	I

113 SPC 114 RTN	Subroutine for space.		

LABELS					FLAGS
A Start	B Print?	C $\nu \rightarrow \Gamma(\nu/2)$	D $x \rightarrow f(x)$	E $x \rightarrow P(x)$	0 Print
a	b	c	d	e f(x)	1 P(x)
0	1 $\nu = ?$ ½	2 Γ(½)	3 P(x)	4	2
5	6	7	8 Space	9 Print	3

SET STATUS			
FLAGS		TRIG	DISP
	ON OFF		
0	☐ ☒	DEG ☒	FIX ☒
1	☐ ☒	GRAD ☐	SCI ☐
2	☐ ☒	RAD ☐	ENG ☐
3	☐ ☒		n 2

t Distribution

001	*LBLA		057	X≠Y	
002	CLRG		058	2	
003	CF0		059	÷	
004	CF1	Initialize	060	STO1	
005	0		061	INT	
006	RTN		062	LSTX	
007	*LBLB	Set flag 0 for print.	063	X≠Y?	
008	SF0		064	GTO1	
009	1		065	1	
010	RTN		066	-	Compute Γ(ν/2).
011	*LBLC		067	N!	
012	STO0	Input ν.	068	STO3	
013	GSB7		069	RTN	
014	GSB5		070	*LBL1	
015	RTN		071	.	
016	*LBLD	Input x, compute f(x).	072	5	
017	GSB7		073	X=Y?	
018	STOA		074	GTO2	
019	RCL0		075	X≠Y	
020	GSBa		076	1	
021	STOE		077	-	
022	RCL0		078	STx3	
023	1		079	GTO1	
024	+		080	*LBL2	Compute R(x).
025	GSBa		081	Pi	
026	STOC		082	√x	
027	RCLA		083	RCL3	
028	RCLC		084	x	
029	RCLB		085	STO3	
030	÷		086	RTN	
031	Pi		087	*LBLE	
032	RCL0		088	STOA	
033	x		089	GSB7	
034	√x		090	ABS	
035	÷		091	RCL0	
036	1		092	RAD	
037	RCLA		093	√x	
038	X²		094	÷	
039	RCL0		095	TAN⁻¹	
040	÷		096	STO2	
041	+		097	RCL0	
042	RCL0		098	2	
043	1		099	÷	
044	+		100	INT	
045	2		101	LSTX	
046	÷		102	X≠Y?	
047	CHS		103	GTO4	
048	Y^x		104	0	
049	x		105	STO5	
050	STOD		106	*LBL6	
051	GSB7		107	RCL2	For even ν.
052	GSB5		108	COS	
053	RTN		109	X²	
054	*LBLa		110	STO3	
055	1		111	RCL2	
056	STO2		112	SIN	

REGISTERS

0 ν	1	2 θ	3 Used	4 sin θ	5 Used	6 Used	7 2θ/π, R	8 Used	9
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9
A x	B Γ(ν/2)	C Γ(ν + 1/2)	D	E	I				

```
113   STO4
114   RCL0
115     2
116   X=Y?
117   GTO8
118     ÷
119   STO1
120     1
121   STO6
122   DSZI
123   GTO3
124   GTO6
125  *LBL3
126   RCL3
127     x
128   RCL5
129     1
130     +
131     x
132   LSTX
133     1
134     +
135   STO5
136     ÷
137   ST+6
138   DSZI
139   GTO3
140   RCL6
141  *LBL6
142   RCL4
143     x
144    F1?
145   GTOe
146   RCLA
147   GTO6
148  *LBLe
149    RTN
150  *LBL4
151   RCL2
152     2
153     x
154     Pi
155     ÷
156   STO7
157   RCL0
158     1
159   STO5
160   X=Y?
161   GTO9
162    SF1
163   GSBb
164    CF1
165   RCL2
166    COS
167     x
168     2
```

For odd ν.

```
169     x
170     Pi
171     ÷
172   RCL7
173     +
174   RCLA
175   GTO6
176    RTN
177  *LBL9
178   RCL7
179   RCLA
180   GTOE
181    RTN
182  *LBL8
183   RCL4
184   RCLA
185   GTOE
186    RTN
187  *LBLE
188   X>0?
189   GTO0
190   XZY
191     1
192     -
193   CHS
194     2
195     ÷
196   GSB7
197   GSB5
198    RTN
199  *LBL0
200   XZY
201     1
202     +
203     2
204     ÷
205   GSB7
206   GSB5
207    RTN
208  *LBL7
209    F0?
210   PRTX
211    RTN
212  *LBL5
213    F0?
214    SPC
215    RTN
```

Compute P(x) from R(x) for $x \leq 0$.

Compute P(x) for $x > 0$.

Subroutine to print.

Subroutine to space

LABELS					FLAGS
A Start	B Print	C $\nu \rightarrow$	D $x \rightarrow f(x)$	E $x \rightarrow P(x)$	0 Print
a $\nu/2$ integer	b even ν	c	d	e Used	1 odd ν
0 P(x), $x > 0$	1 $\nu/2$ no integer	2 $\sqrt{\pi}$ x	3 even ν	4 odd ν	2
5 Space	6 P(x), $x \leq 0$	7 Print	8 For even ν	9 For odd ν	3

SET STATUS

FLAGS			TRIG		DISP	
	ON	OFF				
0	☐	X	DEG	X	FIX	X
1	☐	X	GRAD	☐	SCI	☐
2	☐	X	RAD	☐	ENG	☐
3	☐	X			n	2

F Distribution

001	*LBLA		057	RCL4	
002	CLRG		058	GSB9	
003	CF0	Initialize	059	RTN	
004	CF1		060	*LBL5	
005	0		061	1	
006	RTN		062	STO5	
007	*LBLB	Set flag 0 for print.	063	RCL3	Even ν_1.
008	SF0		064	-	
009	1		065	STO3	
010	RTN		066	RCL2	
011	*LBLC	Input ν_1.	067	2	
012	STO1		068	÷	
013	GSB9		069	x	
014	RTN		070	ST+5	
015	*LBLD		071	DSZI	
016	STO2	Input ν_2.	072	GTO3	
017	GSB9		073	GTO2	
018	RTN		074	*LBL3	
019	*LBLE		075	RCL2	
020	GSB9	Input x.	076	2	
021	STO6		077	+	
022	ENT↑		078	STO2	
023	RCL1		079	RCL7	
024	x		080	2	
025	RCL2		081	+	
026	+		082	STO7	
027	RCL2		083	÷	
028	X≠Y		084	RCL3	
029	÷		085	x	
030	STO3		086	x	
031	RCL1		087	ST+5	
032	2	If ν_1 even then go to LBL d	088	DSZI	
033	÷	else go to LBL e.	089	GTO3	
034	FRC		090	*LBL2	Even ν_1.
035	0		091	RCL5	
036	X=Y?		092	RCL4	
037	GTOd		093	x	
038	GTOe		094	F1?	
039	RTN		095	GSB9	
040	*LBLd		096	F1?	
041	RCL3		097	GSB8	
042	RCL2		098	RTN	
043	2		099	*LBLe	
044	STO7		100	CF1	
045	÷		101	RCL1	
046	Y^x	For even ν_1.	102	RCL2	
047	STO4		103	STO1	
048	RCL1		104	X≠Y	For odd ν_1.
049	2		105	STO2	
050	-		106	1	
051	2		107	RCL3	
052	÷		108	-	
053	STOI		109	STO3	
054	0		110	GSBd	
055	X≠Y?		111	SF1	
056	GTO5		112	1	

REGISTERS

0	1 ν_1 or ν_2	2 ν_2 or ν_1	3 t, 1 − t	4 $t^{\nu_1/2}$ or $t^{\nu_1/2}$	5 Used	6 x	7 Used	8 Used	9 Used
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9

A	B	C	D	E	I Used

```
113   X≠Y
114   -
115   GSB9
116   GSB8
117   R/S
118   *LBL9
119   F0?
120   PRTX
121   RTN
122   *LBL8
123   F0?
124   SPC
125   RTN
```

Subroutine for print.

Subroutine for space.

LABELS					FLAGS
A Start	B Print?	C $v_1 \rightarrow$	D $v_2 \rightarrow$	E x→P(x)	0 Print
a	b	c	d even v_1	e odd v_1	1 v_1 even
0	1	2 output	3 for P(x)	4	2
5 for P(x)	6	7	8 Space	9 Print	3

SET STATUS		
FLAGS	TRIG	DISP
ON OFF 0 ☐ ☒ 1 ☐ ☒ 2 ☐ ☒ 3 ☐ ☒	DEG ☒ GRAD ☐ RAD ☐	FIX ☒ SCI ☐ ENG ☐ n 2

Multiple Linear Regression

001	*LBL A			057	X^2		
002	CLRG			058	GSB 2		
003	CF 0	Initialize		059	ST+ i		
004	CF 1			060	RTN		
005	0			061	*LBL E		
006	RTN			062	RCL 0		
007	*LBL C			063	RCL 4		
008	STO C			064	×		
009	R↓			065	RCL 7		
010	STO B			066	X^2		
011	R↓			067	−		
012	STO A			068	STO D		
013	GSB 7			069	RCL 0		
014	7			070	RCL 3		
015	STO I			071	×		
016	R↓	Input x_i, y_i, z_i.		072	RCL 8		
017	GSB 1			073	RCL 9		
018	8			074	×		
019	STO I			075	−		
020	RCL B	Compute Σx_i, Σy_i, Σz_i;		076	×		
021	GSB 9	Σx_i^2, Σy_i^2, Σz_i^2, $\Sigma x_i y_i$,		077	STO C		
022	GSB 1	$\Sigma y_i z_i$, $\Sigma z_i x_i$.		078	RCL 0	Calculating a, b, c.	
023	9			079	RCL 1		
024	STO I			080	×		
025	RCL C			081	RCL 7		
026	GSB 9			082	RCL 8		
027	GSB 1			083	×		
028	RCL A			084	−		
029	RCL B			085	STO A		
030	×			086	RCL 0		
031	GSB 2			087	RCL 2		
032	ST+ 1			088	×		
033	RCL A			089	RCL 7		
034	RCL C			090	RCL 9		
035	×			091	×		
036	GSB 2			092	−		
037	ST+ 2			093	STO B		
038	RCL B			094	×		
039	RCL C			095	RCL C		
040	×			096	X⇄Y		
041	GSB 2			097	−		
042	ST+ 3			098	RCL D		
043	1			099	RCL 0		
044	GSB 2			100	RCL 5		
045	ST+ 0			101	×		
046	RCL 0			102	RCL 8		
047	GSB 9			103	X^2		
048	RTN			104	−		
049	*LBL 1			105	×		
050	GSB 2	Subroutine for Σx_i, Σx_i^2 ...		106	RCL A		
051	ST+ i			107	X^2		
052	RCL I			108	−		
053	3			109	÷		
054	−			110	STO C		
055	STO I			111	RCL B		
056	R↓			112	RCL A		

REGISTERS

0 n	1 $\Sigma x_i y_i$	2 $\Sigma x_i z_i$	3 $\Sigma y_i z_i$	4 Σx_i^2	5 Σy_i^2	6 Σz_i^2	7 Σx_i	8 Σy_i	9 Σz_i
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9

A Used, a	B Used, b	C z_i, B in b, c	D $[n\Sigma x_j^2 - (\Sigma x_j)^2]$	E $n\Sigma x_i^2 - (\Sigma x_{ij})^2$	I

```
113  RCLC
114    x
115    -
116  RCLD
117    ÷
118  STOB
119  RCL9
120  RCLC
121  RCL8
122    x
123    -
124  RCLB
125  RCL7
126    x
127    -
128  RCL0
129    ÷
130  STOA
131  GSB7
132   R/S
133  RCLB
134  GSB9
135   R/S
136  RCLC
137  GSB9
138   R/S
139 *LBLa
140  RCLA
141  RCL9
142    x
143  RCLB
144  RCL2
145    x
146    +
147  RCLC
148  RCL3
149    x
150    +
151  RCL9
152    X²
153  RCL0
154    ÷
155    -
156  RCL6
157  RCL9
158    X²
159  RCL0
160    ÷
161    -
162    ÷
163  GSB7
164   RTN
165 *LBLb
166   X≷Y
167  GSB7
168   X≷Y
```

a

b

c

Calculating R^2

```
169  GSB9
170  ENT↑
171  RCLC
172    x
173   X≷Y
174  RCLB
175    x
176    +
177  RCLA
178    +
179  GSB9
180   RTN
181 *LBLc
182   F0?
183   SPC
184     6
185  STOI
186  GTOB
187   RTN
188 *LBLd
189     3
190  STOI
191  GSB8
192   RTN
193 *LBLe
194     0
195  STOI
196  GSB8
197   RTN
198 *LBL8
199  ISZI
200  RCLi
201  GSB9
202   R/S
203  GTOB
204   RTN
205 *LBLD
206   SF1
207  GSBC
208   CF1
209   RTN
210 *LBLB
211   SF0
212     1
213   RTN
214 *LBL7
215   F0?
216   SPC
217 *LBL9
218   F0?
219   PRTX
220   RTN
221 *LBL2
222   F1?
223   CHS
224   RTN
```

Calculating approximated z value for given x, y.

Σx_i, Σy_i, Σz_i

Σx_i^2, Σy_i^2, Σz_i^2

$\Sigma x_i y_i$, $\Sigma x_i z_i$, $\Sigma y_i z_i$

Print for LBL a
LBL b
LBL c

For correction of x_k, y_k, z_k.

Set flag 0 for print.

Print x_i, y_i, z_i.

Change sign for correction.

LABELS

A Start	B Print?	C Input	D Correction	E →a; b; c	0 Print
a →R²	b x↑y→z	c →Σx_i, ...	d →Σx_i^2, ...	e →$\Sigma x_i y_i$, ...	1 Correction
0	1 Σx_i, Σx_i^2	2 CHS	3	4	2
5	6	7 Space	8 Print	9 Print	3

FLAGS

SET STATUS

FLAGS	ON	OFF
0	☐	☒
1	☐	☒
2	☐	☒
3	☐	☒

TRIG	
DEG	☒
GRAD	☐
RAD	☐

DISP	
FIX	☒
SCI	☐
ENG	☐
n	2

Polynomial Approximation (Card 1)

001	*LBL*A		057	ST+6	
002	CLRG	Initialize	058	GTO0	
003	CF0		059	*LBLe	
004	CF1		060	RCL8	
005	0		061	ENT↑	
006	RTN		062	+	
007	*LBLC	Input N.	063	1	Calculate $f_n(i)$ $f(x_i)$.
008	F0?		064	+	
009	PRTX		065	x	
010	STO7		066	RCL7	
011	F0?		067	RCL6	
012	SPC		068	-	
013	RTN		069	x	
014	*LBLD	Input y_i.	070	X≠Y	
015	F1?		071	RCL7	
016	GTO7		072	RCL8	
017	SF1		073	+	
018	F0?		074	1	
019	PRTX		075	+	
020	STO1		076	x	
021	STO2		077	RCL8	
022	STO3		078	x	
023	STO4		079	-	
024	STO5		080	RCL7	
025	2		081	RCL8	
026	STO6		082	-	
027	*LBL0		083	÷	
028	1		084	RCL8	
029	STO8		085	1	
030	RCL6		086	+	
031	2		087	STO8	
032	÷		088	÷	
033	F0?		089	RCL9	
034	PRTX		090	X≠Y	
035	F0?		091	STO9	
036	SPC		092	RTN	
037	R/S		093	*LBLE	
038	*LBL7		094	F0?	Input order of polynomial.
039	F0?		095	PRTX	
040	PRTX		096	ENT↑	
041	ST+1		097	2	
042	1		098	X=Y?	
043	RCL6		099	GTO2	
044	RCL7		100	CLX	
045	÷		101	3	
046	-		102	X=Y?	
047	x		103	GTO3	
048	ST+2		104	CLX	
049	STO9		105	4	
050	GSBe		106	X=Y?	
051	ST+3		107	GTO4	
052	GSBe		108	CLX	
053	ST+4		109	0	
054	GSBe		110	÷	
055	ST+5		111	*LBL2	
056	2		112	CLX	

REGISTERS

0	1 (f, f_0), a_0	2 (f, f_1), a_1	3 (f, f_2), a_2	4 (f, f_3), a_3	5 (f, f_4), a_4	6 2j	7 N	8 n	9 $f_n(i)$ $f(x_i)$
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9

A 1 for print	B	C	D	E	I

```
113  STO4
114  *LBL3
115    CLX
116  STO5
117  *LBL4
118  RCL1
119  RCL7
120     1
121  STO8
122     +
123     ÷
124  STO1
125  GSBd
126  ST÷2
127  GSBd
128  ST÷3
129  GSBd
130  ST÷4
131  GSBd
132  ST÷5
133  RCL7
134  RCL7
135  RCL7
136     1
137     -
138     2
139     x
140  STO6
141     ÷
142     3
143     x
144  STO8
145    R↓
146     2
147     +
148  RCL6
149     ÷
150  STO6
151  RCL3
152     x
153  ST-1
154  RCL8
155  RCL3
156     x
157  STO3
158  RCL7
159  RCL7
160  RCL7
161     2
162     -
163     ÷
164     5
165     x
166     3
167     ÷
168  RCL8
```

```
169   X≷Y
170     x
171  STO9
172  LSTX
173  RCL6
174     x
175  STO7
176    CLX
177    R/S
178  *LBLd
179  RCL7
180  RCL8
181     1
182     +
183     +
184     N!
185  RCL7
186  RCL8
187     -
188     N!
189     x
190  RCL8
191  RCL8
192     1
193     +
194  STO8
195     +
196     ÷
197  RCL7
198     N!
199  ENT↑
200     x
201     ÷
202   RTN
203  *LBLB
204    SF0
205     1
206  STOA
207   RTN
```

(f, fi)

Set flag 0 for print.

LABELS					FLAGS
A Start	B Print?	C N→	D y_i→	E n→	0 Print
a	b	c	d	e (f_n, f_n)	1 y_i, i > 1
0 Print i	1	2 Used	3 Used	4 Used	2
5	6	7 i > 1	8	9 $f_n(i)$ $f(x_i)$	3

SET STATUS

FLAGS			TRIG	DISP
	ON	OFF		
0	☐	☒	DEG ☒	FIX ☒
1	☐	☒	GRAD ☐	SCI ☐
2	☐	☒	RAD ☐	ENG ☐
3	☐	☒		n 2

(Card 2)

001	*LBL₀		057	RCL7	
002	R↑		058	X≷Y	
003	R↑		059	×	
004	3		060	ST+8	
005	+		061	LSTX	
006	ENT↑		062	RCL9	
007	ENT↑		063	×	
008	5		064	RCL5	
009	−		065	×	
010	÷		066	STO5	
011	2		067	LSTX	
012	×		068	ST×8	
013	3		069	RCL6	
014	÷		070	×	
015	RCL7	Continue	071	ST+1	
016	+		072	RCL8	
017	STO7		073	ST−3	
018	RCL4		074	*LBLB	
019	×		075	RCLA	
020	ST−2		076	X>0?	
021	LSTX		077	SF0	
022	RCL9		078	1	
023	×		079	R/S	
024	STO4		080	*LBL6	
025	R↑		081	X≷Y	
026	R↑		082	GSB9	
027	4		083	X≷Y	Input x_N, x_0.
028	+		084	GSB8	
029	ENT↑		085	−	
030	ENT↑		086	STO8	
031	7		087	LSTX	
032	−		088	2	
033	÷		089	×	
034	3		090	+	
035	×		091	RCL8	
036	4		092	÷	
037	÷		093	STO6	
038	RCL6		094	2	
039	X≷Y		095	RCL8	
040	×		096	÷	
041	STO6		097	STO8	
042	LSTX		098	RCL1	
043	RCL8		099	RCL6	
044	×		100	RCL2	
045	STO8		101	×	
046	R↑		102	+	
047	ENT↑		103	RCL6	
048	ENT↑		104	ENT↑	
049	ENT↑		105	×	
050	3		106	RCL3	
051	−		107	×	
052	÷		108	+	
053	7		109	STO1	
054	×		110	RCL2	
055	4		111	RCL6	
056	÷		112	RCL3	

REGISTERS

0	1 b_0,c_0,d_0	2 b_1,c_1,d_1	3 b_2,c_2,d_2	4 b_3,c_3,d_3	5 b_4,c_4,d_4	6 Used	7 Used	8 Used	9
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9

A 1 for print	B	C	D	E	I

```
113    x
114    2
115    x
116    +
117    STO2
118    RCL4
119    RCL6
120    x
121    3
122    x
123    ST+3
124    RCL6
125    x
126    ST+2
127    3
128    ÷
129    RCL6
130    x
131    ST+1
132    RCL5
133    RCL6
134    x
135    4
136    x
137    ST+4
138    RCL6
139    x
140    1
141    .
142    5
143    x
144    ST+3
145    LSTX
146    ÷
147    RCL6
148    x
149    ST+2
150    4
151    ÷
152    RCL6
153    x
154    ST+1
155    RCL8
156    CHS
157    ENT↑
158    ENT↑
159    ENT↑
160    STx2
161    x
162    STx3
163    x
164    STx4
165    x
166    STx5
167    R/S
168    *LBLc
```

```
169    RCL1
170    GSB9
171    R/S
172    RCL2
173    GSB9
174    R/S
175    RCL3
176    GSB9
177    R/S
178    RCL4
179    GSB9
180    R/S
181    RCL5
182    GSB9
183    R/S
184    RTN
185    *LBLd
186    GSB9
187    STO7
188    RCL2
189    x
190    RCL7
191    X²
192    RCL3
193    x
194    +
195    RCL7
196    3
197    Yˣ
198    RCL4
199    x
200    +
201    RCL7
202    X²
203    X²
204    RCL5
205    x
206    +
207    RCL1
208    +
209    GSB8
210    RTN
211    *LBL9
212    F0?
213    SPC
214    *LBL8
215    F0?
216    PRTX
217    RTN
```

Output d_0, d_1, d_2, d_3, d_4.

Input x, calculate $\hat{y}$.

Subroutine for space and print.

LABELS					FLAGS	SET STATUS		
A	B Set flag 0	C	D	E	0 Print	FLAGS	TRIG	DISP
a (continue)	b $x_N \uparrow x_0$	c $\rightarrow d_0, d_1,$	d $x \rightarrow \hat{y}$	e	1	ON OFF 0 ☐ X 1 ☐ X	DEG X GRAD ☐	FIX X SCI ☐
0	1	2	3	4	2	2 ☐ X	RAD ☐	ENG ☐
5	6	7	8 Print	9 Used	3	3 ☐ X		n 2

t Statistics

001	*LBLA		057	-	
002	CF0		058	ST-2	
003	CF1	Initialize	059	X²	
004	0		060	ST-3	
005	STO1		061	RCL1	
006	STO2		062	1	
007	STO3		063	-	
008	RTN		064	STO1	
009	*LBLC	Input x_i, y_i for paired t	065	GSB0	
010	F0?	statics.	066	RTN	Set flag 0 for print.
011	GSB9		067	*LBLE	
012	-		068	SF0	
013	ST+2		069	1	
014	X²		070	RTN	
015	ST+3		071	*LBL9	Print x_i, y_i.
016	RCL1		072	X≷Y	
017	1		073	SPC	
018	+		074	PRTX	
019	STO1		075	X≷Y	
020	GSB0		076	PRTX	
021	RTN		077	RTN	
022	*LBLE	$\bar{D}$, S_D, t_1, df_1	078	*LBL6	Input x_i or y_i for t
023	RCL2		079	GSB0	statics for two means.
024	RCL1		080	ST+2	
025	÷		081	X²	
026	GSB1		082	ST+3	
027	GSB0		083	RCL1	
028	R/S		084	1	
029	RCL3		085	+	
030	RCL2		086	STO1	
031	X²		087	GSB0	
032	RCL1		088	GSB1	
033	÷		089	RTN	
034	-		090	*LBL6	
035	RCL1		091	STO7	
036	1		092	GSB0	
037	-		093	GSB1	
038	÷		094	F1?	
039	√X		095	R/S	
040	GSB0		096	RCL1	
041	R/S		097	STO4	d
042	RCL1		098	RCL2	
043	√X		099	STO5	
044	÷		100	RCL3	
045	÷		101	STO6	
046	GSB0		102	0	
047	R/S		103	STO1	
048	RCL1		104	STO2	
049	1		105	STO3	
050	-		106	RCL7	
051	GSB0		107	SF1	
052	GSB1		108	RTN	
053	RTN		109	*LBL7	
054	*LBLD	For correction of x_k, y_k.	110	RCL6	
055	F0?		111	RCL5	
056	GSB9		112	X²	

REGISTERS

0 Used	1 n, n_1, n_2	2 Used	3 Used	4 n_1	5 Σx_i	6 Σx_i^2	7 d	8 n_1+n_2-2	9
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9

A	B	C	D	E	I

113 RCL4 114 ÷ 115 − 116 RCL3 117 + 118 RCL2 119 X² 120 RCL1 121 ÷ 122 − 123 RCL1 124 RCL4 125 + 126 2 127 − 128 STO8 129 ÷ 130 √x 131 1 132 RCL1 133 ÷ 134 1 135 RCL4 136 ÷ 137 + 138 1/x 139) 140 RCL5 141 RCL4 142 ÷ 143 RCL2 144 RCL1 145 ÷ 146 − 147 RCL7 148 − 149 x≷y 150 ÷ 151 GSB0 152 R/S 153 RCL8 154 GSB0 155 GSB1 156 RTN 157 *LBLk 158 GSB0 159 ST-2 160 x² 161 ST-3 162 RCL1 163 1 164 − 165 STO1 166 GSB0 167 GSB1 168 RTN	t_2, df_2	For correction of x_k or y_k.	169 *LBL0 170 F0? 171 PRTX 172 RTN 173 *LBL1 174 F0? 175 SPC 176 RTN	Subroutine for space.

LABELS					FLAGS	SET STATUS		
A Start	B Print	C $x_i \uparrow y_i$	D $x_k \uparrow y_k \rightarrow$	E $\rightarrow \overline{D}, S_D$ ...	0 Print	FLAGS	TRIG	DISP
a x_i or y_i	b x_k or y_k	c $d \rightarrow$	d t_2, df_2	e	1 New d	ON OFF 0 ☐ ☒ 1 ☐ ☒ 2 ☐ ☒ 3 ☐ ☒	DEG ☒ GRAD ☐ RAD ☐	FIX ☒ SCI ☐ ENG ☐ n 2
0 Print	1 Space	2	3	4	2			
5	6	7	8	9 Print x_i, y_i	3			

Chi-Square Evaluation

001	*LBL A		057	RCL 3	
002	CF 0	Initialize	058	÷	
003	CF 1		059	ST- 2	
004	0		060	RCL 1	
005	STO 1		061	1	
006	STO 2		062	-	
007	STO 3		063	STO 1	
008	RTN		064	GSB 8	
009	*LBL C		065	RTN	
010	F0?		066	*LBL a	For equal expected frequency.
011	GSB 9		067	GSB 8	
012	STO 3	$\Sigma+$	068	ST+ 2	
013	-		069	X^2	
014	X^2		070	ST+ 3	
015	RCL 3		071	RCL 1	
016	÷		072	1	
017	ST+ 2		073	+	
018	RCL 1		074	STO 1	
019	1		075	GSB 8	
020	+		076	GSB 7	
021	STO 1		077	SF 1	
022	GSB 8		078	RTN	
023	RTN		079	*LBL b	
024	*LBL E		080	GSB 8	For correction O_j.
025	F1?		081	ST- 2	
026	GTO 1	χ_1^2	082	X^2	
027	RCL 2		083	ST- 3	
028	GSB 7		084	RCL 1	
029	GSB 8		085	1	
030	GSB 7		086	-	
031	RTN		087	STO 1	
032	*LBL 1		088	GSB 8	
033	1		089	GSB 7	
034	RCL 1	χ_2^2	090	RTN	
035	RCL 3		091	*LBL B	Set flag 0 for print.
036	x		092	SF 0	
037	RCL 2		093	1	
038	÷		094	RTN	
039	RCL 2		095	*LBL 9	Print O_i, E_i.
040	-		096	SPC	
041	GSB 7		097	X≠Y	
042	GSB 8		098	PRTX	
043	RTN		099	X≠Y	
044	*LBL d		100	PRTX	
045	RCL 2		101	RTN	
046	RCL 1		102	*LBL c	Set flag 1 for equally expected frequency.
047	÷		103	SF 1	
048	GSB 8		104	GTO E	
049	GSB 7		105	RTN	
050	RTN		106	*LBL 8	Subroutine for print x.
051	*LBL D		107	F0?	
052	F0?	For correction O_j, E_j.	108	PRTX	
053	GSB 9		109	RTN	
054	STO 3		110	*LBL 7	Subroutine for space.
055	-		111	F0?	
056	X^2		112	SPC	

REGISTERS

0	1 n	2 Used	3 Used	4	5	6	7	8	9
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9

A	B	C	D	E	I

```
113  RTN
```

LABELS					FLAGS	SET STATUS		
A Start	B Print	C O_i↑E_i→	D O_k↑E_k	E χ_1^2	0 Print	FLAGS	TRIG	DISP
a O_l (Σ+)	b O_h (Σ−)	c χ_2^2	d E	e	1 Equal E_i	ON OFF 0 ☐ x 1 ☐ x 2 ☐ x 3 ☐ x	DEG x GRAD ☐ RAD ☐	FIX x SCI ☐ ENG ☐ n 2
0	1 χ_2^2	2	3	4	2			
5	6	7 Space	8 Print	9 Print	3			

Contingency Table

001	*LBL A		057	*LBL a	
002	CLRG	Initialize	058	SF 0	Set flag for print.
003	CF 0		059	1	
004	CF 1		060	RTN	Print x_{1j}, x_{2j}.
005	0		061	*LBL 9	
006	RTN		062	X⇄Y	
007	*LBL B	Input x_{1j}, x_{2j} for 2 x k.	063	PRTX	
008	F0?		064	X⇄Y	
009	GSB 9		065	PRTX	
010	0		066	RTN	
011	STO 7		067	*LBL b	3 x k
012	R↓		068	F0?	
013	GSB 0		069	GSB 8	Input x_{1j}, x_{2j}, x_{3j}.
014	RTN		070	STO C	
015	*LBL D		071	F1?	
016	RCL 0		072	CHS	
017	RCL 4	2 x k	073	ST+ 3	
018	x	χ^2	074	STO 0	
019	RCL 1		075	STO 7	
020	÷		076	ENT↑	
021	RCL 0		077	x	
022	RCL 5		078	STO 8	
023	RCL 2		079	R↓	
024	÷		080	*LBL 0	Input x_{1j}, x_{2j}, for 3 x k and 2 x k.
025	x		081	STO B	
026	+		082	F1?	
027	RCL 0		083	CHS	
028	-		084	ST+ 2	
029	STO D		085	ST+ 0	
030	GSB 7		086	ST+ 7	
031	R/S		087	ENT↑	
032	RCL D	2 x k	088	x	
033	RCL 0	C_c	089	STO 9	
034	RCL D		090	R↓	
035	+		091	STO A	
036	÷		092	F1?	
037	∫X		093	CHS	
038	GSB 7		094	ST+ 1	
039	GSB 6		095	ST+ 0	
040	RTN		096	ST+ 7	
041	*LBL E	2 x k	097	ENT↑	
042	RCL 1	R_1, R_2, T	098	x	
043	GSB 7		099	RCL 7	
044	R/S		100	÷	
045	RCL 2		101	ST+ 4	
046	GSB 7		102	RCL 9	
047	R/S		103	RCL 7	
048	+		104	÷	
049	GSB 7		105	ST+ 5	
050	GSB 6		106	RCL 8	
051	RTN		107	RCL 7	
052	*LBL C	2 x k correction.	108	÷	
053	SF 1		109	ST+ 6	
054	GSB B		110	1	
055	CF 1		111	F1?	
056	RTN		112	CHS	

REGISTERS

0 T	1 R_1	2 R_2	3 R_3	4 $\Sigma^2 x_{1j}/C_j$	5 $\Sigma^2 x_{2j}/C_j$	6 $\Sigma^2 x_{3j}/C_j$	7 C_j	8 x_{3j}^2	9 x_{2j}^2
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9

A x_{1j}	B x_{2j}	C x_{3j}	D χ^2	E k	I

113	RCLE			169	RCL0		
114	+			170	GSB7		
115	STOE			171	RTN		
116	GSB7			172	*LBLc		3 x k correction.
117	GSB6			173	SF1		
118	CF1			174	GSB6		
119	R/S			175	CF1		
120	RCL7			176	RTN		
121	GSB7			177	*LBL8		Print x_{1j}, x_{2j}, x_{3j}.
122	GSB6			178	R↓		
123	RTN			179	R↓		
124	*LBLd			180	PRTX		
125	RCL1			181	R↑		
126	RCL2			182	PRTX		
127	RCL3			183	R↑		
128	+			184	PRTX		
129	+			185	RTN		
130	STO0		3 x k	186	*LBL7		
131	RCL4		χ^2	187	F0?		Subroutine for print.
132	RCL1			188	PRTX		
133	÷			189	RTN		
134	STO9			190	*LBL6		
135	RCL5			191	F0?		Subroutine for space.
136	RCL2			192	SPC		
137	÷			193	RTN		
138	ST+9						
139	RCL6						
140	RCL3						
141	÷						
142	ST+9						
143	RCL9						
144	1						
145	−						
146	RCL0						
147	x						
148	GSB7						
149	R/S						
150	ENT↑						
151	ENT↑		3 x k				
152	RCL0		C_c				
153	+						
154	÷						
155	∫X						
156	GSB7						
157	GSB6						
158	R/S						
159	*LBLe						
160	RCL1		3 x k				
161	GSB7		R_1, R_2, R_3, T				
162	R/S						
163	RCL2						
164	GSB7						
165	R/S						
166	RCL3						
167	GSB7						
168	R/S						

LABELS					FLAGS	SET STATUS		
A Start	B x_{1j}, x_{2j}	C Correction	D χ^2; C_c	E R_1; R_2; T	0 Print	FLAGS	TRIG	DISP
a Print	b Input 3 x k	c Correction	d χ^2; C_c	e R_1; R_2	1 Correction	ON OFF 0 ☐ ☒ 1 ☐ ☒ 2 ☐ ☒ 3 ☐ ☒	DEG ☒ GRAD ☐ RAD ☐	FIX ☒ SCI ☐ ENG ☐ n 2
0 Used	1	2	3	4	2			
5	6 Space	7 Print	8 Print	9 Print	3			

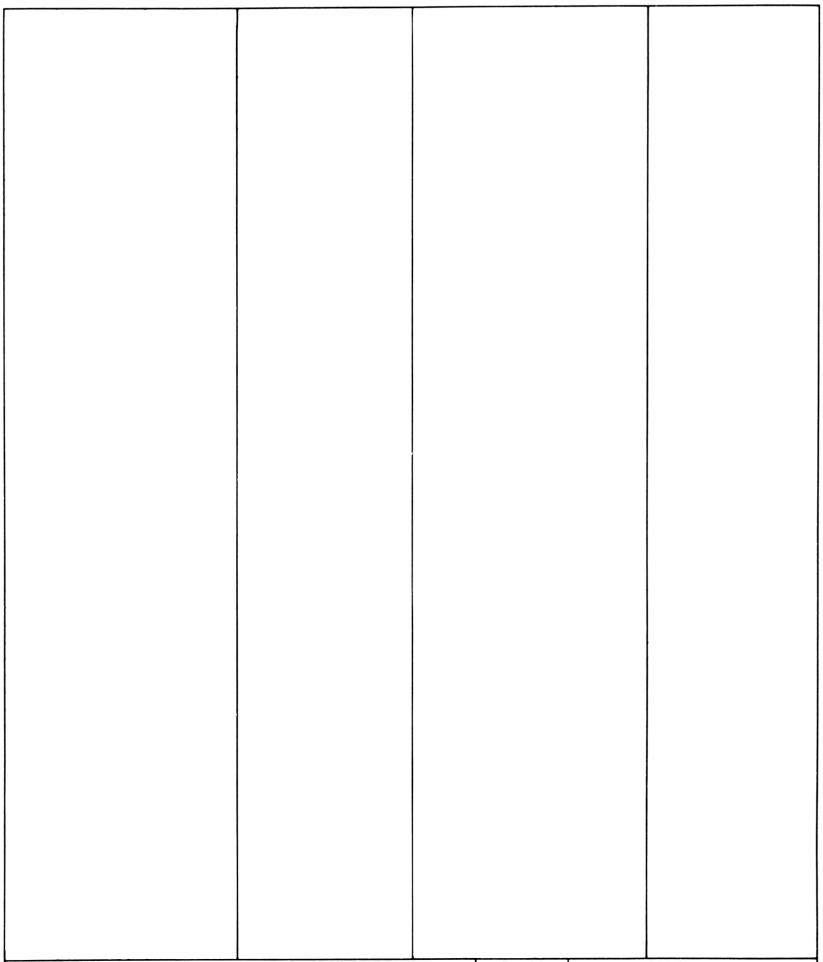
Spearman's Rank Correlation Coefficient

001	*LBLA		057	X≠Y	
002	CLRG		058	PRTX	
003	CF0	Initialize	059	X≠Y	Print R_i, S_i.
004	CF1		060	PRTX	
005	0		061	RTN	
006	RTN		062	*LBL0	Subroutine for print.
007	*LBLC		063	F0?	
008	F0?		064	PRTX	
009	GSB9	Input R_i, S_i.	065	RTN	Subroutine for space.
010	-		066	*LBL1	
011	X²		067	F0?	
012	F1?		068	SPC	
013	CHS		069	RTN	
014	ST+2				
015	RCL1				
016	1				
017	F1?				
018	CHS				
019	+				
020	STO1				
021	GSB0				
022	GSB1				
023	RTN				
024	*LBLE				
025	1				
026	RCL2				
027	6				
028	x				
029	RCL1				
030	X²				
031	1				
032	-				
033	RCL1				
034	x				
035	÷				
036	-	r_s, z			
037	GSB0				
038	R/S				
039	RCL1				
040	1				
041	-				
042	∫X				
043	x				
044	GSB0				
045	GSB1				
046	RTN				
047	*LBLD				
048	SF1				
049	GSBC	For correction.			
050	CF1				
051	RTN				
052	*LBLB				
053	SF0				
054	1	Set flag 0 for print.			
055	RTN				
056	*LBL9				

REGISTERS

0	1	2	3	4	5	6	7	8	9
	n	ΣD_i^2							
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9

A	B	C	D	E	I



LABELS					FLAGS	SET STATUS		
A Start	B Print	C $(\Sigma+)$	D $(\Sigma-)$	E $\rightarrow r_s, z$	0 Print	FLAGS	TRIG	DISP
a	b	c	d	e	1 $(\Sigma-)$	ON OFF 0 ☐ X 1 ☐ X	DEG X GRAD ☐	FIX X SCI ☐
0	1	2	3	4	2	2 ☐ X 3 ☐ X	RAD ☐	ENG ☐ n 2
5	6	7	8	9	3			

$\bar{x}$ and R Control Chart

001	*LBLA		057	R/S	
002	CLRG		058	*LBLE	
003	CF0	Initialize	059	RCL5	
004	CF1		060	GSB9	
005	0		061	RTN	
006	RTN	Store 1 in R_E for print.	062	*LBLa	
007	*LBLB		063	CF1	
008	1		064	RCL6	
009	STOE		065	1	
010	RTN		066	+	
011	*LBLC		067	STO6	
012	STO0		068	RCL2	$\bar{x}_i$
013	RCL4		069	RCL1	
014	STO4		070	÷	
015	RCL5		071	GSB9	
016	STO5		072	ST+7	
017	RCL0		073	R/S	
018	GSB9		074	*LBLa	
019	F1?		075	RCL4	
020	GTO1		076	RCL5	
021	0	Input x_{ij}.	077	-	R_i
022	STO1		078	ST+8	
023	STO2		079	GSB9	
024	STO3		080	GSB7	
025	X≠Y		081	R/S	
026	STO4		082	*LBL6	
027	STO5		083	RCL7	
028	SF1		084	RCL6	$\bar{\bar{x}}$
029	*LBL1		085	÷	
030	RCL4		086	GSB9	
031	X≠Y		087	RTN	
032	X>Y?		088	*LBL6	
033	STO4		089	RCL8	
034	RCL5		090	RCL6	
035	X≠Y		091	÷	
036	X≠Y?		092	STO3	$\bar{R}$
037	STO5		093	GSB9	
038	F0?		094	GSB7	
039	CHS		095	R/S	
040	ST+2		096	*LBLc	
041	X²		097	RCL3	
042	F0?		098	x	
043	CHS		099	RCL7	
044	ST+3		100	RCL6	$L_{\bar{x}}$
045	RCL1		101	÷	
046	1		102	X≠Y	
047	F0?		103	-	
048	CHS		104	GSB9	
049	+		105	R/S	
050	STO1		106	*LBLc	
051	GSB9		107	LSTX	
052	RTN		108	2	$U_{\bar{x}}$
053	*LBLE		109	x	
054	GSB7	x_{max}, x_{min}	110	+	
055	RCL4		111	GSB9	
056	GSB9		112	GSB7	

REGISTERS

0	1	2	3	4	5	6	7	8	9
x_{ij}	n	Σx_{ij}	Σx_{ij}^2, $\bar{R}$	x_{max}	x_{min}	m	$\Sigma \bar{x}_i$	ΣR_i	Used
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9

A	B	C	D	E	I
Last x_{max}	Last x_{min}			1 for print	

113 R/S			
114 *LBL*d*			
115 RCL3	L_R		
116 x			
117 GSB9			
118 GSB7			
119 R/S			
120 *LBLe			
121 RCL3	U_R		
122 x			
123 GSB9			
124 GSB7			
125 RTN			
126 *LBLD	Correction x_{ik}.		
127 STO0			
128 RCLA			
129 STO4			
130 RCLB			
131 STO5			
132 RCL0			
133 SF0			
134 GSBC			
135 RCLA			
136 STO4			
137 RCLB			
138 STO5			
139 CF0			
140 RCL1			
141 RTN			
142 *LBL9	Subroutine for print.		
143 RCLE			
144 X>0?			
145 GTO8			
146 R↓			
147 RTN			
148 *LBL8			
149 R↓			
150 PRTX			
151 RTN			
152 *LBL7	Subroutine for space.		
153 RCLE			
154 X>0?			
155 SPC			
156 R↓			
157 RTN			

LABELS					FLAGS	SET STATUS		
A Start	B Print	C x_{ij} ($\Sigma+$)	D x_{ik} ($\Sigma-$)	E x_{max}, x_{min}	0 Correction	FLAGS	TRIG	DISP
a $\bar{x}_{ij}$, R_i	b $\bar{\bar{x}}$, $\bar{R}$	c $L_{\bar{x}}$, $U_{\bar{x}}$	d L_R	e U_R	1 1st data	ON OFF 0 ☐ ☒ 1 ☐ ☒ 2 ☐ ☒ 3 ☐ ☒	DEG ☒ GRAD ☐ RAD ☐	FIX ☒ SCI ☐ ENG ☐ n 2
0	1 j>1	2	3	4	2			
5	6	7 Space	8 Print	9 Print?	3			

Operating Characteristic Curves

001	*LBLA		057	÷	
002	CF0		058	LSTX	
003	CLRG	Initialize	059	RCL1	
004	0		060	RCL4	
005	RTN		061	−	
006	*LBLB	Set flag 0 for print.	062	RCL2	
007	SF0		063	−	
008	1		064	+	
009	RTN		065	÷	
010	*LBLC	Store N.	066	RCL6	
011	GSB3		067	×	
012	STO1		068	STO6	
013	RTN		069	ST+7	
014	*LBLD		070	RCL3	
015	STO3	Store n, c for finite lot	071	1	
016	X≷Y	size.	072	RCL8	
017	STO2		073	+	
018	GSB3		074	STO8	
019	X≷Y		075	X≠Y?	
020	GSB3		076	GTO0	
021	RTN		077	1	
022	*LBLE		078	RCL7	
023	GSB4		079	X>Y?	
024	GSB3		080	X≷Y	
025	RCL1		081	GSB3	
026	×		082	R/S	
027	INT		083	*LBLa	
028	STO4		084	−	
029	RCL1		085	LSTX	
030	RCL2	Finite lot size.	086	X≤Y?	
031	GSBa		087	GSB9	
032	RCL1		088	STO5	
033	RCL4		089	1	
034	−		090	STO7	
035	RCL2		091	+	
036	GSBa		092	STO6	
037	R↑		093	CLX	
038	÷		094	X=Y?	
039	STO5		095	GTO8	
040	STO6		096	*LBL1	
041	STO7		097	R↓	
042	RCL3		098	1	
043	0		099	RCL7	
044	STO8		100	+	
045	X=Y?		101	STO7	
046	GSB5		102	X>Y?	
047	*LBL0		103	GTO7	
048	RCL4		104	RCL5	
049	−		105	X≷Y	
050	RCL8		106	+	
051	RCL2		107	LSTX	
052	−		108	÷	
053	×		109	RCL6	
054	RCL8		110	×	
055	1		111	STO6	
056	+		112	GTO1	

REGISTERS

0	1 N, n	2 n, p	3 c, f(0)	4 M	5 f(0)	6 , c	7 , counter	8 Used	9
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9

A	B	C	D	E	I

```
113   RTN
114  *LBL7
115   RCL6
116   RTN
117  *LBL8
118     1
119   GSB3
120   RTN
121  *LBL9
122   STO6
123    X≠Y
124   RTN
125  *LBLa
126   STO6
127    X≠Y
128   STO1
129   GSB3
130    X≠Y
131   GSB3
132   RTN
133  *LBLe
134   STO2
135   GSB4
136   GSB3
137   STO2
138   RCL2
139     1
140     -
141   CHS
142     ÷
143   STO8
144   LSTX
145   RCL1
146     Y^
147   STO3
148   RCL6
149     0
150   STO7
151   X=Y?
152   GSB6
153   CLX
154   RCL1
155    X≠Y
156   X>Y?
157   GTO6
158   RCL3
159   STO4
160   STO5
161  *LBL2
162   RCL1
163   RCL7
164     -
165   RCL7
166     1
167     +
168     ÷
```

Input n, c for infinite lot size.

```
169   RCL8
170     x
171   RCL4
172     x
173   STO4
174   ST+5
175   RCL7
176     1
177     +
178   STO7
179   RCL6
180   X≠Y?
181   GTO2
182     1
183   RCL5
184   X>Y?
185    X≠Y
186   GSB3
187    R/S
188  *LBL6
189   RCL3
190   GSB3
191    R/S
192   RTN
193  *LBL5
194   RCL5
195   GSB3
196    R/S
197   RTN
198  *LBL3
199    F0?
200   PRTX
201   RTN
202  *LBL4
203    F0?
204   SPC
205   RTN
```

Subroutine for print.

Subroutine for space.

LABELS					FLAGS
A Start	B Print	C N	D n!c→	E p→P_a	0 Print
a f(x)	b	c	d n!c→	e p→P_a	1
0 P_a	1 f(x + 1)	2 Used	3 Print	4	2
5 c, f(0)	6 f(0)	7 c	8 1	9 STO 6	3

SET STATUS

FLAGS		TRIG	DISP
	ON OFF		
0	☐ ☒	DEG ☒	FIX ☒
1	☐ ☒	GRAD ☐	SCI ☐
2	☐ ☒	RAD ☐	ENG ☐
3	☐ ☒		n 2

Single- and Multi-Server Queues

001	*LBLA		057	RCL1	
002	GSB9		058	RCL3	
003	STO1		059	-	
004	STOI		060	÷	
005	R↓	ρ	061	STO4	Lq, L
006	STO2		062	SPC	
007	X≷Y		063	PRTX	
008	STO5		064	R/S	
009	÷		065	*LBLC	
010	STO3		066	RCL3	
011	PRTX		067	+	
012	R/S		068	STO6	
013	*LBLB		069	PRTX	
014	1		070	R/S	
015	STO4		071	*LBLD	
016	0		072	RCL4	
017	*LBL1		073	RCL2	Tq, T
018	RCL4		074	÷	
019	+		075	SPC	
020	LSTX		076	PRTX	
021	RCL3		077	R/S	
022	x		078	*LBLD	
023	RCL1		079	RCL6	
024	RCLI		080	RCL2	
025	-		081	÷	
026	1		082	PRTX	
027	+		083	R/S	
028	÷		084	*LBLE	
029	STO4	P_0, P_b	085	SPC	P(t)
030	R↓		086	PRTX	
031	DSZI		087	RCL1	
032	GTO1		088	RCL5	
033	1		089	x	
034	RCL3		090	RCL2	
035	RCL1		091	-	
036	÷		092	X	
037	-		093	CHS	
038	RCL4		094	e^	
039	X≷Y		095	RCLI	
040	÷		096	x	
041	STOI		097	PRTX	
042	+		098	SPC	
043	1/X		099	R/S	
044	SPC		100	*LBL9	
045	PRTX		101	R↓	Print u, λ, n.
046	R/S		102	R↓	
047	*LBLB		103	SPC	
048	RCLI		104	PRTX	
049	x		105	R↑	
050	STOI		106	PRTX	
051	PRTX		107	R↑	
052	R/S		108	PRTX	
053	*LBLC		109	RTN	
054	RCLI		110	*LBLa	m, n
055	RCL3		111	GSB8	
056	X		112	STO2	

REGISTERS

0	1 n, m	2 λ, n	3 ρ	4 , Lq, k	5 μ, Q_k, L	6 L, ΣQ_k	7 $\Sigma k Q_k$, -F	8 a	9 Used
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9

A	B	C	D	E	I Used, P_b

113	R↓			169	STO5		
114	STO1			170	SPC		
115	R/S			171	PRTX		
116	*LBLb	ρ		172	R/S		
117	GSB8			173	*LBLc		T
118	STO8			174	RCL8		
119	÷			175	x		
120	STO3			176	PRTX		
121	PRTX			177	R/S		
122	R/S			178	*LBLd		L_q
123	*LBLc			179	RCL5		
124	CLX			180	RCL1		
125	STO7			181	÷		
126	1			182	1		
127	STO4			183	-		
128	STO5			184	RCL3		
129	STO6			185	1		
130	*LBL3			186	+		
131	RCL2			187	x		
132	RCL4			188	STO7		
133	X>Y?			189	1		
134	X≠Y			190	+		
135	RCL3			191	RCL1		
136	X≠Y			192	x		
137	÷			193	SPC		
138	RCL1			194	PRTX		
139	RCL4			195	R/S		
140	-			196	*LBLd		T_q
141	1	L		197	RCL8		
142	+			198	x		
143	x			199	PRTX		
144	RCL5			200	R/S		
145	x			201	*LBLe		
146	STO5			202	RCL7		
147	EEX			203	CHS		
148	CHS			204	SPC		F
149	9			205	PRTX		
150	0			206	SPC		
151	X>Y?			207	R/S		
152	GTO2			208	*LBL8		
153	R↓			209	R↓		Print m, n, s, a.
154	ST+6			210	SPC		
155	RCL4			211	PRTX		
156	x			212	R↑		
157	ST+7			213	PRTX		
158	RCL1			214	RTN		
159	RCL4						
160	1						
161	+						
162	STO4						
163	X≤Y?						
164	GTO3						
165	*LBL2						
166	RCL7						
167	RCL6						
168	÷						

LABELS					FLAGS
A ...→ρ	B →P_o,→P_b	C →L_q,→L	D →T_q,→T	E t→P(t)	0
a m↑n→	b S↑a→ρ	c →L,→T	d →L_q,→T_q	e →F	1
0	1 P_o, P_b	2 L	3 K	4	2
5	6	7	8 Print	9 Print	3

SET STATUS			
FLAGS		TRIG	DISP
	ON OFF		
0	☐ ☒	DEG ☒	FIX ☒
1	☐ ☒	GRAD ☐	SCI ☐
2	☐ ☒	RAD ☐	ENG ☐
3	☐ ☒		n 2

Appendix A
MAGNETIC CARD
SYMBOLS AND CONVENTIONS

SYMBOL OR CONVENTION	INDICATED MEANING
White mnemonic: x A	White mnemonics are associated with the user-definable key they are above when the card is inserted in the calculator's window slot. In this case the value of x could be input by keying it in and pressing A.
Gold mnemonic: y x f E	Gold mnemonics are similar to white mnemonics except that the gold f key must be pressed before the user-definable key. In this case y could be input by pressing f E.
x ↑ y A	↑ is the symbol for ENTER↑. In this case ENTER↑ is used to separate the input variables x and y. To input both x and y you would key in x, press ENTER↑, key in y and press A.
[X] A	The box around the variable x indicates input by pressing STO A.
(x) A	Parentheses indicate an option. In this case, x is not a required input but could be input in special cases.
→x A	→ is the symbol for calculate. This indicates that you may calculate x by pressing key A.
→x, y, z A	This indicates that x, y, and z are calculated by pressing A once. The values would be printed in x, y, z order.
→x; y; z A	The semi-colons indicate that after x has been calculated using A, y and z may be calculated by pressing R/S.
→"x ", y A	The quote marks indicate that the x value will be "paused" or held in the display for one second. The pause will be followed by the display of y.
◁▷ x A	The two-way arrow ◁▷ indicates that x may be either output or input when the associated user-definable key is pressed. If numeric keys have been pressed between user-definable keys, x is stored. If numeric keys have not been pressed, the program will calculate x.

SYMBOLS AND CONVENTIONS (Continued)

SYMBOL OR CONVENTION	INDICATED MEANING
P? A	The question mark indicates that this is a mode setting, while the mnemonic indicates the type of mode being set. In this case a print mode is controlled. Mode settings typically have a 1.00 or 0.00 indicator displayed after they are executed. If 1.00 is displayed, the mode is on. If 0.00 is displayed, it is off.
START A	The word START is an example of a command. The start function should be performed to begin or start a program. It is included when initialization is necessary.
DEL A	This special command indicates that the last value or set of values input may be deleted by pressing A.
→x; ... A	Three dots (...) indicate that additional output follows. See User Instructions for complete description of variables output.

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